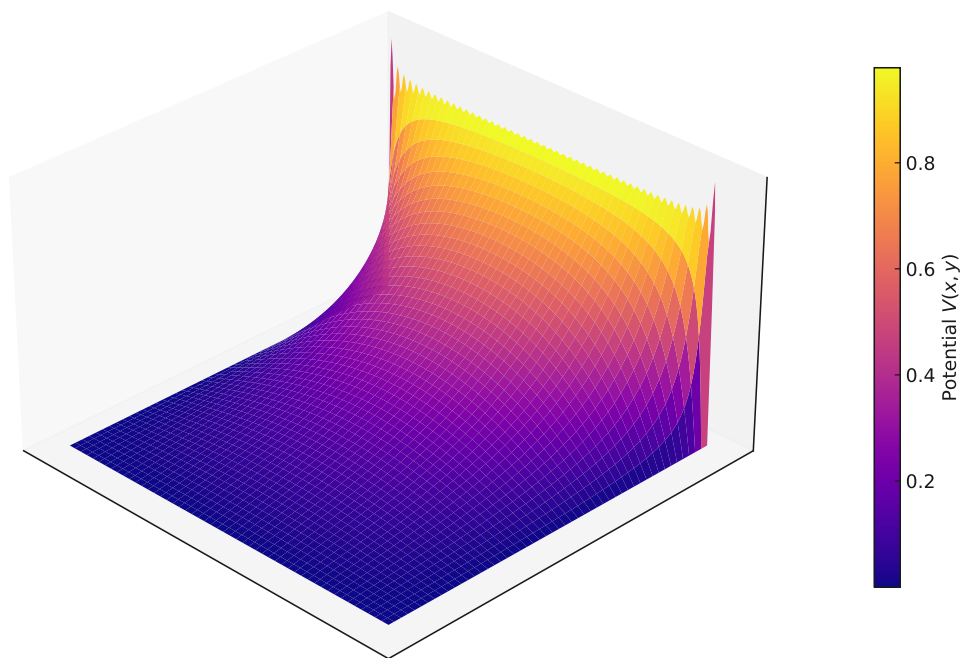

AMATH 453

Partial Differential Equations

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Credit to Dr. Sander Rhebergen for His Lectures on AMATH 453

Preface

A majority of the information presented in this note package is transcribed directly from the *AMATH 453: Partial Differential Equations* lectures during the Spring 2025 term, taught by Dr. Sander Rhebergen.

These notes are by no means exhaustive nor are they guaranteed to be error-free, and I do not make any claims about the reliability or accuracy of these set of notes. They are intended primarily for personal use and for those who may benefit from an alternative presentation of the material.

Please use these notes at your own discretion.

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1 Introduction

1.1 Relevant Notation

1. C : a positive constant. But also C is a space of continuous functions.
2. $U \subset \mathbb{R}^n$ and U is open.
3. If $u : U \rightarrow \mathbb{R}$ is a function, then we write $u(x) = u(x_1, x_2, \dots, x_n)$ for $x \in U$. Here, x is n -dimensional.
4. We say that u is **smooth** provided that u is infinitely differentiable.
5. We will use multi-index notation to denote derivatives. Let $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$ and $|\alpha| = \alpha_1 + \alpha_2 + \dots + \alpha_n$. Then,

$$D^\alpha u(x) = \frac{\partial^{|\alpha|} u}{\partial_{x_1}^{\alpha_1} \partial_{x_2}^{\alpha_2} \dots \partial_{x_n}^{\alpha_n}} = \partial_{x_1}^{\alpha_1} \partial_{x_2}^{\alpha_2} \dots \partial_{x_n}^{\alpha_n} u.$$

6. $\Delta u = \sum_{i=1}^n u_{x_i x_i} = \operatorname{div}(Du) = \operatorname{tr}(D^2 u)$.

Example 1.1. If $\alpha = (3, 1, 5)$, then $|\alpha| = 9$. Thus, multi-index notation gives

$$D^{3,1,5} u(x) = \partial_{x_1}^3 \partial_{x_2} \partial_{x_3}^5 u.$$

In particular, we have

$$D^k u(x) = \{D^\alpha u(x) : |\alpha| = k\}$$

if $k \geq 0$ is an integer.

Case (i) $k = 1$. Then, $Du = D_x u = (u_{x_1}, u_{x_2}, \dots, u_{x_n})$ which is simply the gradient. Notice we do not write $\nabla \cdot u$.

Case (ii) $k = 2$. Then, we arrange the elements of $D^2 u$ in a matrix,

$$D^2 u = \begin{bmatrix} u_{x_1 x_1} & \dots & u_{x_1 x_n} \\ \vdots & \ddots & \vdots \\ u_{x_n x_1} & \dots & u_{x_n x_n} \end{bmatrix}$$

which is the **Hessian matrix**.

1.2 Derivation of a PDE

Consider a body $U \subset \mathbb{R}^3$. Let $U_0 \subset U$ with boundary ∂U_0 . We will assume that U_0 does not change over time (non-critical assumption). Introducing notation, we have $e = e(x, t) \in \mathbb{R}^1$ to be the density of internal energy.

¹**Note** that $x = (x_1, x_2, x_3)$.

The amount of heat in U_0 is defined as

$$\int_{U_0} e \, dx.$$

The heat flux is defined to be $j = j(x, t) \in \mathbb{R}^3$ which is a vector pointing in the direction that heat flows. Now, let n denote the exterior unit normal to ∂U_0 . Then, the net outflow of heat through ∂U_0 is given by

$$\int_{\partial U_0} j \cdot n \, ds.$$

We define $p = p(x, t) \in \mathbb{R}$ to be the power density of heat sources. Heat production in U_0 is defined as

$$\int_{U_0} p \, dx.$$

Conservation of energy within the system $U \subset \mathbb{R}^3$ implies that

rate of change of total energy in U_0 = inflow of heat through ∂U_0 + heat produced in U_0 .

That is,

$$\frac{d}{dt} \int_{U_0} e \, dx = - \int_{\partial U_0} j \cdot n \, ds + \int_{U_0} p \, dx \quad (1)$$

where we attach a negative on the first integral on the RHS due to the inflow of the heat flux. By the Divergence Theorem,

$$\int_{\partial U_0} j \cdot n \, ds = \int_{U_0} \operatorname{div}(j) \, dx$$

and furthermore

$$\frac{d}{dt} \int_{U_0} e \, dx = \int_{U_0} \partial_t e \, dx.$$

Then from 1, we have

$$\int_{U_0} [\partial_t e + \operatorname{div}(j) - p] \, dx = 0, \quad t \geq 0.$$

This is true if the integrand is zero, thus

$$\boxed{\partial_t e + \operatorname{div}(j) = p, \quad \text{in } U \text{ for } t \geq 0.} \quad (2)$$

We need to relate e and j , and to do this we will assume that e depends linearly on temperature, T . Then,

$$e = e_0 + \sigma u$$

where e_0 is an arbitrary constant, $u = T - T_0$ is the reference temperature, and $\sigma = \sigma(x)$ is the specific heat capacity of a given material.

Fourier's Law states that heat flux is proportional to the temperature gradient and that heat is transported by convection with a heat flux be , where $b = b(x, t) \in \mathbb{R}^3$ is a given convective vector field. That is,

$$j = -aDu + be = -aDu + be_0 + b\sigma u \quad (3)$$

where a is the known heat conductivity and Du is the temperature gradient.

Combining (1), (2), and (3), we arrive at the PDE

$$\boxed{\sigma \partial_t u + \operatorname{div}(b\sigma u) - \operatorname{div}(aDu) = p - \operatorname{div}(be_0)} \quad (4)$$

and the RHS is known. We will now define the following,

$$Lu := - \sum_{i,j=1}^n (a^{ij} u_{x_i})_{x_j} + \sum_{i=1}^n b^i u_{x_i} + cu$$

for the given coefficients a^{ij} , b^i , c ($i, j = 1, \dots, n$). The specific forms we will study are

1. Second-order elliptic BVPs:

$$\begin{cases} Lu = f, & \text{in } U \\ u = 0, & \text{on } \partial U_0. \end{cases}$$

2. Second-order IBVPs:

$$\begin{cases} u_t + Lu = f, & \text{for } x \in U, t \in (0, T] \\ u = 0, & \text{on } \partial U_0 \\ u = g, & \text{on } U \text{ for } t = 0. \end{cases}$$

Special cases of BVPs are the Laplace equation ($-\Delta u = 0$), the Poisson equation ($-\Delta u = f$), and the Heat Equation ($u_t - \Delta u = 0$).

1.3 Banach Spaces

Let us assume that X is a real linear space.

Definition 1.3.1. A mapping, denoted by $\|\cdot\| : X \rightarrow [0, \infty)$, is a **norm** if

- (i) $\|u + v\| \leq \|u\| + \|v\| \quad \forall u, v \in X$.
- (ii) $\|\lambda u\| = |\lambda| \|u\| \quad \forall u \in X, \lambda \in \mathbb{R}$.
- (iii) $\|u\| = 0 \iff u = 0$.

Definition 1.3.2. A **normed linear space** is a linear space on which a norm is defined. We sometimes write $(X, \|\cdot\|)$ to indicate what norm is used on X .

Let us now assume that X is a normed linear space as in Def. (1.3.2).

Definition 1.3.3. A sequence $\{u_k\}_{k=1}^\infty \subset X$ is said:

- to **converge** to $u \in X$ and we write $u_k \rightarrow u$ if

$$\lim_{k \rightarrow \infty} \|u_k - u\| = 0.$$

- to be **bounded** if $\exists M > 0 :$

$$\|u_k\| \leq M \quad \forall k = 1, \dots$$

- to be **Cauchy** if

$$\forall \varepsilon > 0, \exists N > 0 : \|u_k - u_l\| < \varepsilon \forall k, l, \geq N$$

where $N \in \mathbb{N}$ and $M\mathbb{R}$.

Theorem 1.3.1. Every convergent sequence in a normed space is a Cauchy sequence.

Definition 1.3.4. X is **complete** if every Cauchy sequence in X converges to an element in X . In other words,

$$\exists u \in X : u_k \rightarrow u \text{ for Cauchy sequence } \{u_k\}_{k=1}^{\infty}.$$

Definition 1.3.5. $(X, \|\cdot\|)$ is a **Banach space** if X is complete w.r.t. $\|\cdot\|$.

Definition 1.3.6. Suppose Y is a subspace of X . That is, $Y \subseteq X$. Then, equivalent definitions of the phrase " Y is **dense** in X " are:

1. $\overline{Y} = X$, where $\overline{Y}$ denotes $(\cdot)$ is dense.
2. For any $x \in X$, $\varepsilon > 0 \exists y \in Y : \|x - y\| < \varepsilon$.
3. For any $x \in X$, $\exists$ sequence $\{x_n\}_{n=1}^{\infty}$ in Y s.t. $\lim_{n \rightarrow \infty} x_n = x$.

Definition 1.3.7. If X is complete (and so is Banach), and Y is dense in X , then X is the **completion** of Y .

Example 1.2. $\mathbb{Q}$ is not complete. Take

$$x_0 = 1, x_{n+1} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right) \in \mathbb{Q}$$

but $\lim_{n \rightarrow \infty} x_n = \sqrt{2} \notin \mathbb{Q}$.

That is, $\mathbb{R}$ is the completion of $\mathbb{Q}$ because $\mathbb{R}$ is complete, and any $x \in \mathbb{R}$ can be written as a limit of a sequence in $\mathbb{Q}$.

Definition 1.3.8. X is **separable** if X contains a countable and dense subset (let $Y \subseteq X$. Y is countable if $Y = \{y : i \in \mathbb{N}\}$).

1.4 Continuous Functions

Let $U \subset \mathbb{R}^n$ be open.

Definition 1.4.1. A function $u : U \rightarrow \mathbb{R}$ is:

- **continuous at a point** $x \in U$ if

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \forall y \in U \text{ with } |x - y| < \delta.$$

- **uniformly continuous** if

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \forall x, y \in U \text{ with } |x - y| < \delta \implies |u(x) - u(y)| < \varepsilon.$$

It should be noted that continuity at a point is a local property as $\delta = \delta(x, y)$, while uniform continuity is a global property, since $\delta = \delta(\varepsilon)$.

Example 1.3. $u(x) = 1/x$ on $(0, 1)$.

Definition 1.4.2.

- $C(U) = \{u : U \rightarrow \mathbb{R} : u \text{ is continuous}\}$
- $C^k(U) = \{u : U \rightarrow \mathbb{R} : u \text{ is } k\text{-times continuously differentiable}\}$
- $u \in C^\infty(U)$ means u has continuous derivatives of all orders. Note that $C^\infty(U)$ does not automatically imply boundedness.
- $C(\overline{U}) = \{u \in C(U) : u \text{ is uniformly continuous on bounded subsets of } U\}$
- $C^k(\overline{U}) = \{u \in C^k(U) : D^\alpha u \text{ is uniformly continuous on bounded subsets of } U \forall |\alpha| < k\}$

Definition 1.4.3. The **support** of a function $u : U \rightarrow \mathbb{R}$ is defined as

$$\text{spt}(u) := \{x \in U : u(x) \neq 0\}$$

and we say that a function has **compact support** in u if $\text{spt}(u)$ is compact subset of U . Functions in $C(U)$ and $C^k(U)$ with compact support in U are denoted by $C_c(U)$ and $C_c^k(U)$.

Example 1.4. Consider the functions $u(x) = e^{-x^2}$ and $v(x) = \max(0, 1 - |x|)$ both on $\mathbb{R}$. $u(x)$ does *not* have compact support since $u(x) \neq 0 \forall x \in \mathbb{R}$. $v(x)$ does have compact support since $v(x) \neq 0$ only within the domain $\text{spt}(u) = [-1, 1]$; that is, $\text{spt}(u)$ is closed and bounded.

1.5 Convergence of Sequences

Definition 1.5.1. Consider a sequence of functions $\{u_m\}_1^\infty$ with $u_m : U \rightarrow \mathbb{R}$ and a function $u : U \rightarrow \mathbb{R}$.

- $u_m \rightarrow u$ **pointwise** on U if

$$\forall \delta > 0, \exists M = M(x, \delta) \in \mathbb{N} \text{ s.t. } m > M \implies |u_m(x) - u(x)| < \delta.$$

- $u_m \rightarrow u$ **uniformly** on U if

$$\forall \delta > 0, \exists M = M(\delta) \in \mathbb{N} \text{ s.t. } m > M \implies |u_m(x) - u(x)| < \delta \forall x \in U.$$

1.6 Lebesgue Spaces

Let us consider a 1D example. We will take a domain $(a, b) \in \mathbb{R}$ and let $C(\overline{(a, b)})$ be the space of continuous functions from $u : [a, b] \rightarrow \mathbb{R}$. Then, by the Boundedness Theorem (calculus), a continuous function on closed and bounded intervals are bounded. So,

$$\int_a^b |u(x)|^p dx < \infty, \quad 1 \leq p < \infty.$$

As a consequence, we can define the L^p -norm for continuous functions

$$\|p\|_{L^p(a, b)} = \begin{cases} \left(\int_a^b |u(x)|^p dx \right)^{1/p}, & 1 \leq p < \infty \\ \max_{x \in [a, b]} |u(x)|, & p = \infty. \end{cases}$$

The L^p -norm is very important when studying PDEs since L^p -norms are norms on $C([a, b])$. However, we need to know what the problem with $C([a, b])$ when doing analysis. In short, for $1 \leq p < \infty$, it can be shown that $C([a, b])$ is **not complete** within the L^p -norm. That is, the Cauchy sequences in $C([a, b])$ can converge to something outside of the space $C([a, b])$.

Example 1.5. Consider the sequence

$$u_n(x) = \begin{cases} 1, & x \geq 1/n \\ nx, & -1/n \leq x \leq 1/n \\ -1, & x \leq -1/n. \end{cases}$$

We know that $u_n(x) \in [-1, 1]$ and is bounded, and so $\{u_n\}_{n=1}^\infty$ is Cauchy. Then, taking the limit, we have

$$\lim_{n \rightarrow \infty} u_n = u(x) = \begin{cases} -1, & x < 0 \\ 1, & x > 0 \end{cases} \notin C[-1, 1].$$

If the space of continuous functions is not complete w.r.t. the L^p -norm, we must figure out what space is complete instead. Rather than focusing on continuous functions, we will consider the space $\mathcal{L}(a, b)$ of all **integrable**² functions $u : [a, b] \rightarrow \mathbb{R}$ s.t.

$$\int_a^b |u(x)|^p dx < \infty.$$

Unfortunately, $\|\cdot\|_{L^p(a, b)}$ is **not** a norm on $\mathcal{L}(a, b)$.

Example 1.6. Let $u : [-1, 1] \rightarrow \mathbb{R}$ s.t.

$$u(x) = \begin{cases} 0, & x \neq 0 \\ 1, & x = 0. \end{cases}$$

Notice that $\|u\|_{L^p(-1, 1)} = 0$ but $u \neq 0$.

Definition 1.6.1. Let $u, v \in \mathcal{L}(a, b)$. These functions belong to the same **equivalence class** if and only if

$$\|u - v\|_{L^p(a, b)} = 0.$$

In particular, $L^p(a, b)$ is the equivalence class of functions from $\mathcal{L}^p(a, b)$. In this way, we can say that $\|\cdot\|_{L^p(a, b)}$ is a norm on $L^p(a, b)$ because all functions $u \in \mathcal{L}^p(a, b)$ with $\|u\|_{L^p(a, b)} = 0$ belong to the equivalence class of the zero function in $L^p(a, b)$.

Definition 1.6.2. Let $u, u^* \in \mathcal{L}(a, b)$. Then, u and u^* are equal **almost everywhere** (a.e) if the set $\{x \in [a, b] : u(x) \neq u^*(x)\}$ has measure zero. We have that

$$\|u - u^*\|_{L^p(a, b)} = 0 \iff u(x) = u^*(x) \text{ a.e}$$

Example 1.7. Let

$$u(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Any countable subset of $u(x)$ has measure zero and that $\mathbb{Q} \subset \mathbb{R}$. That is, $\mathbb{Q}$ has measure zero. Therefore, $u(x) = 0$ a.e.

²Note that we mean **Lebesgue integrable**, though we will assume that all functions we deal with satisfy this property, so we drop the Lebesgue portion of it.

Example 1.8. We say that u^* is a **version** of u provided that $u = u^*$ a.e.

Let $U \subset \mathbb{R}^n$ be open (appendix A.3).

Definition 1.6.3. The **essential supremum** of $u : U \rightarrow \mathbb{R}$ is

$$\text{ess sup}(u) = \inf \{ \mu \in \mathbb{R} : |u > \mu| = 0 \}.$$

We say that $\text{ess sup}(u)$ is the sup of u except for a measurable set with Lebesgue measure zero. That is, $\text{ess sup}(u) = \sup(u)$ a.e.

Example 1.9. Consider the function

$$u(x) = \begin{cases} 1, & x = 0 \\ 0, & x \neq 0, \end{cases} \quad x \in (-1, 1).$$

The graph of $u(x)$ is (insert fig)

Definition 1.6.4. Let $u : U \rightarrow \mathbb{R}$. We define

$$\|u\|_{L^p(U)} = \begin{cases} \left(\int_U |u|^p dx \right)^{\frac{1}{p}}, & 1 \leq p < \infty \\ \text{ess sup}_u |u|, & p = \infty. \end{cases}$$

Definition 1.6.5. The **Lebesgue spaces** $L^p(U)$ are equivalence classes of functions

$$L^p(U) : \left\{ u : U \rightarrow \mathbb{R} : \|u\|_{L^p(U)} < \infty \right\}.$$

Remark. L^p -spaces are **Banach spaces**. That is, $L^p(U)$ is complete w.r.t. the norm $\|\cdot\|_{L^p(U)}$.

Theorem 1.6.1. Let $U \subset \mathbb{R}^n$ be open. Then,

- $C_c(U)$ is dense in $L^p(U)$ for $1 \leq p < \infty$.
- U is bounded provided $C(\overline{U})$ is dense in $L^p(u)$ for $1 \leq p < \infty$.

Any function in $(L^p(U), \|\cdot\|_{L^p(U)})$ with $1 \leq p < \infty$ can be approximated arbitrarily well by a function in $C(\overline{U})$.

Definition 1.6.6. Let U, V be open subsets in $\mathbb{R}^n$. We say that V is **compactly contained** in U if $V \subset \overline{V} \subset U$, and we write $V \subset\subset U$.

Definition 1.6.7. **Locally summable functions** are defined as

$$L^p_{\text{loc}}(U) := \{ u : U \rightarrow \mathbb{R} : u \in L^p(U) \text{ for each } V \subset\subset U \}$$

Now we ask: what is the difference between $L^p(U)$ and $L^p_{\text{loc}}(U)$? A function in $L^p_{\text{loc}}(U)$ can grow arbitrarily fast near the domain boundary but still be "nice enough" away from the boundary.

Example 1.10. Consider the function $u(x) = 1/x$ on $U = (0, 1)$. We know that

$$\int_0^1 |u(x)| dx \rightarrow \infty$$

so $u \notin L^1(U)$ since $u(x)$ is not bounded. However, we can choose a $V \subset\subset U$ such that

$$\int_a^b |u(x)| dx < \infty \rightarrow u \in L^1_{\text{loc}}(U)$$

provided that $0 < a < b < 1$.

Note that for u to live in $L^1(U)$, there needs to be a growth restriction on u near the boundaries. On the other hand, there is no growth restriction for $u \in L^1_{\text{loc}}(U)$ needed.

1.7 Useful Inequalities

Hölder's Inequality. Assume that $1 \leq p, q \leq \infty$ and $1/p + 1/q = 1$. If $u \in L^p(U)$ and $v \in L^q(U)$, then

$$\int_U |uv| \, dx \leq \|u\|_{L^p(U)} \|v\|_{L^q(U)}.$$

Minkowski's Inequality. Let $u, v \in L^p(U)$ for $1 \leq p \leq \infty$. Then,

$$\|u + v\|_{L^p(U)} \leq \|u\|_{L^p(U)} + \|v\|_{L^p(U)}.$$

Discrete Hölder's and Minkowski's Inequalities. Let $a = (a_1, \dots, a_n) \in \mathbb{R}^n$, $b = (b_1, \dots, b_n) \in \mathbb{R}^n$ and let $1 \leq p < \infty$ such that $1/p + 1/q = 1$. Then,

$$\left| \sum_{k=1}^n a_k b_k \right| \leq \left(\sum_{k=1}^n |a_k|^p \right)^{\frac{1}{p}} \left(\sum_{k=1}^n |b_k|^q \right)^{\frac{1}{q}}.$$

$$\left(\sum_{k=1}^n |a_k + b_k|^p \right)^{\frac{1}{p}} \leq \left(\sum_{k=1}^n |a_k|^p \right)^{\frac{1}{p}} + \left(\sum_{k=1}^n |b_k|^q \right)^{\frac{1}{q}}.$$

Young's Inequality. Let $a, b > 0$ and assume $1/p + 1/q = 1$. Then,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Young's Inequality with ϵ . Assume the same conditions as in Young's Inequality. Then,

$$ab \leq \epsilon a^p + C(\epsilon) b^q \quad (a, b > 0, \epsilon > 0)$$

for $C(\epsilon) = (\epsilon p)^{-q/p} q^{-1}$.

1.8 Brief Notes

When can we exchange a limit and integration? That is, when are we allowed to say

$$\lim_{m \rightarrow \infty} \int_U f_m \, dx = \int_U \lim_{m \rightarrow \infty} f_m \, dx.$$

Example 1.11. One case where this fails is to consider $U = [0, 1]$. Let

$$f_n(x) = \begin{cases} 0, & \text{if } x = 0 \\ n, & \text{for } 0 < x < 1/n \\ 0, & \text{for } \frac{1}{n} \leq x \leq 1. \end{cases}$$

Then,

$$\int_U \lim_{n \rightarrow \infty} f_n \, dx = 0.$$

But

$$\int_0^1 f_n dx = \int_0^{1/n} n dx + \int_{1/n}^1 0 dx = 1$$

so

$$\lim_{n \rightarrow \infty} \int_0^1 f_n dx = 1$$

and this is not equal to what we had found before.

Theorem 1.8.1 (The Monotone Convergence Theorem). Assume $\{f_k\}_{k=1}^\infty$ are measurable with

$$0 \leq f_1 \leq f_2 \leq f_3 \leq \cdots \leq f_k \leq f_{k+1} \leq \cdots$$

Then, we have

$$\int_{\mathbb{R}^n} \lim_{k \rightarrow \infty} f_k dx = \lim_{k \rightarrow \infty} \int_{\mathbb{R}^n} f_k dx.$$

Theorem 1.8.2 (Dominated Convergence Theorem). Assume $\{f_m\}_{m=1}^\infty$ are integrable and $f_n \rightarrow f$ a.e. Suppose that for some summable³ function, g , we have $|f_m| \leq g$ a.e. Then, we have

$$\lim_{m \rightarrow \infty} \int_{\mathbb{R}^n} f_m dx = \int_{\mathbb{R}^n} \lim_{m \rightarrow \infty} f_m dx = \int_{\mathbb{R}^n} f dx.$$

1.9 Hilbert Spaces

We shall assume H is a real linear space.

Definition 1.9.1. The mapping $(\cdot, \cdot) : H \times H \rightarrow \mathbb{R}$ is an **inner product** if

- (i) $(u, v) = (v, u) \forall u, v \in H$.
- (ii) $(\alpha u + \beta v, u) = \alpha(u, u) + \beta(v, u) \forall u, v \in H, \alpha, \beta \in \mathbb{R}$.
- (iii) $(u, u) \geq 0 \forall u \in H$.
- (iv) $(u, u) = 0 \iff u = 0$.

We say that $(\cdot, \cdot)$ induces a norm:

$$\|u\| = (u, u)^{\frac{1}{2}}, \quad \forall u \in H.$$

The Cauchy-Schwarz Inequality. Suppose $u, v \in H$. Then,

$$|(u, v)| \leq \|u\| \|v\|.$$

Definition 1.9.2. A **Hilbert space** is a Banach space endowed with an inner product that generates a norm.

Orthogonality. We say that $u, v \in H$ are **orthogonal** if $(u, v) = 0$.

³The integral is finite.

Orthonormality. A countable basis $\{w_k\}_{k=1}^\infty \in H$ is **orthonormal** if

$$\begin{cases} (w_k, w_l) = 0, & \text{for } k, l = 1, \dots, k \neq l \\ \|w_k\| = 1, & k = 1, \dots \end{cases}$$

Some Relations. Let $u \in H$ and $\{w_k\}_{k=1}^\infty \in H$ be an orthonormal basis, then

$$u = \sum_{k=1}^{\infty} (u, w_k) w_k \quad \text{and} \quad \|u\|^2 = \sum_{k=1}^{\infty} (u, w_k)^2$$

Let S be a subspace of H . Then the subspace orthogonal to S is

$$S^\perp = \{u \in H : (u, v) = 0, \quad \forall u \in S\}.$$

Definition 1.9.3 (Linear Operator on Banach Spaces). Assume that X and Y are Banach. Then,

- $A : X \rightarrow Y$ is a **linear operator** if $A[\lambda u + \mu v] = \lambda Au + \mu Av$, $\forall u, v \in X, \mu, \lambda \in \mathbb{R}$.
- The **range** of A is $R(A) : \{v \in Y : v = Au \text{ for } u \in X\}$.
- The **null space** of A is $N(A) := \{u \in X : Au = 0\}$.
- The norm of a linear operator $A : X \rightarrow Y$ is defined as

$$\|A\| = \sup_{\|u\|_X \leq 1} \|Au\|_Y.$$

- The linear operator $A : X \rightarrow Y$ is
 - (i) **bounded** if $\|A\| < \infty$. And we say that A is continuous if and only if it is bounded.
 - (ii) **closed** if whenever $u_k \rightarrow u$ in X and $Au_k \rightarrow v$ in Y then $Au = v$.

Theorem 1.9.1 (Closed Graph Theorem). Let $A : X \rightarrow Y$ be a closed linear operator. Then, A is bounded. This theorem is equivalent to the **Bounded Inverse Theorem**.

Theorem 1.9.2 (Bounded Inverse Theorem). Let $A : X \rightarrow Y$ be a bounded linear operator. If A is bijective, then A^{-1} is continuous, and so bounded.

Remark. A **linear operator** maps a normed space X into a normed space Y .

Remark. A **functional** maps a normed space X into $\mathbb{R}$.

Definition 1.9.4. A bounded linear operator $u^* : X \rightarrow \mathbb{R}$ is a bounded linear functional. We say that u^* is continuous if and only if u^* is bounded. That is,

$$|u^*(u)| \leq C\|u\|_X, \quad \forall u \in X.$$

Definition 1.9.5. The **dual space** X^* of X is the collection of bounded linear functionals on X .

Definition 1.9.6. If $u \in X, u^* \in X^*$, then we write $\langle u^*, u \rangle$ to denote the real number $u^*(u)$. We say that $\langle u^*, u \rangle$ is the action of u^* on some $u \in X$. Also, we say that $\langle \cdot, \cdot \rangle$ is the **pairing** of X^* and X .

Definition 1.9.7. The **dual norm** of u^* is defined as

$$\|u^*\| = \sup_{\|u\| \leq 1} |\langle u^*, u \rangle|.$$

Definition 1.9.8. A Banach space is **reflexive** if $(X^*)^* = X \implies$ for each $u^{**} \in X^{**}$, $\exists u \in X$ s.t.

$$\langle u^{**}, u^* \rangle = \langle u^*, u \rangle, \quad \forall u^* \in X^*.$$

Theorem 1.9.3. Every Hilbert space is reflexive.

Theorem 1.9.4 (Riesz Representation Theorem). For each $u^* \in H^*$, $\exists$ a unique element $u \in H$ s.t.

$$(i) \quad \langle u^*, u \rangle = (u, v)_H \quad \forall v \in H.$$

$$(ii) \quad \|u^*\|_{H^*} = \|u\|_H.$$

2 Sobolev Spaces

We would like to recast a PDE as an operator $A : X \rightarrow Y$, where A contains information of the PDE and X, Y are appropriate function spaces.

Mostly, X and Y are Sobolev spaces.

2.1 Sobolev Spaces

Example 2.1. We want to find the function $u : U \rightarrow \mathbb{R}$ s.t.

$$\begin{cases} u'' = f, & \text{in } U = (-1, 1) \\ u(-1) = u(1) = 0, & \text{in } U = (-1, 1). \end{cases}$$

If $f \in C(\overline{U})$, then $\exists!$ (! denotes a unique element) $u \in C^2(U) \cap C(\overline{U})$. What if $f \notin C(\overline{U})$ now, for example,

$$f(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0. \end{cases}$$

Then, we need u s.t. $u'' \notin C(\overline{U})$.

Question: Instead of seeking "classical" solutions, can we weaken the definition of partial derivatives and so the differentiability requirements on u to solve such problems?

The answer is *yes*, and these are called the **weak derivatives**.

2.1.1 Weak Derivatives

Motivation for the Weak Derivative. Assume $u \in C^1(U)$. Then $\forall \phi \in C_c^\infty(U)$, we have the following equality

$$\int_U u \phi_{x_i} dx = - \int_U u_{x_i} \phi dx + \int_{\partial U} u \phi n_i ds$$

by IBP. However, since ϕ has compact support, the boundary integral is zero, so

$$\int_U u \phi_{x_i} dx = - \int_U u_{x_i} \phi dx.$$

More generally, if $u \in C^k(U)$ and $|x| = \alpha_1 + \dots + \alpha_n = k$, then

$$\int_U u D^\alpha \phi dx = (-1)^{|\alpha|} \int_U D_u^\alpha \phi dx \quad \forall \phi \in C_c^\infty(U).$$

This is for $u \in C^k(U)$. Are we able to relax this condition? Looking at the LHS, we can see that it does **not** need $u \in C^k(U)$ to be finite. In fact, it will be finite for $u \in L_{\text{loc}}^1(U)$. On the RHS, if $u \notin C^k(U)$, then D_u^α has no clear meaning. Can we replace D_u^α by some $v \in L_1^{\text{loc}}(U)$ s.t. the integral equality in U is valid?

Definition 2.1.1. Suppose $u, v \in L_{\text{loc}}^1(U)$ and α is a multi-index. Then v is the α **th-weak partial derivative** of u provided the following holds:

$$\int_U u D^\alpha \phi dx = (-1)^{|\alpha|} \int_U v \phi dx \quad \forall \phi \in C_c^\infty(U). \quad (5)$$

If v exists, we say $D^\alpha u = v$ in the weak sense. Otherwise, if v does not exist, then u has no α th-weak partial derivative.

Theorem 2.1.1. Let $U \subset \mathbb{R}^n$ be open. Let $v \in L^1_{\text{loc}}(U)$ be s.t.

$$\int_U v \phi \, dx = 0 \quad \forall \phi \in C_c^\infty(U).$$

Then $v = 0$ a.e.

Lemma 2.1.2. If the α th-weak derivative exists, then it is uniquely defined up to a set of measure zero.

Proof. Assume $v, \tilde{v} \in L^1_{\text{loc}}(U)$ satisfy

$$\int_U u D^\alpha \phi \, dx = (-1)^{|\alpha|} \int_U v \phi \, dx = (-1)^{|\alpha|} \int_U \tilde{v} \phi \, dx \quad \forall \phi \in C_c^\infty(U).$$

Then,

$$\int_U (v - \tilde{v}) \phi \, dx = 0 \quad \forall \phi \in C_c^\infty(U) \implies v = \tilde{v} \text{ a.e.}$$

By Theorem 2.1.1. □

Example 2.2. Let $U = (0, 1)$ and

$$u(x) = \begin{cases} x, & 0 < x < \frac{1}{2} \\ 1 - x, & \frac{1}{2} < x < 1. \end{cases}$$

The function is not differentiable on U since $u'(x)$ is not defined at $x = 1/2$. However, the weak derivative does exist. We define

$$v(x) = \begin{cases} 1, & 0 < x < \frac{1}{2} \\ -1, & \frac{1}{2} < x < 1. \end{cases}$$

Then $u' = v$ in the weak sense. Indeed $\forall \phi \in C_c^\infty(U)$, we have

$$\begin{aligned} \int_0^1 u \phi' \, dx &= \int_0^{1/2} u \phi' \, dx + \int_{1/2}^1 \phi' \, dx \\ &= - \int_0^{1/2} \phi \, dx + u \phi \Big|_0^{1/2} - \int_{1/2}^1 (-1) \phi \, dx + u \phi \Big|_{1/2}^1 \\ &= - \int_0^1 v \phi \, dx + u \phi \Big|_{1/2}^1 - u \phi \Big|_{1/2}^1 \end{aligned}$$

where the integrals on the RHS are combined by the definition of v , and $u(0) = u(1) = 0$ since u has compact support. Thus,

$$\int_0^1 u \phi' \, dx = - \int_0^1 v \phi \, dx$$

implies that v is the first weak derivative of u .

We should be careful to know that not all functions have weak derivatives.

Example 2.3. Let $U = (0, 1)$ and

$$u(x) = \begin{cases} 2, & x \leq \frac{1}{2} \\ 1, & x > \frac{1}{2}. \end{cases}$$

This function does **not** have a weak derivative.

Proof. Assume $\exists v \in L^1_{\text{loc}}(U)$ s.t.

$$\int_0^1 u \phi' dx = - \int_0^1 v \phi dx \quad \forall \phi \in C_c^\infty(U).$$

Then,

$$- \int_0^1 v \phi dx = \int_0^1 u \phi' dx = \int_0^{1/2} 2 \phi' dx + \int_{1/2}^1 \phi' dx = 2\phi\left(\frac{1}{2}\right) - 2\phi(0) + \phi(1) - \phi\left(\frac{1}{2}\right)$$

but since ϕ has compact support,

$$- \int_0^1 v \phi dx = \phi\left(\frac{1}{2}\right).$$

Define the sequence $\{\phi_m\}_{m=1}^\infty$ of smooth functions s.t. $0 \leq \phi_m \leq 1$, $\text{spt}(\phi) \subset\subset U$, $\phi_m(1/2) = 1$, $\phi_m(x) \rightarrow 0 \forall x \neq 1/2$. Then, we have

$$\int_0^1 v \phi_m dx = \phi_m\left(\frac{1}{2}\right).$$

Taking $m \rightarrow \infty$, then

$$1 = \lim_{m \rightarrow \infty} \phi_m\left(\frac{1}{2}\right) = \lim_{m \rightarrow \infty} \left(- \int_0^1 v \phi_m dx \right).$$

By the Monotone Convergence Theorem,

$$\lim_{m \rightarrow \infty} \left(- \int_0^1 v \phi_m dx \right) = - \int_0^1 \lim_{m \rightarrow \infty} (v \phi_m) dx = 0 \implies 0 = 1.$$

By contradiction, $v \notin L^1_{\text{loc}}(U)$ since $0 \neq 1$. □

2.2 Sobolev Spaces

Definition 2.2.1. Let $k \geq 0$ be an integer, $1 \leq p \leq \infty$, $u \in L^1_{\text{loc}}(U)$. Suppose $D^\alpha u$ exists in the weak sense for each multi-index α s.t. $|\alpha| \leq k$. Then, the **Sobolev norm** is defined as

$$\|u\|_{W^{k,p}(U)} := \begin{cases} \left(\sum_{|\alpha| \leq k} \int_U |D^\alpha u|^p dx \right)^{\frac{1}{p}} = \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p(U)}^p \right)^{\frac{1}{p}}, & 1 \leq p < \infty \\ \sum_{|\alpha| \leq k} \text{ess sup}_U |D^\alpha u| = \max_{|\alpha| \leq k} \|D^\alpha u\|_{L^\infty(U)}, & p = \infty. \end{cases} \quad (6)$$

Definition 2.2.2. The **Sobolev space** is defined as:

$$W^{k,p}(U) := \left\{ v \in L^1_{\text{loc}}(U) : \|v\|_{W^{k,p}(U)} < \infty \right\}. \quad (7)$$

Remark.

- $W^{k,2}(U)$ is a Hilbert space. We write $H^k(U)$ to denote $W^{k,2}(U)$.
- $W^{0,2}(U) = H^0(U) = L^2(U)$.

Example 2.4. • Let $n = 1, k = 0$. Then,

$$\|u\|_{W^{0,p}(U)} = \|u\|_{L^p(U)}.$$

- Let $n = k = 1, |\alpha| = \alpha = 1$. Then,

$$\|u\|_{W^{1,p}(U)} = \begin{cases} \left(\|u\|_{L^p(U)}^p + \|u'\|_{L^p(U)}^p \right)^{\frac{1}{p}}, & 1 \leq p < \infty \\ \max(\|u\|_{L^\infty(U)}, \|u'\|_{L^\infty(U)}), & p = \infty. \end{cases}$$

Definition 2.2.3. Let $\{u_m\}_{m=1}^\infty, u \in W^{k,p}(U)$. Then,

- $u_m \rightarrow u$ in $W^{k,p}(U)$ means

$$\lim_{m \rightarrow \infty} \|u_m - u\|_{W^{k,p}(U)} = 0.$$

- $u_m \rightarrow u$ in $W_{\text{loc}}^{k,p}(U)$ means

$$u_m \rightarrow u \text{ in } W^{k,p}(V) \text{ for each } V \subset\subset U.$$

Definition 2.2.4. $W_0^{k,p}(U)$ is the closure of $C_c^\infty(U)$ in $W^{k,p}(U)$. This means:

$$u \in W_0^{k,p}(U) \iff \exists u_m \in C_c^\infty(U) : u_m \rightarrow u \text{ in } W^{k,p}(U).$$

$u_m \in C_c^\infty(U)$ means u_m and all its derivatives vanish near ∂U . So, $u \in W_0^{k,p}(U)$ are those $u \in W^{k,p}(U)$ s.t. $D^\alpha u = 0$ on ∂U .

We denote $H_0^k(U) = W_0^{k,2}(U)$.

Definition 2.2.5.

$$B(x, r) = \{y \in \mathbb{R}^n : |x - y| \leq r\} \quad (8)$$

is a closed ball in $\mathbb{R}^n$ with center x and radius r .

$$B^0(x, r) = \{y \in \mathbb{R}^n : |x - y| < r\} \quad (9)$$

be an open ball.

Example 2.5. Let $U = B^0(0, 1)$. Let $u(x) = 1/|x|^\alpha$ for $x \in U, x \neq 0$. For what $\alpha > 0, n, p$ is $u \in W^{1,0}(U)$?

Away from $x = 0$, u is smooth and

$$u_{x_i} = -\frac{\alpha x_i}{|x|^{\alpha+1}}, \quad x \neq 0 \implies |Du(x)| = \frac{\alpha}{|x|^{\alpha+1}}, \quad x \neq 0.$$

We want the derivative for $x \neq 0$. We will consider what happens on $U - B(0, \varepsilon)$ in the limit $\varepsilon \rightarrow 0$. Let $\phi \in C_c^\infty(U)$ and fix ε . Then,

$$\int_{U-B(0,\varepsilon)} u \phi_{x_i} \, dx = - \int_{U-B(0,\varepsilon)} u_{x_i} \phi \, dx + \int_{\partial B(0,\varepsilon)} u \phi \nu^i \, ds$$

where ν^i is the normal vector pointing inward to $B(0, \varepsilon)$. Note that ϕ vanishes on ∂U but **not** $\partial B(0, \varepsilon)$. We want to try to bound this function. To do notice, notice that since ϕ is compactly supported and infinitely differentiable, it exists in the L^∞ -norm, so

$$\begin{aligned} \left| \int_{\partial U \cap B(0, \varepsilon)} u \phi \nu^i \, ds \right| &\leq \|\phi\|_{L^\infty(U)} \int_{\partial B(0, \varepsilon)} u \, ds \\ &= \|\phi\|_{L^\infty(U)} \int_{\partial B(0, \varepsilon)} \varepsilon^{-\alpha} \, ds \\ &= \|\phi\|_{L^\infty(U)} \varepsilon^{-\alpha} \int_{\partial B(0, \varepsilon)} \, ds. \end{aligned}$$

We will make use of the fact that the surface area of an n -dimensional ball of radius r is proportional to r^{n-1} . Then,

$$\left| \int_{\partial B(0, \varepsilon)} u \phi \, n u^i \, ds \right| \leq C \varepsilon^{n-1-\alpha}.$$

Observe that if $n - 1 - \alpha > 0$, then

$$\int_{\partial B(0, \varepsilon)} u \phi \nu^i \, ds \rightarrow 0.$$

(a). We will check whether u_{x_i} is integrable on U .

$$\int_U |u_{x_i}| \, dx \leq \int_U |Du| \, dx = \alpha \int_U |x|^{-\alpha-1} \, dx.$$

In spherical coordinates,

$$\begin{aligned} \alpha \int_U |x|^{-\alpha-1} \, dx &= \alpha \int_0^1 \int_{\phi_1}^\pi \cdots \int_{\phi_{n-1}}^{2\pi} \frac{1}{r^{\alpha+1}} r^{n-1} \sin^{n-2}(\phi_1) \sin^{n-3}(\phi_2) \cdots \sin(\phi_{n-2}) \, d\phi_{n-1} \cdots d\phi \, dr \\ &= \alpha \int_0^1 r^{n-1-\alpha-1} \, dr \int_{\phi} (\cdot) \, d\phi_{n-1} \cdots d\phi \\ &= \alpha \int_0^1 r^{n-1-\alpha-1} [C] \, dr. \end{aligned}$$

So,

$$\int_U |u_{x_i}| \, dx \leq \frac{C\alpha}{n-\alpha-1} r^{n-\alpha-1} \Big|_0^1$$

which is finite if $n - \alpha - 1 > 0$.

(b). Using the same steps as in (a), we can show that $u \in L^p(U)$ if $\alpha < n/p$ (replace $|x|^{-\alpha-1}$ by $|x|^{-p\alpha}$).

(c). We now show that u_{x_i} is a weak derivative of u on U if $n - \alpha - 1 > 0$. Let χ_A be an indicator function: $\chi_A = 1$ if $x \in A$ and $\chi_A = 0$ otherwise. Then,

$$\int_U u \phi_{x_i} \, dx = \int_U \lim_{\varepsilon \rightarrow 0} [u \phi_{x_i} \chi_{U-B(0, \varepsilon)}] \, dx.$$

By step (b), notice that $u \in L^1(U)$ if $\alpha < n$. We assume that $\alpha < n - 1 < n$, so $u \in L^1(U)$. Then,

$$|u\phi_{x_j}\chi_{U-B(0,\varepsilon)}| \leq |u|\|D\phi\|_{L^\infty(U)}.$$

Thus, $u\phi_{x_j}\chi_{U-B(0,\varepsilon)}$ is dominated, so we can apply the Dominated Convergence Theorem (Theorem 1.8.2). Thus,

$$\int_U u\phi_{x_i} dx = \lim_{\varepsilon \rightarrow 0} \int_{U-B(0,\varepsilon)} u\phi_{x_j} dx = -\lim_{\varepsilon \rightarrow 0} \int_{U-B(0,\varepsilon)} u_{x_i}\phi dx + \lim_{\varepsilon \rightarrow 0} \int_{\partial B(0,\varepsilon)} u\phi \nu^i ds$$

where the last integral cancels since ϕ has compact support. Note we also use IBP. Then,

$$\int_U u\phi_{x_i} dx = -\lim_{\varepsilon \rightarrow 0} \int_U u_{x_i}\phi\chi_{U-B(0,\varepsilon)} dx.$$

We must use the Dominated Convergence Theorem (Theorem 1.8.2) again. In step (a), we showed that $Du \in L^1(U)$ if $\alpha < n - 1$. Therefore, we also have

$$|u_{x_j}\phi\chi_{U-B(0,\varepsilon)}| \leq |Du|\|\phi\|_{L^\infty(U)} \in L^1(U)$$

so that

$$-\lim_{\varepsilon \rightarrow 0} \int_U u_{x_i}\phi\chi_{U-B(0,\varepsilon)} dx = -\int_U \lim_{\varepsilon \rightarrow 0} u_{x_i}\phi\chi_{U-B(0,\varepsilon)} dx = -\int_U u_{x_i}\phi dx.$$

Thus, we have proven that u has a weak derivative on U .

(d). We know that $Du \in L^1(U)$, but we need also $Du \in L^p(U)$ since we want to show that $u \in W^{1,p}(U)$. Similar steps as in (a) can be used to prove that $Du \in L^p(U)$ if $\alpha < (n-p)/p$ (that is, replace $|x|^{-\alpha-1}$ by $|x|^{-p(\alpha+1)}$).

(e). We found three conditions for relevant integrals to be finite. That is,

$$\begin{aligned} \alpha &< n - 1 \\ \alpha &< \frac{n}{p} \\ \alpha &< \frac{n-p}{p}. \end{aligned}$$

The most strict condition is $\alpha < (n-p)/p$. Thus, we can conclude that for $\alpha < (n-p)/p$, we have $u \in L^p(U)$ (by step (b)); the weak derivative exists (by step (c)); $Du \in L^p(U)$ (by step (d)). So, $u \in W^{1,p}(U)$ if $\alpha < (n-p)/p$.

2.2.1 Properties of Weak Derivatives

Theorem 2.2.1 (Properties of Weak Derivatives). Assume $u, v \in W^{k,p}(U)$ and that $|\alpha| \leq k$. Then,

- (i) $D^\alpha u \in W^{k-|\alpha|,p}(U)$.
- (ii) $D^\beta(D^\alpha u) = D^\alpha(D^\beta u) = D^{\alpha+\beta}u \ \forall \ \alpha, \beta \text{ s.t. } |\alpha| + |\beta| \leq k$.
- (iii) $\lambda u + \mu v \in W^{k,p}(U)$ and $D^\alpha(\lambda u + \mu v) = \lambda D^\alpha u + \mu D^\alpha v \ \forall \ \lambda, \mu \in \mathbb{R}$.

(iv) If $V \subset U$ is open, then $u \in W^{k,p}(V)$.

(v) If $\zeta \in C_c^\infty(U)$, then $\zeta u \in W^{k,p}(U)$ and the Leibniz's formula holds for Sobolev functions

$$D^\alpha(\zeta u) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} D^\beta \zeta D^{\alpha-\beta} u, \quad \text{where } \binom{\alpha}{\beta} = \frac{\alpha!}{\beta!(\alpha-\beta)!}.$$

Indeed,

$$\binom{\alpha}{\beta} = \binom{\alpha_1}{\beta_1} \binom{\alpha_2}{\beta_2} \cdots \binom{\alpha_n}{\beta_n} \quad \text{and} \quad \alpha! = \alpha_1! \alpha_2! \cdots \alpha_n!. \quad (10)$$

Proof. (i) is by the definition of a weak derivative. (iii) and (iv) are clear. We shall prove (ii) and (v).

(ii). By definition, $D^\beta u$ is the β -th weak derivative of u , which means

$$\int_U u D^\beta \phi \, dx = (-1)^{|\beta|} \int_U D^\beta u \phi \, dx \quad \forall \phi \in C_c^\infty(U).$$

Therefore, $D^\beta(D^\alpha u)$ means

$$\int_U D^\alpha u D^\beta \phi \, dx = (-1)^{|\beta|} \int_U D^\beta(D^\alpha u) \phi \, dx \quad \forall \phi \in C_c^\infty(U).$$

Performing IBP on the LHS gives

$$(-1)^{|\alpha|} \int_U u D^{\alpha+\beta} \phi \, dx = (-1)^{|\alpha|} (-1)^{|\alpha|+|\beta|} \int_U D^{\alpha+\beta} u \phi \, dx, \quad (u \in W^{k,p}(U), |\alpha| + |\beta| \leq k).$$

with the boundary term cancelled since ϕ has compact support. We also use the definition of the weak derivative on β . Notice that $|\alpha| + |\alpha| + |\beta| = 2|\alpha| + |\beta|$, so comparing this to the RHS, we have

$$(-1)^{2|\alpha|+|\beta|} \int_U D^{\alpha+\beta} u \phi \, dx = (-1)^{|\beta|} \int_U D^{\alpha+\beta} u \phi \, dx.$$

Thus

$$D^\beta(D^\alpha u) = D^{\alpha+\beta} u$$

as required.

(v). By definition, $D^\alpha(\zeta u)$ is the α -th weak derivative of ζu , meaning

$$\int_U \zeta u D^\alpha \phi \, dx = (-1)^{|\alpha|} \int_U D^\alpha(\zeta u) \phi \, dx, \quad \forall \phi \in C_c^\infty(U).$$

We shall now prove (v) by induction.

Base Case: First, $|\alpha| = 1$. Choose any $\phi \in C_c^\infty(U)$. Then, since $\zeta, \phi \in C_c^\infty(U)$, we can apply Leibniz rule to the functions

$$D^\alpha(\zeta \phi) = \zeta D^\alpha \phi + \phi D^\alpha \zeta \implies \int_U \zeta u D^\alpha \phi \, dx = \int_U [u D^\alpha(\zeta \phi) - u(D^\alpha \zeta) \phi] \, dx.$$

However, we know that $u \in W^{1,p}(U)$ and so for $|\alpha| = 1$,

$$\int_U u D^\alpha \xi \, dx = - \int_U D^\alpha u \xi \, dx \quad \forall \xi \in C_c^\infty(U).$$

Choose $\xi = \zeta \phi$, then

$$\int_U u D^\alpha (\zeta \phi) \, dx = - \int_U \zeta (D^\alpha u) \phi \, dx.$$

Plugging this into the Leibniz-ruled-applied functions, we have

$$\int_U \zeta u D^\alpha \phi \, dx = - \int_U [\zeta D^\alpha u + u D^\alpha \zeta] \phi \, dx.$$

Comparing this to the definition of a weak derivative, we can see that

$$D^\alpha (\zeta u) = \zeta D^\alpha u + u D^\alpha \zeta$$

in the weak sense.

Induction Step: Assume for $l < k$ and that (v) holds $\forall |\alpha| \leq l$ and all functions ζ . Choose a multi-index α with $|\alpha| = l + 1$. Then, $\alpha = \beta + \gamma$ for some $|\beta| = l$ and $|\gamma| = 1$. Then,

$$\begin{aligned} \int_U \zeta u D^\alpha \phi \, dx &= \int_U \zeta u D^\beta (D^\alpha \phi) \, dx \\ &= (-1)^{|\beta|} \int_U D^\beta (\zeta u) D^\gamma \phi \, dx, \quad (\zeta u \in W^{l,p}(U)). \\ &= (-1)^{|\beta|} \int_U \sum_{\sigma \leq \beta} \binom{\beta}{\sigma} D^\sigma \zeta D^{\beta-\sigma} u D^\gamma \phi \, dx \quad (\text{Leibniz rule for } |\beta| = l). \\ &= (-1)^{|\beta|+|\gamma|} \int_U \sum_{\sigma \leq \beta} \binom{\beta}{\sigma} D^\gamma (D^\sigma \zeta D^{\beta-\sigma} u) \phi \, dx \\ &= (-1)^{|\beta|+|\gamma|} \int_U \sum_{\sigma \leq \beta} \binom{\beta}{\gamma} \left[D^{\sigma+\gamma} \zeta D^{\beta-\gamma} u + D^\sigma \zeta D^{\beta-\sigma+\gamma} u \right] \phi \, dx \quad (|\gamma| = 1 \text{ case step 1}) \\ &= (-1)^{|\alpha|} \int_U \sum_{\sigma \leq \beta} \binom{\beta}{\sigma} \left[D^\rho \zeta D^{\alpha-\rho} u + D^\sigma \zeta D^{\alpha-\sigma} u \right] \phi \, dx \quad (\rho = \sigma + \gamma) \\ &= (-1)^{|\alpha|} \int_U \left[\sum_{\sigma \leq \beta} \binom{\alpha}{\sigma} D^\sigma \zeta D^{\alpha-\sigma} u \right] \phi \, dx \end{aligned}$$

□

Theorem 2.2.2. $W^{k,p}(U)$ is a Banach space for each $k = 1, 2, \dots$ and $1 \leq p \leq \infty$.

2.2.2 Convolution and Smoothing

We will assume $U \subset \mathbb{R}^n$ be open and $\varepsilon > 0$. Then,

$$U^\varepsilon = \{x \in U : \text{dist}(x, \partial U) > \varepsilon\}.$$

Definition 2.2.6. The **standard mollifier** $\eta(x) \in C^\infty(\mathbb{R}^n)$ is defined as

$$\eta(x) = \begin{cases} C \exp\left(\frac{1}{|x|^2-1}\right), & \text{if } |x| < 1 \\ 0, & \text{if } |x| \geq 1. \end{cases} \quad (11)$$

where C is s.t.

$$\int_{\mathbb{R}^n} \eta(x) dx = 1.$$

Then, for each $\varepsilon > 0$:

$$\eta_\varepsilon(x) = \frac{1}{\varepsilon^n} \eta\left(\frac{x}{\varepsilon}\right)$$

with the properties

(i) $\eta_\varepsilon \in C^\infty(\mathbb{R}^n)$

(ii) $\int_{\mathbb{R}^n} \eta_\varepsilon dx = 1.$

(iii) $\text{spt}(\eta_\varepsilon) \subset B(0, \varepsilon).$

Definition 2.2.7. Let $f \in L^1_{\text{loc}}$. Its **mollification** $f^\varepsilon : U^\varepsilon \rightarrow \mathbb{R}$ is defined for every $x \in U^\varepsilon$. In particular,

$$\begin{aligned} f^\varepsilon(x) &= (\eta_\varepsilon * f)(x) = \int_U f(y) \eta_\varepsilon(x-y) dy = \int_{B(x, \varepsilon)} f(y) \eta_\varepsilon(x-y) dy = \int_{B(0, \varepsilon)} f(x-z) \eta_\varepsilon(z) dz \\ &= \text{spt}(f^\varepsilon) = \text{spt}(\eta_\varepsilon * f) \subset \text{spt}(f) + B(0, \varepsilon). \end{aligned} \quad (12)$$

Theorem 2.2.3 (Properties of Mollification). 1. $f^\varepsilon \in C^\infty(U^\varepsilon)$, $D^\alpha f^\varepsilon = D^\alpha_{\eta_\varepsilon} * f$ in U^ε .

2. $f^\varepsilon \rightarrow f$ a.e. as $\varepsilon \rightarrow 0$:

$$\left| \left\{ x \in U : \lim_{\varepsilon \rightarrow 0} f^\varepsilon(x) \neq f(x) \right\} \right| = 0.$$

3. If $f \in C(U)$ then $f^\varepsilon \rightarrow f$ uniformly on compact subsets of U :

$$\forall x \in V \subset\subset U : \forall \delta > 0, \exists \varepsilon_0 = \varepsilon_0(\delta) > 0 : |f(x) - f^\varepsilon(x)| \leq \delta \quad \forall \varepsilon < \varepsilon_0.$$

4. If $1 \leq p < \infty$ and $f \in L^p_{\text{loc}}(U)$, then $f^\varepsilon \rightarrow f$ in $L^p_{\text{loc}}(U)$.

2.3 Approximation

With approximations, we shall:

- (1) prove that smooth functions are dense in $W^{k,p}_{\text{loc}}(U)$.
- (2) prove that smooth functions are dense in $W^{k,p}(U)$.
- (3) approximate $u \in W^{k,p}(U)$ by functions in $C^\infty(\overline{U})$ rather than $C^\infty(U)$.

Theorem 2.3.1. Assume $u \in W^{k,p}(U)$ (with $k > 0$ and k is an integer and $1 \leq p < \infty$). Set $u^\varepsilon = \eta_\varepsilon * u$ in U^ε . Then,

- (i) $u^\varepsilon \in C^\infty(U^\varepsilon)$ for each $\varepsilon > 0$.
- (ii) $u^\varepsilon \rightarrow u$ in $W_{\text{loc}}^{k,p}(U)$ as $\varepsilon \rightarrow 0$.

Proof. (i). We can prove this by simply using the properties of mollification.

(ii). To prove (ii) we shall first prove that if $|\alpha| \leq k$, then $D^\alpha u^\varepsilon = \eta_\varepsilon * D^\alpha u$ in U^ε . Fix $x \in U_\varepsilon$. Then,

$$\begin{aligned} D^\alpha u^\varepsilon(x) &= D^\alpha(u * \eta_\varepsilon)(x) = (u * D^\alpha \eta_\varepsilon)(x) \quad (\text{prop. (i) of mollification}). \\ &= \int_U D_x^\alpha \eta_\varepsilon(x-y) u(y) dy \quad (\text{by def of convolution}). \\ &= (-1)^{|\alpha|} \int_U D_y^\alpha \eta_\varepsilon(x-y) u(y) dy \end{aligned}$$

and note that

$$\partial_{x_j} \eta\left(\frac{x-y}{\varepsilon}\right) = -\partial_{y_j} \eta\left(\frac{x-y}{\varepsilon}\right) \quad (\text{a})$$

applied α -times nets a $(-1)^{|\alpha|}$. Then, by definition of the α -th weak derivative of u , we have

$$\int_U D_y^\alpha u(y) dy = (-1)^{|\alpha|} \int_U \phi(y) D^\alpha u(y) dy \quad \forall \phi \in C_c^\infty(U).$$

If $x \in U^\varepsilon$ is fixed, then $\phi(y) := \eta_\varepsilon(x-y) \in C_c^\infty(U)$ by the property of η_ε . Thus,

$$\int_U [D_y^\alpha \eta_\varepsilon(x-y)] u(y) dy = (-1)^{|\alpha|} \int_U \eta_\varepsilon(x-y) D^\alpha u(y) dy. \quad (\text{b})$$

Combining both integrals (a) and (b), we have

$$D^\alpha u_\varepsilon(x) = (-1)^{|\alpha|+|\alpha|} \int_U \eta_\varepsilon(x-y) D^\alpha u(y) dy = (\eta_\varepsilon * D^\alpha u)(x) = \int_U \eta_\varepsilon(x-y) D^\alpha u(y) dy = (\eta_\varepsilon * D^\alpha u)(x).$$

(iii). Let $V \subset\subset U$ be open and choose $\varepsilon > 0$ s.t. $V \subset U^\varepsilon$. In (ii), we proved that $D^\alpha u^\varepsilon = \eta_\varepsilon * D^\alpha u$ in U^ε if $|\alpha| \leq k$. Therefore, $D^\alpha u^\varepsilon \rightarrow D^\alpha u$ in $L^p(U)$ if $|\alpha| \leq k$ by property (iv) of mollification).

Thus,

$$\|u^\varepsilon - u\|_{W^{k,p}(V)}^p = \sum_{|\alpha| \leq k} \|D^\alpha u^\varepsilon - D^\alpha u\|_{L^p(V)}^p \rightarrow 0$$

which proves property (ii). □

Theorem 2.3.2. Assume U is bounded and $u \in W^{k,p}(U)$ ($k > 0$ is an integer and $1 \leq p < \infty$). Then, there exists functions $u_m \in C^\infty(U) \cap W^{k,p}(U)$ s.t. $u_m \rightarrow u$ in $W^{k,p}(U)$.

Proof. (i). The notation we shall use is as follows:

- $U_i = \{x \in U : \text{dist}(x, \partial U) > 1/i\}$ for $i = 1, 2, \dots$. Then $U = \bigcup_{i=1}^\infty U_i$.

- $V_i = U_{i+3} - \overline{U}_{i+1}$ for $i = 1, 2, \dots$.
- Choose open set V_0 s.t. $V_0 \subset\subset U$ and $U = \bigcup_{i=0}^{\infty} V_i$.
- Let $\{\zeta_i\}_{i=0}^{\infty}$ be s.t.:

$$\begin{cases} 0 \leq \zeta_i \leq 1, & \zeta_i \in C_c^{\infty}(V_i) \\ \sum_{i=0}^{\infty} \zeta_i(x) = 1, & \text{for each } x \in U. \end{cases}$$

- Choose any $u \in W^{k,p}(U)$. Then, by Theorem (2.2.1), $\zeta_i u \in W^{k,p}(U)$. Moreover, $\text{spt}(\zeta_i u) \subset V_i$ since $\text{spt}(\zeta_i) \subset V_i$.

(ii). Given $\varepsilon > 0$, let $u^\varepsilon = \eta_\varepsilon * (\zeta_i u)$. We know from Theorem (2.3.1) that $u^\varepsilon \rightarrow \zeta_i u$ in $W_{\text{loc}}^{k,p}(U)$. This means, given $Z \subset\subset U$, $\delta > 0$, and $C > 0$, we can find $\varepsilon > 0$ s.t. $\|u^\varepsilon - \zeta_i u\|_{W^{k,p}(Z)} < C\delta$.

We want $\|u^\varepsilon - \zeta_i u\|_{W^{k,p}(U)} \rightarrow 0$. If we choose a Z s.t. $\text{spt}(\zeta_i u) \subset Z$ and $\text{spt}(u^\varepsilon) \subset Z$ and $Z \subset\subset U$, then

$$\|u^\varepsilon - \zeta_i u\|_{W^{k,p}(U)} = \|u^\varepsilon - \zeta_i u\|_{W^{k,p}(Z)} < C\delta.$$

To find such Z , first note that $\text{spt}(\zeta_i u) \subset V_i$ since $\zeta_i \in C_c^\infty(U)$, but $\text{spt}(u^\varepsilon) \not\subset V_i$:

$$\text{spt}(u^\varepsilon) = \text{spt}(\eta_\varepsilon * \zeta_i u) \subset \text{spt}(\eta_\varepsilon) + \text{spt}(\zeta_i u) + B(0, \varepsilon) + V_i.$$

So, we cannot pick $Z = V_i$, and we must instead choose Z slightly larger than V_i . For ε small enough,

$$V(0, \varepsilon) + V_i = V(0, \varepsilon) + (U_{i+3} - \overline{U}_{i+1}) \subset U_{i+3} - \overline{U}_i = W_i.$$

Take $Z = W_i$ and choose $C = 1/2^i$. By Theorem 2.3.1, we can find $\varepsilon_i > 0$ s.t.

$$\left\| u^{\varepsilon_i - \zeta_i u} \right\| = \|u^{\varepsilon_i} - \zeta_i u\|_{W^{k,p}(W_i)} < \frac{\delta}{2^{i+1}}.$$

We want to show $u_m \rightarrow u$ in $W^{k,p}(U)$ for $u_m \in C^\infty(U) \cap W^{k,p}(U)$. Set $v = \sum_{i=0}^{\infty} u^{\varepsilon_i}$. Take any $x \in U$. Then, there exists an open set $V \subset\subset U$ s.t. $x \in V$. Note that $v(y) = \sum_{i=0}^k u^{\varepsilon_i}(y) \forall y \in V$ with k finite. This is because $\text{spt}(u^{\varepsilon_i}) \subset W_i$ and $V \cap W \neq \emptyset$ only for a finite number i .

Since u^{ε_i} being infinitely differentiable at x implies v being infinitely differentiable at x , and since x is arbitrary, we have $v \in C^\infty(U)$. Note also that $u = \sum_{i=0}^{\infty} \zeta_i u$. Then,

$$\|v - u\|_{W^{k,p}(U)} = \left\| \sum_{i=0}^{\infty} (\zeta_i u - u^{\varepsilon_i}) \right\|_{W^{k,p}(U)} \leq \sum_{i=0}^{\infty} \|\zeta_i u - u^{\varepsilon_i}\|_{W^{k,p}(U)} \leq \sum_{i=0}^{\infty} \frac{\delta}{2^{i+1}} = \delta.$$

Thus, $\forall \delta > 0$, $\exists v \in W^{k,p}(U) \cap C^\infty(U)$ s.t. $\|u - v\|_{W^{k,p}(U)} < \delta$. $\square$

Definition 2.3.1. Let $U \subset \mathbb{R}^n$ be bounded. ∂U is C^k if for every point $x^0 \in \partial U$, $\exists r > 0$ and C^k -function $\gamma : \mathbb{R}^{n-1} \rightarrow \mathbb{R}^n$ s.t. we have

$$U \cap B(x^0, r) = \{x \in B(x^0, r) : x_n > \gamma(x_1, \dots, x_{n-1})\}$$

and furthermore ∂U is C^∞ if ∂U is C^k for $k = 1, 2, \dots$.

Theorem 2.3.3. Assume U is bounded and ∂U is C^1 . Suppose $u \in W^{k,p}(U)$ for some $1 \leq p < \infty$. Then there exist functions $u_m \in C^\infty(\overline{U})$ s.t. $u_m \rightarrow u$ in $W^{k,p}(U)$.

(If the boundary is "nice enough" we have that the density of $C^\infty(\overline{U})$ is in $W^{k,p}(U)$).

2.4 Extensions

Can we extend $u \in W^{k,p}(U)$, $U \subset \mathbb{R}^n$, to a function $u \in W^{1,p}(\mathbb{R}^n)$? It is possible, but we should be careful.

Example 2.6. Consider the function $U = \cos(\pi x)$ on $U = (0, 1)$. If we extend U by 0 to $\mathbb{R}$, then simply extending the function results in a discontinuity, and so $u \notin W^{1,p}(U)$.

Definition 2.4.1. Let $U \subset \mathbb{R}^n$ be open and bounded. Let ∂U be C^k . Then, locally in a neighbourhood at each point $x^0 \in \partial U$, there exists a C^k function $\Phi(\cdot)$ s.t.:

- (i) The mapping $x \rightarrow \Phi(x) = y$ straightens out ∂U near x^0 .
- (ii) $\Psi = \Phi^{-1}$.
- (iii) $\det(D\Psi) = \det(D\Phi) = 1$.

Theorem 2.4.1. Let $W \subset \mathbb{R}^n$ and Φ is a C^1 -mapping with inverse Ψ . Let $u'(y) = u(\Psi(y))$.⁴ Then, there exists constants $c_0, c_1 > 0$ s.t.

$$c_0 \|u\|_{W^{1,p}(W)} \leq \|u'\|_W^{1,p}(\Phi(W)) \leq c_1 \|u\|_{W^{1,p}(W)}.$$

Theorem 2.4.2 (The Extension Theorem). Let $1 \leq p < \infty$. Assume U is bounded and ∂U is C^1 . Choose a bounded open set V s.t. $U \subset\subset V$. Then, there exists a bounded linear operator $E : W^{1,p}(U) \rightarrow W^{1,p}(\mathbb{R}^n)$ s.t. $\forall u \in W^{1,p}(U)$, we have:

- (i) $Eu = u$ a.e. in U .
- (ii) $\text{spt}(Eu) \subset V$.
- (iii) $\exists C$ s.t. $\|Eu\|_{W^{1,p}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(U)}$.

Proof. (1). Fix $x^0 \in \partial U$ and assume ∂U is flat near x^0 lying in the plane $\{x_n = 0\}$. Then there exists an open ball $B = B^0(x^0, r)$ s.t.

$$\begin{aligned} B^+ &= B \cap \{x_n \geq 0\} \subset \bar{U} \\ B^- &= B \cap \{x_n \leq 0\} \subset \mathbb{R}^n - U. \end{aligned}$$

(2). In addition to $u \in W^{1,p}(U)$, assume $u \in C^\infty(\bar{U})$. Define

$$\bar{u}(x) := \begin{cases} u(x), & \text{if } x \in B^+ \\ -3u(x_1, \dots, x_{n-1}, -x_n) + 4u(x_1, \dots, x_{n-1}, -x_n/2), & \text{if } x \in B^-. \end{cases} \quad (13)$$

(3). We will show that $\bar{u}$ is C^1 on B . Let $u^- := \bar{u}|_{B^-}$ and $u^+ := \bar{u}|_{B^+}$. By the definition of $\bar{u}$,

- (a) $u^-|_{x_n=0} = u^+|_{x_n=0}$.
- (b) $u_{x_i}^-|_{x_n=0} = u_{x_i}^+|_{x_n=0}$ for $i = 1, \dots, n-1$.

⁴Note prime is not a derivative.

(c)

$$\begin{aligned}
u_{x_n}^-(x) &= 3u_{x_n}(x_1, \dots, x_{n-1}, -x_n) - 2u_{x_n}(x_1, \dots, x_{n-1}, -x_n/2) \\
\implies u_{x_n}^-(x) \Big|_{x_n=0} &= u_{x_n}^+(x) \Big|_{x_n=0} \\
\implies D^\alpha u^- \Big|_{x_n=0} &= D^\alpha u^+ \Big|_{x_n=0} \quad \text{for each } |\alpha| \leq 1.
\end{aligned}$$

(4).

$$\|\bar{u}\|_{W^{1,p}(B)} \leq C\|u\|_{W^{1,p}(B^+)}.$$

The proof of this is in assignment 2.

(5). We have assumed ∂U is flat near x^0 to find the result of (4). Now in the case where ∂U is not flat, we have that the boundary ∂U is C^1 , so there exists a C^1 -mapping Φ with an inverse Ψ s.t. Φ straightens out ∂U near x^0 ,

$$u = \Phi(x), \quad x = \Psi(y), \quad u'(y) = u(\Psi(y)).$$

Choose an open ball $B := B^0(y^0, r)$ in the y -coordinate system and set

$$\begin{aligned}
B^+ &= B \cap \{y_n \geq 0\} \\
B^- &= B \cap \{y_n \leq 0\}.
\end{aligned}$$

As in steps (2) and (3), we can extend u' from B^+ to $\bar{u}'$ in B , and then using step (4),

$$\|\bar{u}'\|_{W^{1,p}(B)} \leq C\|u'\|_{W^{1,p}(B^+)}.$$

Mapping this back to x -coordinates, define $W = \Psi(B)$. Using our norm equivalence theorem (Theorem 2.4.1),

$$\|\bar{u}_{W^{1,p}(W)}\| \leq C\|u\|_{W^{1,p}(\Psi(B^+))} \leq C\|u\|_{W^{1,p}(U)}.$$

(6). We can bound $\bar{u}$ on W by u on U in the $W^{1,p}$ -norm *in the neighbourhood of a point on the boundary*. However, we need this result to hold for all points on the boundary. Since ∂U is compact and C^1 , there exist finitely many points that $x_i^0 \in \partial U$, open sets W_i , and extensions $\bar{u}_i$ of u to W_i ($i = 1, \dots, N$), such that $\partial U \subset \bigcup_{i=1}^N W_i$.

Choose $W_0 \subset \subset U$ s.t. $U \subset \bigcup_{i=0}^N W_i$. Let $\{\zeta_i\}_{i=0}^N$ be the corresponding partition of unity:

$$\begin{cases} 0 \leq \zeta_i \leq 1, & \zeta_i \in C_c^\infty(W_i) \\ \sum_{i=0}^N \zeta_i(x) = 1, & x \in U. \end{cases}$$

Define $\bar{u} = \sum_{i=0}^N \zeta_i \bar{u}_i$, where $\bar{u}_0 = u$. Then,

$$\|\bar{u}_{W^{1,p}(\mathbb{R}^n)}\| \leq \sum_{i=0}^N \|\zeta_i \bar{u}_i\|_{W^{1,p}(\mathbb{R}^n)} = \sum_{i=0}^N \|\zeta_i \bar{u}_i\|_{W^{1,p}(W_i)} \leq \sum_{i=0}^N \|\bar{u}_i\|_{W^{1,p}(W_i)} \leq C\|u\|_{W^{1,p}(U)}.$$

It is possible to arrange for the support of $\bar{u}$ to lie in $V \supset \supset U$.

(7). Define the extension of u to $\mathbb{R}^n$ as $Eu = \bar{u}$. Note that $E[u + v] = Eu + Ev$ (so E is linear).

We have proven the result for $u \in C^\infty(\bar{U})$ in step (2), but we need $u \in W^{1,p}(U)$ for $1 \leq p < \infty$. We know that ∂U is C^1 so $C^\infty(\bar{U})$ is dense in $W^{1,p}(U)$. We can thus pick a sequence $\{u_m\}_{m=1}^\infty \subset C^\infty(\bar{U})$ s.t.

$$\|u - u_m\|_{W^{1,p}(U)} \leq \frac{1}{m}$$

for all m . Since E is linear, it also holds that

$$\|Eu_m - Eu_l\|_{W^{1,p}(\mathbb{R}^n)} \leq_{\text{step (6)}} \|u_m - u_l\|_{W^{1,p}(U)} \leq C \left(\frac{1}{m} + \frac{1}{l} \right).$$

Thus, $\{Eu_m\}_{m=1}^\infty$ is a Cauchy sequence in $W^{1,p}(\mathbb{R}^n)$. Recall that $W^{1,p}(\mathbb{R}^n)$ is Banach, so $\exists u^* \in W^{1,p}(\mathbb{R}^n)$ s.t. $Eu_m \rightarrow y^*$ in $W^{1,p}(\mathbb{R}^n)$.

Define $Eu = u^*$. To check this is well-defined, suppose that $\{u_m\}_{m=1}^\infty \subset C^\infty(\bar{U})$ is another sequence s.t. $\|u - u_m\|_{W^{1,p}(U)} \leq 1/m$. Then,

$$\begin{aligned} \|Eu - Ev_m\|_{W^{1,p}(\mathbb{R}^n)} &\leq \|Eu - Eu_m\|_{W^{1,p}} + \|Eu_m - Ev_m\|_{W^{1,p}} \\ &\leq \|Eu - Eu_m\|_{W^{1,p}} + C\|u_m - v_m\|_{W^{1,p}} \\ &\leq C(\|u_m - u\|_{W^{1,p}} + \|u - v_m\|_{W^{1,p}}). \end{aligned}$$

So, we are left to show that $\|Eu\| \leq C\|u\|_{W^{1,p}(U)}$. In fact,

$$\|Eu\|_{W^{1,p}(\mathbb{R}^n)} = \lim_{m \rightarrow \infty} \|Eu_m\|_{W^{1,p}(\mathbb{R}^n)} \leq_{\text{step (6)}} \lim_{m \rightarrow \infty} C\|u_m\|_{W^{1,p}(U)} = C\|u\|_{W^{1,p}(U)}.$$

□

2.5 Traces

Let $U \subset \mathbb{R}^n$ be open and bounded. If

- $u \in C(\bar{U})$, then u is defined on ∂U .
- $u \in W^{1,p}(U)$, then u is not continuous and u is only defined a.e.

To give meaning to u restricted to ∂U , we introduce the trace operator.

Theorem 2.5.1 (Trace Theorem). Assume U is bounded and ∂U is C^1 . Let $1 \leq p < \infty$. Then, there exists a bounded linear operator $T : W^{1,p}(U) \rightarrow L^p(\partial U)$ s.t.

- (i) if $u \in W^{1,p}(U) \cap C(\bar{U})$, then $Tu = u|_{\partial U}$.
- (ii) $\|Tu\|_{L^p(\partial U)} \leq C\|u\|_{W^{1,p}(U)} \quad \forall u \in W^{1,p}(U)$.

We call Tu the trace of u on ∂U .

Proof. (1). Assume $u \in C^1(\bar{U})$, let $x^0 \in \partial U$, and assume ∂U is *flat* near x^0 lying in the plane where $\{x_n = 0\}$. Choose an open ball $B = B(x^0, r)$ and let $B^+ = B \cap \{x_n \geq 0\} \subset \bar{U}$ and $B^- = B \cap \{x_n < 0\} \subset \mathbb{R}^n - U$. Let $\hat{B} = B(x^0, r/2)$ and let $\Gamma = \hat{B} \cap \partial U$. We shall choose $\zeta \in C_c^\infty(B)$ with

$$\begin{cases} \zeta \geq 0, & \text{in } B \\ \zeta \equiv 1, & \text{in } \hat{B}. \end{cases}$$

Further, we note that $\text{spt}(\zeta u) \subseteq B^+$ and $\zeta u = u$ on Γ . We let $x' = (x_1, \dots, x_{n-1}) \in \mathbb{R}^{n-1} = \{x_n = 0\}$. By the Fundamental Theorem of Calculus, we have

$$\int_0^\infty (\zeta|u|^p)_{x_n}(x't) dt = -\zeta|u|^p(x', 0).$$

Then,

$$\begin{aligned} \int_\Gamma |u|^p(x', 0) dx' &\leq \int_{\mathbb{R}^{n-1}} |u|^p(x', 0) dx' \\ &= - \int_{\mathbb{R}^{n-1}} \int_0^\infty (\zeta|u|^p)(x', t) dt dx' \\ &= - \int_{B^+} (\zeta|u|^p)_{x_n}(x) dx \\ &= - \int_{B^+} \left[|u|^p \zeta_{x_n} + p|u|^{p-1}(\text{sgn}(u))u_{x_n}\zeta \right] dx \\ &\leq \int_{B^+} \left[|u|^p |\zeta_{x_n}| + p|u|^{p-1}|u_{x_n}| \right] dx \\ &= c \int_{B^+} \left[|u|^p + |u|^{p-1}|u_{x_n}| \right] dx \\ &\leq c(|u_{x_n}^p| + |u|^p). \end{aligned}$$

By Young's inequality, we let $a = |u_{x_n}|$ and $b = |u|^{p-1}$, $p = p$ and $q = p/(p-1)$ so that

$$\int_\Gamma |u|^p dx' \leq c \int_{B^+} (|u|^p + |Du|^p) dx \implies \|u\|_{L^p(\Gamma)} \leq c\|u\|_{W^{1,p}(\Omega)}.$$

(2). Similar to the proof of the Extension Theorem (Theorem 2.4.2), if ∂U is not flat near x^0 , we can straighten the boundary and apply the estimate from step (1), and further apply a change of variable. We thus have

$$\|u\|_{L^p(\Gamma)} \leq c\|u\|_{W^{1,p}(U)}$$

where $\Gamma \subset \partial U$ which is continuous on x^0 .

(3). Again, similar to the Extension Theorem (Theorem 2.4.2), since ∂U is compact, there exists a finite number of points $x_i^0 \in \partial U$ and open subsets $\Gamma_i \subset \partial U$ (for $i = 1, \dots, n$) s.t. $\partial U \subset \bigcup_{i=1}^N \Gamma_i$, and by step (2), we have the

$$\|u\|_{L^p(\Gamma_i)} \leq c\|u\|_{W^{1,p}(U)}.$$

If we define $Tu := u|_{\partial U}$ (remember $u \in C^1(\overline{U})$), we have

$$\|Tu\|_{L^p(\partial U)}^p = \int_{\partial U} |u|^p dx \leq \sum_{i=1}^N \int_{\Gamma_i} |u|^p dx \leq c\|u\|_{W^{1,p}(U)}^p.$$

We have proven the result for $u \in C^1(\overline{U})$, and so we will now prove it for $u \in W^{1,p}(U)$.

(4). To show for $u \in W^{1,p}(U)$, we use Extension Theorem (Theorem 2.4.2) to invoke that smooth functions are dense in $W^{1,p}(U)$. We can choose a sequence $\{u_m\}_{m=1}^\infty \subset C^\infty(\overline{U})$ s.t. $\|u - u_m\|_{W^{1,p}(U)} \leq 1/m \ \forall m$. Then,

$$\|Tu_m - Tu_n\|_{L^p(\partial U)} = \|T(u_m - u_n)\|_{L^p(\partial U)} \leq c\|u_m - u_n\|_{W^{1,p}(U)} \leq \frac{1}{m} + \frac{1}{n} \quad \forall m, n$$

and so $\{Tu_m\}_{m=1}^\infty$ is a Cauchy sequence in $L^p(\partial U)$. Since $L^p(\partial U)$ is complete, $\exists v \in L^p(\partial U)$ s.t. $\|v - Tu_m\|_{L^p(\partial U)} \rightarrow 0$ as $m \rightarrow \infty$. Define $Tu := v$

We must check that this definition does not depend on the sequence. Let $\{u_m\}_{m=1}^\infty \subset C^\infty(\bar{U})$ s.t. $\|u_m - u\|_{W^{1,p}(U)} \rightarrow 0$. Then, by the triangle inequality,

$$\begin{aligned} \|Tu - Tv_m\|_{L^p(\partial U)} &\leq \|Tu - Tu_m\|_{L^p(\partial U)} + \|Tu_m - Tv_m\|_{L^p(\partial U)} \\ &\leq \|Tu - Tu_m\|_{L^p(\partial U)} + c\|u_m - v_m\|_{W^{1,p}(U)} \\ &\leq \|Tu - Tu_m\|_{L^p(\partial U)} + c\left(\|u_m - u\|_{W^{1,p}(U)} + \|u - v_m\|_{W^{1,p}(U)}\right) \rightarrow 0. \end{aligned}$$

where we have used the result from step (3). So, Tu is well-defined in $L^p(\partial U)$ and we have

$$\|Tu\|_{L^p(\partial U)} = \|v\|_{L^p(\partial U)} = \lim_{m \rightarrow \infty} \|Tu_m\|_{L^p(\partial U)} \leq \lim_{m \rightarrow \infty} c\|u_m\|_{W^{1,p}(U)} = c\|u\|_{W^{1,p}(U)}.$$

□

Theorem 2.5.2. Let U be bounded and ∂U be C^1 . Let $u \in W^{1,p}(U)$. Then, $u \in W_0^{1,p}(U) \implies Tu = 0$ on ∂U . That is, $\|Tu\|_{L^p(\partial U)} = 0$, or that $W_0^{1,p}(U)$ is the closure of $C^\infty(U)$ in $W^{1,p}(U)$.

2.6 Sobolev Inequalities

If $u \in W^{1,p}(U)$, does it automatically belong to any other space? The answer is yes, but we must be careful as the space depends on the dimension of our domain.

2.6.1 Gagliardo-Nirenberg-Sobolev Inequality

We will consider the case $1 \leq p < n$.

Motivation. Let $1 \leq p < n$ be true. Can the following estimate

$$\|u\|_{L^q(\mathbb{R}^n)} \leq c\|Du\|_{L^p(\mathbb{R}^n)} \quad \forall u \in C_c^\infty(\mathbb{R}^n)$$

hold for any $1 \leq q < \infty$? To check this, we use an example. Let $u \in C_c^\infty(\mathbb{R}^n)$, $u \not\equiv 0$, and define for $\lambda > 0$

$$u_\lambda(x) = u(\lambda x), \quad \text{for } x \in \mathbb{R}^n.$$

We shall apply the estimate to $u_\lambda(x)$, so that

$$\|u_\lambda\|_{L^q(\mathbb{R}^n)} \leq c\|Du_\lambda\|_{L^p(\mathbb{R}^n)}.$$

By definition, we have on the LHS,

$$\|u_\lambda\|_{L^q(\mathbb{R}^n)}^q = \int_{\mathbb{R}^n} |u_\lambda|^q dx = \int_{\mathbb{R}^n} |u(\lambda x)|^q dx = \frac{1}{\lambda^n} \int_{\mathbb{R}^n} |u(y)|^q dy$$

where $\lambda x = y$. On the RHS, we have by definition

$$\|Du_\lambda\|_{L^p(\mathbb{R}^n)}^p = \int_{\mathbb{R}^n} |D_\lambda u_\lambda|^p dx = \frac{1}{\lambda^n} \lambda^p \int_{\mathbb{R}^n} |D_y u(y)|^p dy.$$

Then, we have

$$\frac{1}{\lambda^{n/q}} \|u\|_{L^q(\mathbb{R}^n)} \leq \frac{c\lambda}{\lambda^{n/p}} \|Du\|_{L^p(\mathbb{R}^n)} \implies \|u\|_{L^q(\mathbb{R}^n)} \leq c\lambda^{1-n/p+n/q} \|Du\|_{L^p(\mathbb{R}^n)}$$

for all $\lambda > 0$. Observe that if

$$1 - \frac{n}{p} + \frac{n}{q} > 0$$

and $\lambda \rightarrow 0$, the upper bound vanishes which will lead to a contradiction for $u \neq 0$. If

$$1 - \frac{n}{p} + \frac{n}{q} < 0$$

and $\lambda \rightarrow \infty$, we have the same problem. Thus, the only way for the estimate on $\|u\|_{L^p(\mathbb{R}^n)}$ to hold is to have

$$1 - \frac{n}{p} + \frac{n}{q} = 0 \implies q = \frac{np}{n-p}.$$

Definition 2.6.1. If $1 \leq p < n$, the **Sobolev conjugate** of p is

$$p^* := \frac{np}{n-p}.$$

Note that

$$\frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}, \quad p^* > p.$$

Theorem 2.6.1 (Estimates for $W^{1,p}$, $1 \leq p < n$). Assume $1 \leq p < n$ and $U \subset \mathbb{R}^n$ is open and bounded.

- If ∂U is C^1 , then

$$\|u\|_{L^{p^*}(U)} \leq c\|u\|_{W^{1,p}(U)} \quad \forall u \in W^{1,p}(U).$$

Remark. Since U is bounded, we actually have:

$$\|u\|_{L^q(U)} \leq c\|u\|_{W^{1,p}(U)} \quad \forall u \in W^{1,p}(U), q \in [1, p^*].$$

- If $u \in W_0^{1,p}(U)$, then

- $\|u\|_{L^q(U)} \leq c\|Du\|_{L^p(U)}$ for $q \in [1, p^*]$.
- $\|u\|_{L^p(U)} \leq c\|Du\|_{L^p(U)}$ for $1 \leq p \leq \infty$.

Proof. This shall be proved in A3. □

Definition 2.6.2.

- $u : U \rightarrow \mathbb{R}$ is **Hölder-continuous** with exponent $0 < \gamma \leq 1$ if

$$|u(x) - u(y)| \leq c|x - y|^\gamma, \quad x, y \in U.$$

If $\gamma = 1$, then u is **Lipschitz continuous**.

- If $u : U \rightarrow \mathbb{R}$ is bounded and continuous, then

$$\|u\|_{C(\bar{U})} = \sup_{x \in \bar{U}} |u(x)|.$$

- The γ **th-Hölder semi-norm** of $u : U \rightarrow \mathbb{R}$ is

$$[u]_{C^{0,\gamma}(\bar{U})} = \sup_{x,y \in U, x \neq y} \left\{ \frac{|u(x) - u(y)|}{|x - y|^\gamma} \right\}$$

and the γ **th-Hölder norm** is

$$\|u\|_{C^{0,\gamma}(\bar{U})} = \|u\|_{C(\bar{U})} + [u]_{C^{0,\gamma}(\bar{U})}.$$

- Define

$$\|u\|_{C^{k,\gamma}(\bar{U})} := \sum_{|\alpha| \leq k} \|D^\alpha u\|_{C(\bar{U})} + \sum_{|\alpha| \leq k} [D^\alpha u]_{C^{0,\gamma}(\bar{U})}.$$

The **Hölder space** is defined as

$$C^{k,\gamma}(\bar{U}) = \left\{ u \in C^k(\bar{U}) : \|u\|_{C^{k,\gamma}(\bar{U})} < \infty \right\}.$$

- $C^{k,\gamma}(\bar{U})$ is a Banach space.

Theorem 2.6.2. Let $n < p \leq \infty$, $U \subset \mathbb{R}^n$ be open and bounded, ∂U be C^1 , and $u \in W^{1,p}(U)$. Then, u has a version $u^* \in C^{0,\gamma}(\bar{U})$ for $\gamma = 1 - n/p$ s.t.

$$\|u^*\|_{C^{0,\gamma}(\bar{U})} \leq C \|u\|_{W^{1,p}(U)}.$$

Note in the case $p = \infty$, we have $\gamma = 1$ so that

$$\|u^*\|_{C^{0,1}(\bar{U})} \leq C \|u\|_{W^{1,\infty}(U)}.$$

That is, every $u \in W^{1,p}(U)$ has a Lipschitz continuous version u^* .

Theorem 2.6.3. Let $U \subset \mathbb{R}^n$ be open and bounded, ∂U be C^1 , and $u \in W^{k,p}(U)$.

- (i) If $k < n/p$, define q s.t. $1/q = 1/p - k/n$. Then,

$$\|u\|_{L^q(\Omega)} \leq C \|u\|_{W^{k,p}(\Omega)}.$$

- (ii) If $k > n/p$, define⁵ $r = k - [n/p] - 1$. Then, $u \in C^{r,\gamma}(\bar{U})$, where

$$\gamma = \begin{cases} \left[\frac{n}{p} \right] + 1 - \frac{n}{p}, & \text{if } \frac{n}{p} \text{ is not an integer.} \\ \text{any positive } nr < 1, & \text{if } \frac{n}{p} \text{ is an integer} \end{cases}$$

$$\text{and } \|u\|_{C^{r,\gamma}(\bar{U})} \leq C \|u\|_{W^{k,p}(U)}.$$

⁵We will define $[\cdot]$ to be the operator returning the integer-part of $(\cdot)$ (i.e. $[2.3] = 2$).

2.7 Compact Embeddings

Theorem 2.7.1 (Arzela-Ascoli Compactness Criterion). Let $\{f_k\}_{k=1}^\infty$ be a sequence of real-valued functions on $\mathbb{R}^n$ s.t.

- (i) $|f_k(x)| \leq M$ ($k = 1, \dots, x \in \mathbb{R}^n$), and M be a constant.
- (ii) $\{f_k\}_{k=1}^\infty$ be **equicontinuous**, i.e.,

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } |x - y| < \delta \implies |f_k(x) - f_k(y)| < \varepsilon, \quad \forall k \in \mathbb{N}, x, y \in \mathbb{R}^n.$$

Then, there exists a subsequence $\{f_{k_j}\}_{j=1}^\infty \subseteq \{f_k\}_{k=1}^\infty$ and continuous function f s.t. $f_{k_j} \rightarrow f$ uniformly on compact subsets of $\mathbb{R}^n$.

Definition 2.7.1. Let X, Y be Banach spaces and $X \subset Y$. Then, X is **compactly embedded** in Y , which is denoted by $X \subset\subset Y$ ⁶ if:

- (i) $\|u\|_Y \leq C\|u\|_X, \forall u \in X$.
- (ii) each bounded sequence in X is **precompact** in Y , meaning: $\{u_k\}_{k=1}^\infty$ in X is bounded, then some subsequence $\{u_{k_j}\}_{j=1}^\infty \subseteq \{u_k\}_{k=1}^\infty$ converges in Y to some limit $u \in Y$. That is, $\lim_{j \rightarrow \infty} \|u_{k_j} - u\|_Y = 0$.

Definition 2.7.2 (The Interpolation Inequality for L^p -norms). Assume $1 \leq s \leq r \leq t \leq \infty$ and $1/r = \theta/s + (1 - \theta)/t$ for $0 \leq \theta \leq 1$. Suppose $u \in L^s(U) \cap L^t(U)$. Then, $u \in L^r(U)$, and

$$\|u\|_{L^r(U)} \leq \|u\|_{L^s(U)}^\theta \|u\|_{L^t(U)}^{1-\theta}.$$

Theorem 2.7.2 (Rellich-Kondrachov Compactness Theorem). Let $U \subset \mathbb{R}^n$ be open and bounded and ∂U be C^1 . Let $1 \leq p < n$. Then, $W^{1,p}(U) \subset\subset L^q(U)$ for each $1 \leq q < p^*$ ($= np/(n - p)$).

Proof. $W^{1,p}(U) \subset\subset L^q(U)$ means:

- (i) $W^{1,p}(U) \subset L^q(U)$ with $\|u\|_{L^q(U)} \leq C\|u\|_{W^{1,p}(U)} \forall u \in W^{1,p}(U)$.
- (ii) If $\{\hat{u}_m\}_{m=1}^\infty \subset W^{1,p}(U)$ is bounded, then there exists a subsequence $\{\hat{u}_{m_j}\}_{j=1}^\infty$ and $u \in L^q(U)$ s.t. $\hat{u}_{m_j} \rightarrow u$ in $L^q(U)$.

(1). Since U is bounded, (i) follows from Theorem (2.6.1).

(2). By the Extension Theorem (Theorem ??), we may assume:

- (i) $\{u_m\}_{m=1}^\infty \subset W^{1,p}(\mathbb{R}^n)$ with $u_m|_U = \hat{u}_m$.
- (ii) for all $m = 1, \dots$, there exists V s.t. $U \subset\subset V$ and $\text{spt}(u_m) \subset V$.
- (iii) $\sup_m \|u_m\|_{W^{1,p}(\mathbb{R}^n)} = \sup_m \|u_m\|_{W^{1,p}(V)} \leq \sup_m C\|\hat{u}_m\|_{W^{1,p}(U)} < \infty$.

⁶While using the same notation, note that $X \subset\subset Y$ in the context of compactness deals with Banach spaces while $X \subset\subset Y$ in the context of compact containment deals with domains.

(3). Consider the smoothed functions $u_m^\varepsilon = \eta_\varepsilon * u_m$ ($\varepsilon > 0$, $m = 1, 2, \dots$). We may also assume $\{u_m^\varepsilon\}_{m=1}^\infty$ have support in V .

(4). We claim that $u_m^\varepsilon \rightarrow u_m$ in $L^q(V)$ as $\varepsilon \rightarrow 0$ uniformly in m ($\sup_m \|u_m^\varepsilon - u_m\|_{L^q(V)} \rightarrow 0$). To show this, we shall first consider the case u_m is smooth. We have

$$u_m^\varepsilon(x) - u_m(x) = (\eta_\varepsilon * u_m)(x) - u_m(x) = \int_{B(0,\varepsilon)} \eta_\varepsilon(y) u_m(x-y) dy - u_m(x) \int_{B(0,\varepsilon)} \eta_\varepsilon(y) dy.$$

The last integral is one, so we have

$$\int_{B(0,\varepsilon)} \eta_\varepsilon(y) u_m(x-y) dy - u_m(x) \int_{B(0,\varepsilon)} \eta_\varepsilon(y) dy = \int_{B(0,\varepsilon)} \eta_\varepsilon(y) [u_m(x-y) - u_m(x)] dy.$$

We know $\eta_\varepsilon(y) = \eta(y/\varepsilon)/\varepsilon^n$ by change of variables $\varepsilon z = y \rightarrow dy_1 dy_2 \dots dy_n = \varepsilon^n dz_1 \dots dz_n$. Then,

$$\begin{aligned} \int_{B(0,\varepsilon)} \eta_\varepsilon(y) (u_m(x-y) - u_m(x)) dy &= \int_{B(0,1)} \eta(z) [u_m(x - \varepsilon z) - u_m(x)] dz \\ &= \int_{B(0,1)} \eta(y) \left[\int_0^1 \frac{d}{dt} (u_m(x - \varepsilon ty)) dt \right] dy \end{aligned}$$

where $z = ty$ (not exactly sure). We shall use the chain-rule relation

$$\frac{d}{dt} u_m(x - \varepsilon ty) = Du_m(x - \varepsilon ty)[- \varepsilon y]$$

so that

$$\int_{B(0,1)} \eta(y) \left[\int_0^1 \frac{d}{dt} (u_m(x - \varepsilon ty)) dt \right] dy = \int_{B(0,1)} \eta(y) \int_0^1 Du_m(x - \varepsilon ty)[- \varepsilon y] dt dy.$$

We will take the absolute value on both sides, we integrate over V , use the relation $0 \leq y \leq 1$, and use $|\int f dx| \leq \int |f| dx$ and swap integrals.

$$\begin{aligned} \int_V |u_m^\varepsilon(x) - u_m(x)| dx &\leq \varepsilon \int_{B(0,1)} \eta(y) \int_0^1 \int_V |Du_m(x - \varepsilon ty)| dx dt dy \\ &= \varepsilon \int_{B(0,1)} \eta(y) \int_0^1 \left(\int_{\mathbb{R}^n} |Du_m(x - \varepsilon ty)| dx \right) dt dy. \end{aligned}$$

Applying the change of variable $z = x - \varepsilon ty$, we have

$$\varepsilon \int_{B(0,1)} \eta(y) \int_0^1 \left(\int_{\mathbb{R}^n} |Du_m(x - \varepsilon ty)| dx \right) dt dy = \varepsilon \int_{B(0,1)} \eta(y) dy \int_V |Du_m(z)| dz.$$

Notice that the integral in $\eta(y)$ is one, and the $\mathbb{R}^n$ can be reduced to V which depends only on x , so that we have

$$\int_V |u_m^\varepsilon(x) - u_m(x)| dx \leq \varepsilon \int_V |Du_m(x)| dx.$$

So, for $u_m \in C_c^\infty(V)$ and $\text{spt}(u_m^\varepsilon) \subset V$, we have

$$\|u_m^\varepsilon - u_m\|_{L^1(V)} \leq \varepsilon \|Du_m\|_{L^1(V)}.$$

However, smooth functions are dense in $W^{1,p}(V)$ (this same argument was used to prove the Extension Theorem), and so by a density argument, we have that the above inequality holds for all $u_m \in W^{1,p}(V)$. We also know that $L^q(U) \subset L^p(U)$, and thus

$$\|u_m^\varepsilon - u_m\|_{L^1(V)} \leq \varepsilon \|Du_m\|_{L^1(V)} \leq C\varepsilon \|Du_m\|_{L^p(V)} \quad \forall u_m \in W^{1,p}(V).$$

By Step (2) item (iii), we have that $\|Du_m\|_{L^p(V)} < \infty$ for all m , and so

$$\|u_m^\varepsilon - u_m\|_{L^1(V)} \leq C\varepsilon \quad (C \text{ indep. of } m, \varepsilon)$$

i.e., $u_m^\varepsilon \rightarrow u_m$ in $L^1(V)$ uniformly in m . But, $1 \leq q < p^*$, so using the interpolation inequality for L^p -norms with $s = 1$, $r = q$, $t = p^*$, we have

$$\|u_m^\varepsilon - u_m\|_{L^q(V)} \leq \|u_m^\varepsilon - u_m\|_{L^q(V)}^\theta \|u_m^\varepsilon - u_m\|_{L^{p^*}(V)}^{1-\theta} \leq \|u_m^\varepsilon - u_m\|_{L^q(V)}^\theta C \|u_m^\varepsilon - u_m\|_{W^{1,p}(V)}^{1-\theta}.$$

Since the last norm is finite ($\|u_m^\varepsilon - u_m\|_{W^{1,p}(V)}^{1-\theta} < C$), we thus have

$$\|u_m^\varepsilon - u_m\|_{L^q(V)} \leq C \|u_m^\varepsilon - u_m\|_{L^1(V)}^\theta \leq C\varepsilon^\theta.$$

That is,

$$\|u_m^\varepsilon - u_m\|_{L^q(V)} \leq C\varepsilon^\theta \rightarrow 0 \text{ as } \varepsilon \rightarrow 0.$$

(5). For $\varepsilon > 0$ fixed, we have $u_m^\varepsilon \in C_c^\infty(U)$ and so $\{u_m^\varepsilon\}_{m=1}^\infty$ is a sequence in $C_c^\infty(\mathbb{R}^n)$. We claim that for each fixed $\varepsilon > 0$, the sequence $\{u_m^\varepsilon\}_{m=1}^\infty$ is uniformly bounded and equicontinuous:

$$(a) \quad \sup_m \|u_m^\varepsilon\|_{L^\infty(\mathbb{R}^n)} < \infty.$$

$$(b) \quad \forall \eta > 0, \exists \delta > 0 \text{ s.t. } |x - y| < \delta \implies |u_m^\varepsilon(x) - u_m^\varepsilon(y)| < \eta \quad \forall m \in \mathbb{N}, \forall x, y \in \mathbb{R}^n.$$

We prove first (a). Let $x \in \mathbb{R}^n$. Then,

$$|u_m^\varepsilon(x)| = \left| \int_V \eta_\varepsilon(x - y) u_m(y) dy \right| \leq \|\eta_\varepsilon\|_{L^\infty(\mathbb{R}^n)} \int_V |u_m(y)| dy.$$

We can change the region of integration from V to $\mathbb{R}^n$, and notice that the last integral is finite. Indeed,

$$\|u_m\|_{L^1(V)} \leq C \|u_m\|_L$$

Further, we apply the change of variable $\eta_\varepsilon(x) = \eta(x/\varepsilon)/\varepsilon^n$ and $\eta(x/\varepsilon) \leq C$, so we have

$$|u_m^\varepsilon(x)| \leq \frac{C}{\varepsilon^n} \quad \text{for } n \in \mathbb{N}.$$

We have thus proven (a). To prove (b), we shall notice that since $D^\alpha u_m^\varepsilon = u_m * D^\alpha \eta_\varepsilon$, we have by the property of the mollifiers

$$|Du_m^\varepsilon(x)| = \left| \int_V D\eta_\varepsilon(x - y) u_m(y) dy \right| \leq \|D\eta_\varepsilon\|_{L^\infty(\mathbb{R}^n)} \int_V |u_m(y)| dy.$$

We change the region of integration from V to $\mathbb{R}^n$ and notice the specific last integral is finite ($< C$), and so we find from the change of variables

$$D\eta_\varepsilon(x) = \frac{D\eta(x/\varepsilon)}{\varepsilon^n} = \frac{1}{\varepsilon^{n+1}} D\eta(y) \quad \text{with } y = \frac{x}{\varepsilon}$$

so that we have

$$|Du_m^\varepsilon(x)| \leq \frac{C}{\varepsilon^{n+1}} \quad \forall m \in \mathbb{N}.$$

Since $u_m^\varepsilon \in C^\infty(c)(\mathbb{R}^n)$, it is also differentiable everywhere and thus Lipschitz continuous with a Lipschitz constant $\|Du_m^\varepsilon\|_{L^\infty(\mathbb{R}^n)}$. Thus,

$$\|u_m^\varepsilon(x) - u_m^\varepsilon(y)\| \leq \|Du_m^\varepsilon\|_{L^\infty(\mathbb{R}^n)}|x - y| \leq \frac{C}{\varepsilon^{n+1}}|x - y|$$

and so we have proven (b).

(6). By step (5), $\{u_m^\varepsilon\}_{m=1}^\infty$ satisfies the conditions to apply the Arzela-Ascoli Compactness Criterion (Theorem 2.7.1). Thus, there exists a subsequence $\{u_{m_k}\}_{k=1}^\infty$ that converges uniformly to a continuous function u^ε on V .

Let $\mathbb{N}_\varepsilon \subset \mathbb{N}$ be the set s.t. $\{u_{m_k}\}_{k=1}^\infty = \{u_k^\varepsilon\}_{k \in \mathbb{N}_\varepsilon}$. Then, $\forall \varepsilon, \exists \mathbb{N}_\varepsilon \subseteq \mathbb{N}$ s.t. $\{u_k^\varepsilon\}_{k \in \mathbb{N}_\varepsilon}$ converges uniformly on V . However, uniform convergence implies convergence in $L^1(V)$ ($1 \leq q \leq \infty$). Therefore, $\{u_k^\varepsilon\}_{k \in \mathbb{N}_\varepsilon}$ is a Cauchy-sequence in $L^1(V)$, and so we have

$$\forall \varepsilon > 0, \exists \mathbb{N}_\varepsilon \subset \mathbb{B} \text{ s.t. } \forall l \in \mathbb{N}, \exists \mathbb{N}_l \subset \mathbb{N}_\varepsilon \text{ s.t. } \forall i, j \in \mathbb{N}_l, \|u_i^\varepsilon - u_j^\varepsilon\| \leq \frac{1}{l}.$$

From step (4), we have that $u_m^\varepsilon \rightarrow u_m$ as $\varepsilon \rightarrow 0$ uniformly, so

$$\forall l \in \mathbb{N}, \exists \varepsilon_l > 0 \text{ s.t. } \forall \varepsilon < \varepsilon_l \text{ and } m \in \mathbb{N}, \|u_m^\varepsilon - u_m\|_{L^q(V)} \leq \frac{1}{l}.$$

Combining these two statements, we thus have $\forall l \in \mathbb{N} \exists \mathbb{N}_l \subseteq \mathbb{N}$ s.t. $\forall i, j \in \mathbb{N}_l$,

$$\begin{aligned} \|u_i - u_j\|_{L^q(V)} &= \|u_i - u_i^\varepsilon + u_i^\varepsilon - u_j^\varepsilon + u_j^\varepsilon - u_j\|_{L^q(V)} \\ &\leq \|u_i - u_i^\varepsilon\|_{L^q(V)} + \|u_i^\varepsilon - u_j^\varepsilon\|_{L^q(V)} + \|u_j^\varepsilon - u_j\|_{L^q(V)} \\ &\leq \frac{3}{l}. \end{aligned}$$

(7). Note that

$$\mathbb{N}_1 > \mathbb{N}_2 > \mathbb{N}_3 > \dots > \mathbb{N}_l > \dots$$

Let $N_i = \min(\mathbb{N}_i)$. Then, $N_j \in \mathbb{N}_i$ for all $j \geq 1$. Therefore, if $j \geq i$ for $i, j \in \mathbb{N}$, we have by the inequality above that

$$\|u_{N_i} - u_{N_j}\|_{L^q(V)} \leq \frac{3}{i}.$$

If $i \geq j$, the same equality holds instead for $3/j$. Thus, $\forall i, j \in \mathbb{N}$, we have that

$$\|u_{N_i} - u_{N_j}\|_{L^q(V)} \leq \max\left(\frac{3}{i}, \frac{3}{j}\right).$$

So, $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. $\forall i, j > N$,

$$\|u_{N_i} - u_{N_j}\|_{L^q(V)} < \varepsilon.$$

Thus, $\{u_{N_i}\}_{i \in \mathbb{N}}$ is a Cauchy sequence in $L^q(V)$ with limit $u \in L^q(V)$. That is,

$$\exists u \in L^q(V) \text{ s.t. } \|u_{N_i} - u\|_{L^q(V)} \rightarrow 0 \text{ as } i \rightarrow \infty.$$

But $U \subseteq V$, and so

$$\|u_{N_i} - u\|_{L^q(U)} \rightarrow 0 \text{ as } i \rightarrow \infty.$$

We have thus proven (ii). □

Remark. Let $U \subset \mathbb{R}^n$ be open and bounded. We showed that $W^{1,p}(U) \subset\subset L^q(U)$ if $1 \leq q < p^*$. It can also be shown that for $1 \leq p \leq \infty$:

- (i) $W^{1,p}(U) \subset\subset L^p(U)$ if ∂U is C^1 .
- (ii) $W_0^{1,p}(U) \subset\subset L^p(U)$ (no need for ∂U to be C^1).

2.8 Additional Topics

2.8.1 Poincaré Inequalities

Notation.

$$(u)_U = \int_U u \, dx = \frac{1}{|U|} \int_U u \, dx.$$

Definition 2.8.1. A domain U is **connected** if any two points in U can be connected by a curve completely contained in U .

Lemma 2.8.1. Let $U \subset \mathbb{R}^n$ be open and connected. Let $u \in W^{1,p}(U)$ for $1 \leq p \leq \infty$. Then, $Du = 0$ a.e if and only if u is a constant.

Theorem 2.8.2 (Poincaré's Inequality). Let $U \subset \mathbb{R}^N$ be open, bounded, connected, and such that ∂U is C^1 . Let $1 \leq p \leq \infty$. Then, $\exists C > 0$ s.t.

$$\|u - (u)_U\|_{L^p(U)} \leq C \|Du\|_{L^p(U)} \quad \forall u \in W^{1,p}(U).$$

Proof. (1). Assume the statement is false, i.e.,

$$\forall k \in \mathbb{N}, \exists u_k \in W^{1,p}(U) \text{ s.t. } \|u_k - (u_k)_U\|_{L^p(U)} > k \|Du_k\|_{L^p(U)}.$$

Define

$$v_k := \frac{u_k - (u_k)_U}{\|u_k - (u_k)_U\|_{L^p(U)}}.$$

We show that $\{v_k\}_{k=1}^\infty$ is bounded in $W^{1,p}(U)$. Note that $\|v_k\|_{L^p(U)} = 1$ and that $(v_k)_U = 0$. Also,

$$Dv_k = \frac{Du_k}{\|u_k - (u_k)_U\|_{L^p(U)}}.$$

Then,

$$\|Dv_k\|_{L^p(U)} = \frac{\|Du_k\|_{L^p(U)}}{\|u_k - (u_k)_U\|_{L^p(U)}} < \frac{1}{k}.$$

This means that

$$\|v_k\|_{W^{1,p}(U)} \leq C \left(\|Dv_k\|_{L^p(U)} + \|v_k\|_{L^p(U)} \right) \leq C \left(\frac{1}{k} + 1 \right).$$

That is, $\{u_k\}_{k=1}^\infty$ is a bounded sequence in $W^{1,p}(U)$. Therefore, using $W^{1,p}(U) \subset\subset L^p(U)$, we know there exists a subsequence $\{v_{k_j}\}_{j=1}^\infty \subset \{v_k\}_{k=1}^\infty$ and a $v \in L^p(U)$ such that $v_{k_j} \rightarrow v$ in $L^p(U)$ as $j \rightarrow \infty$. By construction, we have that $(v)_U = 0$ and $\|v\|_{L^p(U)} = 1$.

(2). We will show that $v \in W^{1,0}(U)$ with $Dv = 0$. First note that $\forall \phi \in C_c^\infty(U)$, we have

$$\begin{aligned} \|\phi_{x_i} v_{k_j} - \phi_{x_i} v\|_{L^p(U)}^p &= \int_U |\phi_{x_i} v_{k_j} - \phi_{x_i} v|^p dx \\ &= \int_U |\phi_{x_i}|^p |v_{k_j} - v|^p dx \\ &\leq \|\phi_{x_i}\|_{L^\infty(U)}^p \int_U |v_{k_j} - v|^p dx \\ &= \|\phi_{x_i}\|_{L^\infty(U)}^p \|v_{k_j} - v\|_{L^p(U)}^p \rightarrow 0 \end{aligned}$$

so that $\phi_{x_i} v_{k_j} \rightarrow \phi_{x_i} v$ in $L^p(U)$. By Hölder's inequality and since U is bounded, $\phi_{x_i} v_{k_j} \rightarrow \phi_{x_i} v$ in $L^1(U)$, so

$$\begin{aligned} \lim_{j \rightarrow \infty} \int_U |\phi_{x_i} v_{k_j} - \phi_{x_i} v| dx &= 0 \implies \lim_{j \rightarrow \infty} \int_U \phi_{x_i} v_{k_j} dx = \int_U \phi_{x_i} v dx \\ &\implies - \lim_{j \rightarrow \infty} \int_U \phi v_{k_j, x_i} dx = \int_U \phi_{x_i} v dx. \end{aligned}$$

But also,

$$\left| \lim_{j \rightarrow \infty} \int_U \phi v_{k_j, x_i} dx \right| \leq \lim_{j \rightarrow \infty} \int_U |\phi v_{k_j, x_i}| dx \leq_{\text{Hölder}} \lim_{j \rightarrow \infty} \|Dv_{k_j}\|_{L^p(U)} \|\phi\|_{L^q(U)} \leq_{\text{step (1)}} \lim_{j \rightarrow \infty} \frac{c}{k_j}.$$

Thus, the integral in U and the limit in j show that $Dv = 0$ a.e. We have shown that $v \in W^{1,p}(U)$ and $Dv = 0$, so v must be a constant. However, from step (1) we have $(v)_U = 0$ which is possible only if $v = 0$, but simultaneously $\|v\|_{L^p(U)} = 1$, so v also cannot be zero. Thus, we have reached a contradiction, so the initial statement must hold. $\square$

2.8.2 The H^{-1} Space.

Definition 2.8.2. $H^{-1}(U)$ denotes the dual space of $H_0^1(U)$:

$$H^{-1}(U) = \left\{ f : H_0^1(U) \rightarrow \mathbb{R} : |\langle f, v \rangle| \leq c \|v\|_{H^1(U)} \quad \forall v \in H_0^1(U) \right\}.$$

Recall that the notation $\langle f, v \rangle = f(v)$ and that

$$\|f\|_{H^{-1}(U)} = \sup_{u \in H_0^1(U), \|u\|_{H^1(U)} \leq 1} |\langle f, u \rangle| \equiv \sup_{0 \neq u \in H_0^1(U)} \frac{\langle f, u \rangle}{\|u\|_{H^1(U)}}.$$

Theorem 2.8.3.

(i) If $f \in H^{-1}(U)$, then we can identify it with a tuple $(f^0, \dots, f^n)$ of functions in $L^2(U)$ via:

$$\langle f, v \rangle = (f^0, v)_{L^2(U)} + \sum_{i=1}^n (f^i, v_{x_i})_{L^2(U)} \quad \forall v \in H_0^1(U).$$

(ii) If $f \in H^{-1}(U)$, then the H^{-1} -norm can also be written as:

$$\|f\|_{H^{-1}(U)} = \inf \left\{ \left(\sum_{i=1}^n \|f^i\|_{L^2(U)}^2 \right)^{\frac{1}{2}} : f \text{ satisfies (i) for } f^0, \dots, f_n \in L^2(U) \right\}.$$

(iii) $\forall v \in L^2(U) \subset H^{-1}(U)$ and $\forall u \in H_0^1(U)$,

$$\langle v, u \rangle = (v, u)_{L^2(U)}.$$

Indeed, we have that $H_0^1(U) \subset L^2(U) \subset H^{-1}(U)$.

Proof. (1). We shall prove (i). Let $f \in H^{-1}(U)$. By the Riesz-Representation Theorem (Theorem 1.9.4), $\exists! u \in H_0^1(U)$ s.t.

$$(u, v)_{H^1(U)} = \langle f, v \rangle \quad \forall v \in H_0^1(U).$$

Set $f^0 = u \in L^2(U)$ and $f^i = u_{x_i} \in L^2(U)$ for $i = 1, \dots, n$. Then,

$$(f^0, v)_{L^2(U)} + \sum_{i=1}^n (f^i, v_{x_i})_{L^2(U)} = (u, v)_{L^2(U)} + \sum_{i=1}^n (u_{x_i}, v_{x_i})_{L^2(U)} = (u, v)_{H^1(U)} = \langle f, v \rangle \quad \forall v \in H_0^1(U).$$

By the Riesz-Representation Theorem, we also have $\|f\|_{H^{-1}(U)} = \|u\|_{H^1(U)}$. If (ii) holds, then we write

$$f = f^0 - \sum_{i=1}^n f_{x_i}^i$$

to condense the notation.

(2). We shall prove (ii). For $g^0, \dots, g^n \in L^2(U)$, consider the functional g defined by

$$g : H_0^1(U) \rightarrow \mathbb{R} : (g^0, v)_{L^2(U)} + \sum_{i=1}^n (g^i, v_{x_i})_{L^2(U)}.$$

This functional is linear since $g(\alpha v + \beta w) = \alpha g(v) + \beta g(w)$. Furthermore, g is bounded on $H_0^1(U)$:

$$|g(v)| = \left| (g^0, v)_{L^2(U)} + \sum_{i=1}^n (g^i, v_{x_i})_{L^2(U)} \right| \leq \|g^0\|_{L^2(U)} \|v\|_{L^2(U)} + \sum_{i=1}^n \|g^i\|_{L^2(U)} \|v_{x_i}\|_{L^2(U)}.$$

From the Discrete Hölder inequality, we have

$$\|g(v)\| \leq \left(\sum_{i=0}^n \|g^i\|_{L^2(U)}^2 \right)^{1/2} \left(\|v\|_{L^2(U)}^2 + \sum_{i=1}^n \|v_{x_i}\|_{L^2(U)}^2 \right)^{1/2} = C \|v\|_{H^1(U)}$$

and so $g \in H^{-1}(U)$. Then, the norm of g in $H^{-1}(U)$ is

$$\|g\|_{H^{-1}(U)} = \sup_{0 \neq v \in H_0^1(U)} \frac{|\langle g, v \rangle|}{\|v\|_{H^1(U)}} \leq \left| \sum_{i=0}^n \|g^i\|_{L^2(U)}^2 \right|^{1/2}.$$

We must show that that if this norm is true, that the norm is equal to the infimum of all possible norms of g_i in $L^2(U)$. By Step (1), we saw that we could choose a representation of g to be Riesz representation:

$$(\bar{g}^0, \dots, \bar{g}^n) = (u, u_{x_1}, \dots, u_{x_n})$$

then $\|g\|_{H^{-1}(U)} = \|u\|_{H^1(U)}$. Therefore,

$$\|g\|_{H^{-1}(U)}^2 = \|u\|_{H^1(U)}^2 = \|u\|_{L^2(U)}^2 + \sum_{i=1}^n \|u_{x_i}\|_{L^2(U)}^2 = \sum_{i=0}^n \|\bar{g}^i\|_{L^2(U)}^2$$

by definition of the H^1 -norm. We combine this result with the bound in $\|h\|_{H^{-1}(U)}$ so that

$$\sum_{i=0}^n \|\bar{g}^i\|_{L^2(U)}^2 \leq \sum_{i=0}^n \|g^i\|_{L^2(U)}^2$$

which proves (ii).

(3). Given $v \in L^2(U)$, $f \in (L^2(U))^*$ can be defined as

$$\langle f, u \rangle_{(L^2(U))^*, L^2(U)} = (v, u)_{L^2(U)}.$$

The converse is also true: by the Riesz-Representation Theorem (Theorem 1.9.4) for every $f \in (L^2(U))^*$, $\exists! v \in L^2(U)$ s.t.

$$\langle f, u \rangle_{(L^2(U))^*, L^2(U)} = (v, u)_{L^2(U)} \quad \forall u \in L^2(U)$$

and $\|f\|_{(L^2(U))^*} = \|v\|_{L^2(U)}$. This defines a linear isomorphism $R : L^2(U) \rightarrow (L^2(U))^*$ by

$$\langle f, u \rangle_{(L^2(U))^*, L^2(U)} = \langle Rv, u \rangle_{(L^2(U))^*, L^2(U)} = (v, u)_{L^2(U)} \quad \forall u \in L^2(U).$$

We can identify $L^2(U)$ with $(L^2(U))^*$ such that

$$L^2(U) \cong RL^2(U) = (L^2(U))^*.$$

Notice that the statement $\langle f, u \rangle_{(L^2(U))^*, L^2(U)}$ holds for all $u \in L^2(U)$. Thus, it must also hold for all $u \in H_0^1(U) \subset L^2(U)$. So, define $T : (L^2(U))^* \rightarrow H^{-1}(U)$ to be the restriction of $f \in (L^2(U))^*$ to $H_0^1(U)$ by

$$\langle Tf, u \rangle_{H^{-1}(U), H_0^1(U)} = \langle f, u \rangle_{(L^2(U))^*, L^2(U)}.$$

Then, note that for any $v \in L^2(U)$ and $u \in H_0^1(U)$ we have:

$$\langle (T \circ R)v, u \rangle_{H^{-1}(U), H_0^1(U)} = \langle Rv, u \rangle_{(L^2(U))^*, L^2(U)} = (v, u)_{L^2(U)}.$$

The duality pairing on $H_0^1(U)$, when restricted to $L^2(U)$ coincides with the $L^2(U)$ inner product. Indeed,

$$(v, u)_{L^2(U)} = (v, u)_{L^2(U)} \quad \forall v \in L^2(U), u \in H_0^1(U)$$

where v is implied to be acted on by both T and R . To show that $(L^2(U))^* \subset H^{-1}(U)$, note that $(T \circ R)v \in H^{-1}(U)$. So by (i), we can write

$$\langle (T \circ R)v, u \rangle = (f^0, u)_{L^2(U)} + \sum_{i=1}^n (f^i, u_{x_i})_{L^2(U)}$$

but also, we have that

$$\langle (T \circ R)v, u \rangle = (v, u)_{L^2(U)}.$$

So, for these relations to be equal, we must have that

$$(f^0, u)_{L^2(U)} = (v, u)_{L^2(U)}$$

since the sum (H^1 inner product part) must be zero (since the derivatives u_{x_i} do not play a role and thus do not have any weak derivatives). That is, we have $(L^2(U))^* \subset H^{-1}(U)$ follows since the L^2 -functional $(T \circ R)v$ is represented by $\{f^0, 0, \dots, 0\}$, but there exist many other representations in H^{-1} where $(f^1, \dots, f^n)$ are nonzero. So, the inclusion is strict, and we have that $(L^2(U))^* \subset H^{-1}(U)$. Indeed, we have

$$H_0^1(U) \subset L^2(U) \cong RL^2(U) = (L^2(U))^* \simeq T(L^2(U))^* \subset H^{-1}(U).$$

□

3 Second-Order Elliptic Equations

3.1 Elliptic Equations

Model Problem. We shall assume that $U \subset \mathbb{R}^n$ is open and bounded, and assume that $f : U \rightarrow \mathbb{R}$ is given. We wish to find $u : \bar{U} \rightarrow \mathbb{R}$ such that the BVP is of the form

$$\begin{cases} Lu = f, & \text{in } U \\ u = 0, & \text{on } \partial U. \end{cases} \quad (14)$$

We will study the most general case where L is the second-order differential operator

$$Lu = - \sum_{i,j=1}^n (a^{ij}(x)u_{x_i})_{x_j} + \sum_{i=1}^n b^i(x)u_{x_i} + c(x)u.$$

Some assumptions on a^{ij} we shall make is that:

- $a^{ij} = a^{ji}$ for $i, j = 1, \dots, n$
- a^{ij} are such that L is **uniformly elliptic**:

$$\exists \theta > 0 \text{ s.t. } \sum_{i,j=1}^n a^{ij}(x)\xi_i\xi_j \geq \theta|\xi|^2$$

for a.e $x \in U$ and all $\xi \in \mathbb{R}^n$.

Example 3.1. Suppose $a^{ij} = \delta_{ij}$, $b_i = 0$, and $c = 0$. Then, $Lu = -\Delta u$ (Laplace/Poisson equation depending on f).

Suppose $a^{ij} = \delta_{ij}\mu$ for $\mu > 0$ is a constant, $b^i \neq 0$, and $c \neq 0$. Then, $Lu = -\mu\Delta u + bDu + cu$ (advection-diffusion-reaction equation).

3.1.1 Weak Solutions

Recall we discussed the example where $u : U \rightarrow \mathbb{R}$ such that

$$\begin{cases} -u'' = f, & \text{on } U = (-1, 1) \\ u(-1) = u(1) = 0, & \text{on } \partial U. \end{cases}$$

When $f \notin C(\bar{U})$, there is no classical solution $u \in C^2(\bar{U})$. To give meaning to a solution of such a problem, we introduce the **weak formulation**.

Motivating Problem. Let us assume that $a^{ij}, b^i, c \in L^\infty(U)$ and $f \in L^2(U)$. Assume also that u , the solution to our BVP, is smooth. Multiplying $Lu = f$ by a test function $v \in C_c^\infty(U)$ and integrating over U , we find

$$\int_U \left(- \sum_{i,j=1}^n (a^{ij}u_{x_i})_{x_j} + \sum_{i=1}^n b^i u_{x_i} + cu \right) v \, dx = \int_U f v \, dx.$$

IBP on the first term shows that

$$\int_U \left(\sum_{i,j=1}^n a^{ij} u_{x_i} v_{x_j} + \sum_{i=1}^n b^i u_{x_i} v + cuv \right) dx - \int_{\partial U} \sum_{i,j=1}^n a^{ij} u_{x_i} n_j v ds = \int_U f v dx.$$

The integral on the boundary is zero since v has compact support. By density, the identity also holds for $v \in H_0^1(U)$. The integral form "makes sense" (integrals are finite) for any $u \in H_0^1(U)$.

Definition 3.1.1. The **bilinear form** $B[\cdot, \cdot]$ associated with L is given by

$$B[u, v] = \int_U \left(\sum_{i,j=1}^n a^{ij} u_{x_i} v_{x_j} + \sum_{i=1}^n b^i u_{x_i} v + cuv \right) dx.$$

We call $B[\cdot, \cdot]$ bilinear since it is linear in both of its arguments.

Definition 3.1.2. $u \in H_0^1(U)$ is called a **weak solution** to the BVP if

$$f \in L^2(U) : \begin{cases} Lu = f, & \text{in } U \\ u = 0, & \text{on } \partial U \end{cases} \quad \text{if } f \in H^{-1}(U) : \begin{cases} Lu = f^0 - \sum_{i=1}^n f_{x_i}^i \\ u = 0. \end{cases}$$

If $u \in H_0^1(U)$ satisfies the weak formulation, then

$$\begin{aligned} B[u, v] &= (f, v) \quad \forall v \in H_0^1(U) \text{ (if } f \in L^2(U)) \\ B[u, v] &= \langle f, v \rangle \quad \forall v \in H_0^1(U) \text{ (if } f \in H^{-1}(U)) \end{aligned}$$

where

$$(\cdot, \cdot) = (\cdot, \cdot)_{L^2(U)}, \quad \langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_{H^{-1}(U), H_0^1(U)}, \quad \langle f, v \rangle = (f^0, v) + \sum_{i=1}^n (f_{x_i}^i, v_{x_i}).$$

3.2 Weak Solutions

We shall assume that H is a real Hilbert space with a defined inner product $(\cdot, \cdot)$, norm $\|\cdot\| = (\cdot, \cdot)^{1/2}$, and duality pairing $\langle \cdot, \cdot \rangle$.

Theorem 3.2.1 (Lax-Milgram Theorem). Assume $B : H \times H \rightarrow \mathbb{R}$ is a bilinear form such that $\exists \alpha, \beta > 0$, with

- (i) (Boundedness) $|B[u, v]| \leq \alpha \|u\| \|v\| \quad \forall u, v \in H$.
- (ii) (Coercitivity) $B[u, u] \geq \beta \|u\|^2 \quad \forall u \in H$.

Then, for $f \in H^*$, $\exists! u \in H$ such that $B[u, v] = \langle f, v \rangle$ for all $v \in H$.

Proof. (1). Fix some $u \in H$. Then, $L[uv] = B[u, v]$ is a bounded linear functional on H such that $|L[uv]| = |B[u, v]| \leq C \|v\|$, and $C = \alpha \|u\|$. By the Riesz Representation Theorem, we know that $\exists! w \in H$ such that $B[u, v] = (w, v) \quad \forall v \in H$. We define the functional $A : H \rightarrow H$ so that $B[u, v] = (Au, v)$ is valid.

(2). We claim that $A \in H^*$. To show this is true, we observe that the following is true:

(i) (Linearity) For $\lambda_1, \lambda_2 \in \mathbb{R}$ and $u_1, u_2 \in H$, $\exists v \in H$ such that

$$\begin{aligned} (A(\lambda_1 u_1 + \lambda_2 u_2), v) &= B[\lambda_1 u_1 + \lambda_2 u_2, v] \\ &= \lambda_1 B[u_1, v] + \lambda_2 B[u_2, v] \\ &= \lambda_1 (Au_1, v) + \lambda_2 (Au_2, v) \\ &= (\lambda_1 Au_1 + \lambda_2 Au_2, v) \end{aligned}$$

and so A is linear.

(ii) (Boundedness) We see immediately that the squared norm,

$$\|Au\|^2 = (Au, Au) = B[u, Au] \leq \alpha \|u\| \|Au\| \implies \|Au\| \leq \alpha \|u\|$$

and so A is bounded.

(3). We now make the following assertions. (i) A is injective, and (ii) $R(A)$ is closed in H . To prove this, we make use of the following inequality

$$\beta \|u\|^2 \leq B[u, u] = (Au, u) \leq \|Au\| \|u\| \implies \beta \|u\| \leq \|Au\| \quad \forall (u_1, u_2) \in H.$$

We shall now prove (i). For the functional A , observe that

$$\begin{aligned} Au_1 = Au_2 &\implies A(u_1 - u_2) = 0 \\ &\implies \beta \|u_1 - u_2\| \leq \|A(u_1 - u_2)\| \\ &\implies u_1 - u_2 = 0 \\ &\implies u_1 = u_2. \end{aligned}$$

We will now prove (ii). Let $\{u_i\}_{i=1}^\infty \subset H$ with $Au_j \rightarrow y \in H$. For Au_j to be convergent, we must have that Au_j is Cauchy. Observe that

$$\beta \|u_j - u_k\| \leq \|A(u_j - u_k)\| = \|Au_j - u_k\| \rightarrow 0$$

and so $\{u_j\}_{j=1}^\infty$ is Cauchy. Since H is a Hilbert space, we thus have that $u_j \rightarrow u \in H$. Indeed, we find

$$\|Au - y\| = \lim_{j \rightarrow \infty} \|Au - Au_j\| \leq \alpha \lim_{j \rightarrow \infty} \|u - u_j\| = 0$$

and so $R(A)$ is closed in H .

(4). We note that for a closed subspace Y in the Hilbert space H , we have⁷

$$H = Y \oplus Y^\perp = \{z \in H : (z, y) = 0 \quad \forall y \in Y\}.$$

We now claim that $R(A) = H$. Suppose, for a contradiction, that $R(A) \neq H$. Then, we necessarily have that

$$H = R(A) \oplus R(A)^\perp \implies \exists w \in R(A)^\perp \text{ s.t. } 0 \neq \beta \|w\|^2 \leq B[w, w] = (Aw, w) = 0.$$

By contradiction, we find a non-zero element $w \in H$ is indeed zero if $R(A)$ is closed. Thus, A is a bijection and by the Bounded Inverse Theorem, there exists a bounded linear operator A^{-1} .

⁷Note that for W, Z that are subspaces, the space $V = W \oplus Z$ is the direct sum of the subspaces.

(5). Since $f \in H^*$, we have by the Riesz-Representation Theorem that $\exists! w \in H$ such that $\langle f, v \rangle = (w, v) \forall v \in H$. Let $u = A^{-1}w$, then we have that

$$B[u, v] = (Au, v) = (w, v) = \langle f, v \rangle.$$

(6). We finally prove uniqueness of the element $u \in H$. Suppose $\exists(u, \hat{u})$ such that

$$B[u, v] = \langle f, v \rangle = B[\hat{u}, v] \quad \forall v \in H.$$

Indeed, we have $B[u - \hat{u}, v] = 0$ for all $v \in H$. Let $v = u - \hat{u}$, and so we find

$$\beta \|u - \hat{u}\|^2 \leq B[u - \hat{u}, u - \hat{u}] = 0 \implies u = \hat{u} \text{ a.e.}$$

proving the theorem. $\square$

Theorem 3.2.2 (Energy Estimates). There exist constants $\alpha, \beta > 0$ and some $\gamma \geq 0$ such that the following hold:

- (i) $|B[u, v]| \leq \alpha \|u\|_{H_0^1(U)} \|v\|_{H_0^1(U)}$ for all $u, v \in H_0^1(U)$.
- (ii) $\beta \|u\|_{H_0^1(U)}^2 \leq B[u, u] + \gamma \|u\|_{L^2(U)}^2$ for all $u \in H_0^1(U)$.

Proof. (1). Computing explicitly (i), we find

$$\begin{aligned} |B[u, v]| &\leq \sum_{i,j=1}^n \|a^{ij}\|_{L^\infty(U)} \int_U |Du| |Du| \, dx + \sum_{i=1}^n \|b^i\|_{L^\infty(U)} \int_U Du |v| \, dx \\ &\quad + \|c\|_{L^\infty(U)} \int_U |u| |v| \, dx \\ &\leq \sum_{i,j=1}^n \|a^{ij}\|_{L^\infty(U)} \|Du\|_{L^2(U)} + \sum_{i=1}^n \|b^i\|_{L^\infty(U)} \|Du\|_{L^2(U)} \|v\|_{L^2(U)} \\ &\quad + \|c\|_{L^\infty(U)} \|u\|_{L^2(U)} \|v\|_{L^2(U)} \\ &\leq \left(\sum_{i,j=1}^\infty \|a^{ij}\|_{L^\infty(U)} + \sum_{i=1}^n \|b^i\|_{L^\infty(U)} + \|c\|_{L^\infty(U)} \right) \|v\|_{H_0^1(U)} \|u\|_{H_0^1(U)} \end{aligned}$$

where we have invoked Cauchy-Schwarz in the third inequality result. The first term is simply α by Lax-Milgram, and therefore we have

$$|B[u, v]| \leq \alpha \|u\|_{H_0^1(U)} \|v\|_{H_0^1(U)}.$$

(2). For L being elliptic, we know $\exists \theta > 0$ such that

$$\sum_{i,j=1}^n a^{ij} \xi_i \xi_j \geq \theta |\xi|^2 \quad \forall \xi \in \mathbb{R}^n.$$

Let $\xi = Du$, then we have

$$\begin{aligned}
\theta |Du|^2 &\leq \sum_{i,j=1}^n a^{ij} u_{x_i} u_{x_j} \\
\theta \int_U |Du|^2 dx &\leq \int_U \sum_{i,j=1}^n a^{ij} u_{x_i} u_{x_j} dx \\
&= B[u, u] - \int_U \left(\sum_{i=1}^n b^i u_{x_i} u + cu^2 \right) dx \\
&\leq B[u, u] + \sum_{i=1}^n \|b^i\|_{L^\infty(U)} \int_U |Du| |u| dx + \|c\|_{L^\infty(U)} \int_U u^2 dx \\
&\leq B[u, u] + \epsilon \sum_{i=1}^n \|b^i\|_{L^\infty(U)} \int_U |Du|^2 dx + \left(\frac{1}{4\epsilon} \sum_{i=1}^n \|b^i\|_{L^\infty(U)} + \|c\|_{L^\infty(U)} \right) \int_U u^2 dx
\end{aligned}$$

by Cauchy-Schwarz on the last inequality, and also by Young's Inequality with ϵ . The last constant we equate to γ , and so we have

$$\left(\theta - \epsilon \sum_{i=1}^n \|b^i\|_{L^\infty(U)} \right) \|Du\|_{L^2(U)}^2 \leq B[u, u] + \gamma \|u\|_{L^2(U)}^2.$$

Choosing $\epsilon > 0$ so small that

$$\epsilon \sum_{i=1}^n \|b^i\|_{L^\infty(U)} < \frac{\theta}{2}$$

and noting that

$$\|Du\|_{L^2(U)} \approx \|u\|_{H^1(U)} \quad \forall u \in H_0^1(U)$$

we can see that

$$C \frac{\theta}{2} \|u\|_{H^1(U)}^2 = \beta \|u\|_{H^1(U)}^2 \leq B[u, u] + \gamma \|u\|_{L^2(U)}^2$$

thus proving (ii). $\square$

Notice that for $\gamma = 0$, we arrive at the Lax-Milgram result. However, for $\gamma \neq 0$, we arrive at the modified BVP which we may model as

$$\begin{cases} Lu + \mu u = f, & \text{in } U \\ u = 0, & \text{on } \partial U. \end{cases}$$

Theorem 3.2.3 (First Existence Theorem for Weak Solutions). There exists a number $\gamma \geq 0$ such that for each $\mu \geq \gamma$ and each function $f \in L^2(U)$, there exists a unique weak solution $u \in H_0^1(U)$ of the modified BVP (mBVP)

$$\begin{cases} Lu + \mu u = f, & \text{in } U \\ u = 0, & \text{on } \partial U. \end{cases}$$

Proof. Let $\mu \geq \gamma$ and define the bilinear form

$$B_\mu[u, v] = B[u, v] + \mu(u, v)_{L^2(U)}$$

which is correlated to $L_\mu u = Lu + \mu u$. By the Energy Estimates Theorem, we compute directly (i),

$$|B_\mu[u, v]| \leq |B[u, v]| + \mu \|u\|_{L^2(U)} \|v\|_{L^2(U)} \leq (\alpha + \mu) \|u\|_{H^1(U)} \|v\|_{H^1(U)}$$

and (ii),

$$B_\mu[u, u] = B[u, u] + \mu \|u\|_{L^2(U)}^2 \geq \beta \|u\|_{H^1(U)}^2 + (\mu - \gamma) \|u\|_{L^2(U)}^2 \geq \beta \|u\|_{H^1(U)}^2$$

since $(\mu - \gamma) \|u\|_{L^2(U)}^2 \geq 0$ if $\mu - \gamma \geq 0$. Since both conditions are fulfilled, $B_\mu[\cdot, \cdot]$ satisfies the hypothesis if it is Lax-Milgram. $\square$

Where does mBVP fit in? We know $\exists!$ weak solution to mBVP for μ large enough, such that $L_\mu : H_0^1(U) \rightarrow L^2(U)$ is bijective. Then, L_μ^{-1} exists. We may consider for μ sufficiently large, that the following statements are equivalent

- (i) $u \in H_0^1(U)$ uniquely solves $B[u, v] = (f, v) \forall v \in H_0^1(U)$.
- (ii) $u \in H_0^1(U)$ uniquely solves $B[u, v] + \mu(u, v) = (f, v) + \mu(u, v) \forall v \in H_0^1(U)$.
- (iii) $u \in H_0^1(U)$ uniquely solves $B_\mu[u, v] = (f + \mu u, v) \forall v \in H_0^1(U)$.
- (iv) $u = L_\mu^{-1}(f + \mu u) = L_\mu^{-1}f + \mu L_\mu^{-1}u$.

Then, we write $h = L_\mu^{-1}f \in H_0^1(U)$ and $K = \mu L_\mu^{-1}$, so that the following statement holds: $u \in H_0^1(U)$ is the unique solution to $B[u, v] = (f, v) \forall v \in L^2(U)$ if and only if u uniquely solves $u - Ku = h$.

We will show that K is a bounded compact linear operator on $L^2(U)$.

3.2.1 Compact Operators

Let X be a Banach space.

Definition 3.2.1. A sequence $\{u_k\}_{k=1}^\infty \subset X$ **converges weakly** to $u \in X$ if $\langle u^*, u_k \rangle \rightarrow \langle u^*, u \rangle \forall u^* \in X^*$. Specifically, we write $u_k \rightharpoonup u$ to denote weak convergence, while $u_k \rightarrow u$ denotes strong convergence.

Remark. We make use of the following facts that are left unproven.

- (i) Strong convergence ($u_k \rightarrow u$) implies weak convergence ($u_k \rightharpoonup u$).
- (ii) The weak limit u of $\{u_k\}_{k=1}^\infty$ is unique.
- (iii) Any weakly convergent sequence $\{u_k\}_{k=1}^\infty$ is bounded.
- (iv) Every subsequence of $\{u_k\}_{k=1}^\infty$ converges to the same weak limit.

Theorem 3.2.4 (Weak Compactness). Let X be a reflexive Banach space and suppose $\{u_k\}_{k=1}^\infty \subset X$ is bounded. Then, there exists a subsequence $\{u_{k_j}\}_{j=1}^\infty \subset \{u_k\}_{k=1}^\infty$ and $u \in X$ s.t. $u_{k_j} \rightharpoonup u$ (bounded sequences in reflexive Banach spaces are weakly precompact).

Remark. The Weak Compactness theorem is useful (and important) since all Hilbert spaces are reflexive, and so all bounded sequences contain a weakly convergent subsequence.

Definition 3.2.2. A bounded linear operator $K : X \rightarrow Y$ (where X and Y are Banach spaces) is called **compact** provided for each bounded sequence $\{u_k\}_{k=1}^\infty \subset X$, the sequence $\{Ku_k\}_{k=1}^\infty$ is precompact in Y (i.e., $\exists$ a subsequence $\{u_{k_j}\}_{j=1}^\infty$ s.t. $\{Ku_{k_j}\}_{j=1}^\infty$ converges in Y).

Lemma 3.2.5. Let H be a Hilbert space with inner product $(\cdot, \cdot)$ and let $K : H \rightarrow H$ be a bounded linear operator. If $K : H \rightarrow H$ is compact and $u_k \rightharpoonup u$, then $Ku_k \rightarrow Ku$ (a compact operator maps weakly convergent sequences to strongly convergent sequences).

Lemma 3.2.6. The **adjoint** $K^* : H \rightarrow H$ is defined by $(Ku, v) = (u, K^*v) \forall u, v \in H$.

Lemma 3.2.7. If $K^* = K$, then K is **symmetric**.

Lemma 3.2.8. If K is bounded, then so is K^* .

Theorem 3.2.9 (Compactness of Adjoints). If $K : H \rightarrow H$ is compact, then so is $K^* : H \rightarrow H$.

Theorem 3.2.10 (Fredholm Alternative). Let $K : H \rightarrow H$ be a compact linear operator and H be a Hilbert space. Denote by $N(K)$ the kernel of K and by $R(K)$ the range of K . Then, we have the following five properties:

- (i) $N(I - K)$ is finite dimensional.
- (ii) $R(I - K)$ is closed.
- (iii) $R(I - K) = N(I - K^*)^\perp$.
- (iv) $N(I - K) = \{0\} \iff R(I - K) = H$.
- (v) $\dim N(I - K) = \dim N(I - K^*)$.

Let us consider (iv). In particular, we will consider the solvability of the problem: "given $f \in H$, find u such that $(I - K)u = f$." If $R(I - K) = H$, we implied existence of u . Indeed, if the range can reach *any* $f \in H$, then we must have some operator $I - K$ which is surjective.

If we consider now $N(I - K) = \{0\}$, we have implied uniqueness of u . Indeed, if u_1, u_2 both solve $(I - K)u = f$, then we have $(I - K)u_1 = f$ and $(I - K)u_2 = f$ so that $(I - K)(u_1 - u_2) = 0 \implies u_1 = u_2$ since $N(I - K) = \{0\}$.

Thus, (α) if $N(I - K) = \{0\}$ and $R(I - K) = H$, then $\exists! u \in H$ s.t. $(I - K)u = f \forall f \in H$. However, (iv) also states that (β) if $N(I - K) \neq \{0\}$, then $(I - K)u = 0$ has one or more solutions $u \neq 0$.

The **Fredholm Alternative** states that exactly one of the statements (α) or (β) must hold. So, if (α) holds, then (β) does not, and vice versa.

Remark. If (β) holds, then by (iii) we can still solve the equation $(I - K)u = f$ if and only if $f \in N(I - K^*)^\perp$ (this result only implies existence and not uniqueness).

3.2.2 Fredholm Alternative for Weak Solutions

In what follows, we need the operator L^* defined by $(Lu, v) = (u, L^*v)$. L , in general, is not bounded so L^* is not called the adjoint of L . We instead call it the **formal adjoint**. To find L^* , we may assume as much smoothness as necessary. Then, $\forall v \in C_c^\infty(U)$, we have

$$\begin{aligned} (Lu, v)_{L^2(U)} &= \int_U \left[- \sum_{i,j=1}^n (a^{ij} u_{x_i})_{x_j} + \sum_{i=1}^n b_i u_{x_i} + cu \right] v \, dx \\ &= \text{IBP twice} \int_U u \left[- \sum_{i,j=1}^n (a^{ij} v_{x_j})_{x_i} - \sum_{i=1}^n b^i v_{x_i} + \left(c - \sum_{i=1}^n b_{x_i}^i \right) v \right] dx \\ &= (u, L^*v)_{L^2(U)} \end{aligned}$$

where

$$L^* = - \sum_{i,j=1}^n (a^{ij} v_{x_j})_{x_i} - \sum_{i=1}^n b^i v_{x_i} + \left(c - \sum_{i=1}^n b_{x_i}^i \right) v.$$

Definition 3.2.3. The **adjoint bilinear form** $B^*[u, v]$ is defined by $B^*[v, u] := B[u, v] \, \forall u, v \in H_0^1(U)$.

Definition 3.2.4. For $f \in H^{-1}(U)$, we say that $v \in H_0^1(U)$ is a weak solution of the **adjoint problem**

$$\begin{cases} L^*v = f, & \text{in } U \\ v = 0, & \text{on } \partial U \end{cases}$$

provided that $B^*[v, u] = \langle f, u \rangle \, \forall u \in H_0^1(U)$.

Theorem 3.2.11 (The Second Existence Theorem for Weak Solutions).

(i) Precisely **one** of the following two statements must be true:

(a) for each $f \in L^2(U)$, $\exists! u \in H_0^1(U)$ of

$$\begin{cases} Lu = f, & \text{in } U \\ u = 0, & \text{on } \partial U. \end{cases}$$

(b) There exists a weak solution $0 \neq u \in H_0^1(U)$ of the homogeneous problem

$$\begin{cases} Lu = 0, & \text{in } U \\ u = 0, & \text{on } \partial U. \end{cases}$$

(ii) Assume that (b) holds. Let $N \subset H_0^1(U)$ be the solution space of weak solutions to the homogeneous problem. Let $N^* \subset H_0^1(U)$ be the solution space of weak solutions to the homogeneous adjoint problem. In other words,

$$\begin{cases} L^*v = 0, & \text{on } U \\ v = 0, & \text{on } \partial U. \end{cases}$$

Then, $\dim(N) = \dim(N^*) < \infty$.

(iii) The original BVP has a weak solution if and only if $(f, v) = 0 \forall v \in N^*$. In other words,

$$f \in (N^*)^\perp = \{g \in L^2(U) : (g, v) = 0 \forall v \in N^*\}.$$

The dichotomy (a), (b) is the Fredholm alternative discussed before.

Proof. (1). Let us choose $\mu = \gamma$ as in the First Existence Theorem. We define

$$B_\gamma[u, v] := B[u, v] + \gamma(u, v)$$

correlated to $L_\gamma u = Lu + \gamma u$. By the First Existence Theorem, $\exists! u \in H_0^1(U)$ for all $g \in L^2(U)$ such that $b_\gamma[u, v] = (g, v)$ for all $v \in H_0^1(U)$. Indeed, it follows that the operator $L_\gamma : H_0^1(U) \rightarrow L^2(U)$ is bijective, so L_γ^{-1} exists. That is,

$$u = L_\gamma^{-1}g \implies B_\gamma[L_\gamma^{-1}g, v] = (g, v) \quad \forall v \in H_0^1(U).$$

Thus, $B_\gamma[u, v]$ satisfies the hypothesis of the Lax-Milgram Theorem.

(2). Observe now that $u \in H_0^1(U)$ is the unique solution to (a) if and only if u is the unique solution to the problem $u - Ku = 0$.

(3). We claim that $K : L^2(U) \rightarrow L^2(U)$ is a bounded linear operator. To prove this, we note that for all $g \in L^2(U)$, $\exists! u \in H_0^1(U)$ such that

$$B_\gamma[u, v] = (g, v) \quad \forall v \in H_0^1(U).$$

Then, we have that

$$\beta \|u\|_{H^1(U)}^2 \leq B_\gamma[u, u] = (g, u) \leq \|g\|_{L^2(U)} \|u\|_{L^2(U)} \leq \|g\|_{L^2(U)} \|u\|_{H^1(U)}.$$

by both Coercitivity and Cauchy-Schwarz. It follows that

$$\|u\|_{H^1(U)} = \|L_\gamma^{-1}g\|_{H^1(U)} = \|Kg\|_{H^1(U)} \leq \frac{1}{\beta} \|g\|_{L^2(U)} = \frac{1}{\beta} \|g\|_{L^2(U)} = \frac{\gamma}{\beta} \|g\|_{L^2(U)}$$

where we have asserted that $\gamma L_\gamma^{-1}g = Kg$. Thus, we see that $K : L^2(U) \rightarrow H_0^1(U)$ is bounded. Since K is bounded, we observe that it maps $\{u_k\}_{k=1}^\infty \subset L^2(U)$ to $\{Ku_k\}_{k=1}^\infty \subset H_0^1(U)$. We know also that $H_0^1(U) \subset \subset L^2(U)$, so the bounded sequence $\{Ku_k\}_{k=1}^\infty \subset H_0^1(U)$ is precompact in $L^2(U)$. Indeed, K is compact.

(4). By the Fredholm Alternative, we have either (â) $\forall h \in L^2(U)$, $\exists! u \in L^2(U)$ such that $u - Ku = h$, or; (b) $u - Ku = 0$ has non-zero solutions $u \in L^2(U)$. In our setting, we have $h = L_\gamma^{-1}g \in H_0^1(U)$ and $Ku = \gamma L_\gamma^{-1}g \in H_0^1(U)$ to give $u - Ku = h$. From (2), we see that (â) $\iff$ (a) and (b) $\iff$ (b), which proves (i).

We now consider the two cases of γ . In particular, if $\gamma = 0$, then from (â), for $K = \gamma L_\gamma^{-1}$, we have

$$u - Ku = u = 0 \quad (\gamma L_\gamma^{-1} = K).$$

In the case $\gamma \neq 0$, we have from (b) that the solution space $N \subset H_0^1(U)$ for (b) is also the solution space for (b). Indeed, it follows that

$$u - Ku = 0 \implies (I - K)u = 0.$$

By the Fredholm Alternative (Theorem variant, Theorem 3.2.10), $N(I-K)$ is finite dimensional and so $\dim(N) < \infty$. Invoking property (v) of the Fredholm Alternative, we find $\dim(N) = \dim(N^*)$ and so we conclude that N^* is also the solution space of the alternate equation

$$u - K^*u = 0.$$

We claim that, in particular, that if L^* is the formal adjoint of L and $L_\gamma^* = L^* + \gamma$, then $K^* = \gamma(L_\gamma^*)^{-1}$. From (1), we take the adjoint of the bilinear form,

$$B_\gamma^*[u^*, v^*] = B_\gamma[v^*, u^*]$$

and taking $v^* = u$, we have

$$(g^*, u) = B_\gamma^*[u^*, u] = B_\gamma[u, u^*] = (g, u^*).$$

For $u = L_\gamma^{-1}g$ and $u^* = (L_\gamma^*)^{-1}g^*$, we find that

$$(g^*, L_\gamma^{-1}g) = (g, (L_\gamma^*)^{-1}g^*) = ((L_\gamma^*)^{-1}g^*, g) \quad \forall (g, g^*) \in L^2(U).$$

Indeed, we have

$$(g^*, Kg) = (\gamma(L_\gamma^*)^{-1}g^*, g) \implies K^* = \gamma(L_\gamma^*)^{-1}$$

which proves our claim. We take similar steps as in (2) to conclude that for a solution v to $v - K^*v = 0$, we necessarily have v as a weak solution to the problem

$$\begin{cases} L^*v = 0, & \text{in } U \\ v = 0, & \text{on } \partial U. \end{cases}$$

So, for N^* to be a solution space to $v - K^*v = 0$, it must also be a solution space to the above problem, proving (ii).

(5). Assume now that (a) holds. Then, by the Fredholm Alternative, we have

$$N(I - K) = \{0\} \implies R(I - K) = L^2(U) \implies N(I - K^*)^\perp = L^2(U)$$

so (iii) agrees with (i). Assume now that (b) holds. Then from (2), if (a) has a weak solution then $u - Ku = h$ has a solution. Invoking the Fredholm Alternative, we must necessarily have

$$h \in (N^*)^\perp = N(I - K^*)^\perp = \{f \in L^2(U) : (f, v) = 0 \forall v \in N(I - K^*)\}.$$

Thus if $u - Ku = h$ has a solution, then

$$(h, v) = 0 \forall v \in N(I - K^*) \iff (f, v) = 0 \forall v \in N(I - K^*) \implies f \in (N^*)^\perp.$$

We note that if $h = L_\gamma^{-1}f$, then we have for all $v \in N(I - K^*)$ that

$$(h, v) = (L_\gamma^{-1}f, v) = \frac{1}{\gamma}(Kf, v) = \frac{1}{\gamma}(f, K^*v) = \frac{1}{\gamma}(f, v).$$

Thus we have proven (iii). □

Definition 3.2.5. Let X be a Banach space and $A : X \rightarrow X$ be a bounded linear operator. Then, we make the following definitions:

(i) The **resolvent set** of A is defined as

$$\rho(A) = \{\eta \in \mathbb{R} : (A - \eta I) : X \rightarrow X \text{ is bijective}\}.$$

Let $\eta \in \rho(A)$. By the Bounded inverse theorem, $(A - \eta I)^{-1} : X \rightarrow X$ is a bounded linear operator.

(ii) The **spectrum** of A is

$$\sigma(A) = \mathbb{R} - \rho(A).$$

(iii) $\eta \in \sigma(A)$ is an **eigenvalue** of A provided that

$$N(A - \eta I) \neq \{0\}.$$

The **point spectrum** of A , denoted by $\sigma_p(A)$, is the collection of eigenvalues of A .

(iv) If $\eta \in \sigma_p(A)$ and if $(A - \eta I)w = 0$ for some $w \neq 0$, then w is the associated **eigenvector**.

Theorem 3.2.12 (Spectrum of Compact Operators). Assume H is a Hilbert space, $\dim(H) = \infty$, and $K : H \rightarrow H$ is compact. Then,

(i) $0 \in \sigma(K)$,

(ii) $\sigma(K) - \{0\} = \sigma_p(K) - \{0\}$,

(iii) Either:

$$\begin{cases} \sigma(K) - \{0\} \text{ is finite, or} \\ \sigma(K) - \{0\} \text{ is a countable sequence tending to 0.} \end{cases}$$

Theorem 3.2.13 (The Third Existence Theorem for Weak Solutions).

(i) There exists an, at most, countable set $\Sigma \subset \mathbb{R}$ such that the BVP

$$\begin{cases} Lu = \lambda u + f, & \text{in } U \\ u = 0, & \text{on } \partial U \end{cases}$$

has a unique weak solution for each $f \in L^2(U)$ if and only if $\lambda \notin \Sigma$.

(ii) If Σ from (i) is infinite, then $\Sigma = \{\lambda_k\}_{k=1}^{\infty}$ is a non-decreasing sequence with

$$\lambda_k \rightarrow \infty.$$

Proof. (1). Let γ be from the Energy Estimates Theorem. Then, we have

$$B[u, u] \geq \beta \|u\|_{H^1(U)}^2 - \gamma \|u\|_{L^2(U)}^2 \quad \forall u \in H_0^1(U).$$

If γ satisfies this inequality, then the inequality holds for any $\tilde{\gamma} > \gamma$. Without loss of generality, we shall assume $\gamma > 0$. We will attempt to identify which values of λ may lie in Σ . By the First Existence Theorem, if $\mu \geq \gamma$, then

$$\begin{cases} Lu + \mu u = f, & \text{in } U \\ u = 0, & \text{on } \partial U \end{cases}$$

has a unique weak solution $u \in H_0^1(U)$ for each $f \in L^2(U)$. If $\mu = -\lambda$, then the above BVP becomes precisely (i), so a unique solution to (i) exists if $-\lambda \geq \gamma$. Then, if $\lambda \leq -\gamma$, λ cannot lie in Σ , and so only λ such that $\lambda > -\gamma$ can lie in Σ .

(2). Consider the case where $\lambda > -\gamma$. By the Second Existence Theorem, (i) has a unique weak solution for each $f \in L^2(U)$ if and only if $u = 0$ is the only weak solution to the homogeneous problem

$$\begin{cases} Lu - \lambda u = 0, & \text{in } U \\ u = 0, & \text{on } \partial U \end{cases} \iff \begin{cases} Lu + \gamma u = (\lambda + \gamma)u, & \text{in } U \\ u = 0, & \text{on } \partial U. \end{cases}$$

Recall from the proof of the Second Existence Theorem that

$$L_\gamma u = Lu + \gamma u, \quad Ku = \gamma L_\gamma^{-1} u.$$

Thus, the homogeneous problem holds if and only if

$$u = L_\gamma^{-1}(\lambda + \gamma)u = \frac{\gamma + \lambda}{\gamma} Ku.$$

We showed previously that $K : L^2(U) \rightarrow L^2(U)$ is a compact operator. Therefore, the above equation is the eigenvalue problem for the compact operator K ,

$$Ku = \frac{\gamma}{\gamma + \lambda} u.$$

Since K is a compact operator, we have by the Spectrum of Compact Operators Theorem, that the collection of eigenvalue of K (which have the form $\gamma/(\gamma + \mu)$) is a set that is finite or else a countable sequence tending to zero.

We shall write the collection of eigenvalues to be $\mu_k = \gamma/(\gamma + \lambda_k)$. We have that either $\{u_k\}$ is finite, which implies that $\Sigma = \{\lambda_k\}$, where $\lambda_k = \gamma(1 - \mu_k)/\mu_k$ is finite; or, $\{\mu_k\}$ is a countable sequence such that $\mu_k \rightarrow 0$ which implies that $\Sigma = \{\lambda_k\}$ is a countable sequence such that $\lambda_k \rightarrow \infty$.

We have shown that Σ is either finite or countable, and if countable, then $\lim_{k \rightarrow \infty} \lambda_k = \infty$. Thus, we have proven (i) and part of (i). To prove (i), we consider first

($\Leftarrow$) that if $\lambda \notin \Sigma$, (i.e., μ is not an eigenvalue of K), then $u = 0$ is the only solution to the compact operator $Ku = (\gamma/(\gamma + \lambda))u$, which we saw is equivalent to (i) having a unique weak solution.

($\Rightarrow$) If (i) has a unique weak solution, we saw that it is equivalent to $u = 0$ being the only solution to Ku . As such, μ is not an eigenvalue of K so that $\lambda \notin \Sigma$. $\square$

Remark. A consequence of (i) in Theorem 3.2.13 is that for each $\lambda \notin \Sigma$,

$$\begin{cases} (L - \lambda I)u = f, & \text{in } U \\ u = 0, & \text{on } \partial U \end{cases}$$

has a unique weak solution. Then, $(L - \lambda I) : H_0^1(U) \rightarrow L^2(U)$ is bijective and therefore the also the resolvent operator $(L - \lambda I)^{-1} : L^2(U) \rightarrow H_0^1(U)$ is bijective.

Definition 3.2.6. We call Σ the (real) spectrum of the operator L ; and, the BVP

$$\begin{cases} Lu = \lambda u, & \text{in } U \\ u = 0, & \text{on } \partial U \end{cases}$$

has a nontrivial solution $w \neq 0 \iff \lambda \in \Sigma$. (If $\lambda \notin \Sigma$, then (i) shows that $\exists!$ solution to the BVP. Since $u = 0$ is a solution also, there must be no nontrivial solution).

Theorem 3.2.14 (Boundedness of the Inverse). Let Σ be as in the Third Existence Theorem and let $\lambda \notin \Sigma$. Let $f \in L^2(U)$ and let $u \in H_0^1(U)$ be the unique weak solution to

$$\begin{cases} Lu = \lambda u + f, & \text{in } U \\ u = 0, & \text{on } \partial U. \end{cases}$$

Then, there exists a constant C such that

$$\|u\|_{L^2(U)} \leq C \|f\|_{L^2(U)}.$$

Indeed, we show that $(L - \lambda I)^{-1} : L^2(U) \rightarrow L^2(U)$ is bounded.

3.3 Regularity

We have proven the existence and uniqueness of weak solution $u \in H_0^1(U)$ to BVP's under minimal assumptions on the domain U and the problem parameters a^{ij} , b^i , c , and f . It is now our new goal to prove the existence and uniqueness of a classical solution.

Motivating Example. Are we able to find a solution smoother than $H_0^1(U)$? For example, consider the problem $-\Delta u = f$ in $\mathbb{R}^n$ for $f \in L^2(\mathbb{R}^n)$. We shall assume that u is smooth and that u vanishes sufficiently fast as $|x| \rightarrow \infty$. Observe that

$$\|f\|_{L^2(\mathbb{R}^n)}^2 = \|\Delta u\|_{L^2(\mathbb{R}^n)}^2 = \sum_{i,j=1}^n \int_{\mathbb{R}^n} u_{x_i x_i} u_{x_j x_j} \, dx = \text{IBP twice} \sum_{i,j=1}^n \int_{\mathbb{R}^n} (u_{x_i x_j})^2 \, dx.$$

We see that if $f \in L^2(\Omega)$, then $u_{x_i x_j} \in L^2(\mathbb{R}^n)$ also for $i, j = 1, 2, \dots, n$. What this means is that if there exists a weak solution $u \in H_0^1(\mathbb{R}^n)$ to our example (meaning that $u \in L^2(\mathbb{R}^n)$, $u_{x_i} \in L^2(\mathbb{R}^n)$), then we have also that $u_{x_i x_j} \in L^2(\mathbb{R}^n)$. In other words, $u \in H^2(\mathbb{R}^n) \cap H_0^1(\mathbb{R}^n)$ implying more regularity than just $u \in H_0^1(\mathbb{R}^n)$.

Differentiating our given problem, we find

$$-\Delta u_{x_i} = f_{x_i} \quad \text{for } i = 1, 2, \dots, n.$$

Set, in particular, $\tilde{u} = u_{x_i}$, $\tilde{f} = f_{x_i}$. Then, we have $-\Delta \tilde{u} = \tilde{f}$. Notice that if $f \in H^1(\mathbb{R}^n)$, then $\tilde{f} \in L^2(\mathbb{R}^n)$ to give $\tilde{u} \in H^2(\mathbb{R}^n)$. Indeed, $u \in H^3(\mathbb{R}^n)$. Repeating this process, we observe that if $u \in H_0^1(\mathbb{R}^n)$ is a weak solution to our Poisson problem, and if $f \in H^m(\mathbb{R}^n)$ for $m = 0, 1, \dots, \infty$. Then, in fact, $u \in H^{m+2}(\mathbb{R}^n)$.

Definition 3.3.1. Let $U \subset \mathbb{R}^n$ be open, assume $u \in L_{\text{loc}}^1(U)$, and $V \subset\subset U$. Then, we have the following:

- (i) The i th-difference quotient of size h is

$$D_i^h u(x) = \frac{u(x + h e_i) - u(x)}{h} \quad (i = 1, 2, \dots, n)$$

for all $x \in V$ and $h \in \mathbb{R}$, such that $0 < |h| < \text{dist}(V, \partial U)$.

- (ii) The difference gradient of size h is

$$D^h u(x) := (D_1^h u, D_2^h u, \dots, D_n^h u).$$

Theorem 3.3.1 (Difference Quotients and Weak Derivatives). Let $U \subset \mathbb{R}^n$ be open and assume $u \in L^1_{\text{loc}}(U)$. Then, we have the following results:

- (i) for $1 \leq p < \infty$ and for $V \subset\subset U$, $\exists > 0$ such that

$$\left\| D^h u \right\|_{L^p(V)} \leq C \|Du\|_{L^p(U)}$$

for all $u \in W^{1,p}(U)$ and for all h such that $0 < |h| < \text{dist}(V, \partial U)/2$.

- (ii) for $1 < p < \infty$, for $V \subset\subset U$, for $u \in L^p(V)$, and $\exists C > 0$ such that

$$\left\| D^h u \right\|_{L^p(V)} \leq C \quad \forall h \text{ s.t. } 0 < |h| < \frac{1}{2} \text{dist}(V, \partial U)$$

it follows that $u \in W^{1,p}(V)$ with $\|Du\|_{L^p(V)} \leq C$.

Theorem 3.3.2 (Characterization of $W^{1,\infty}$).

- (i) Let $U \subset \mathbb{R}^n$ be open and bounded and ∂U be C^1 . Then, $u : U \rightarrow \mathbb{R}$ is Lipschitz continuous if and only if $u \in W^{1,\infty}(U)$. Indeed,

$$u \in C^{0,1}(\bar{U}) \iff u \in W^{1,\infty}(U).$$

- (ii) If $U \subset \mathbb{R}^n$ is open, then u is **locally** Lipschitz continuous in U if and only if $u \in W^{1,\infty}_{\text{loc}}(U)$.

Theorem 3.3.3. Assume $U \subset \mathbb{R}^n$ is open and $u \in W^{1,p}_{\text{loc}}(U)$ for $n < p \leq \infty$. Then u is (classically) differentiable a.e in U , and its classical derivative equals the weak derivative a.e in U .

Theorem 3.3.4 (Rademacher). Assume $U \subset \mathbb{R}^n$ be open and let u be locally Lipschitz in U . Then u is (classically) differentiable a.e in U .

Theorem 3.3.5 (Interior H^2 -Regularity). Assume that $U \subset \mathbb{R}^n$ is open and bounded, $a^{ij} \in C^1(U)$ (in addition to $a^{ij} \in L^\infty(U)$), and that $b^i, c \in L^\infty(U)$. Assume also that L is uniformly elliptic, $f \in L^2(U)$ and $u \in H^1(U)$ is a weak solution to the elliptic PDE $Lu = f$ in U . Then, we have that

$$u \in H^2_{\text{loc}}(U)$$

and for each open $V \subset\subset U$, $\exists C > 0$ such that

$$\|u\|_{H^2(V)} \leq C \left(\|f\|_{L^2(U)} + \|u\|_{L^2(U)} \right).$$

Proof. How the proof works: (1) we shall first show that for any $V \subset\subset U$, $\exists C > 0$ such that for $|h|$ small enough,

$$\left\| D_k^h D_u \right\|_{L^2(V)} \leq C_1 \left(\|f\|_{L^2(U)} + \|u\|_{H^1(U)} \right).$$

- (2) By the Different Quotients and Weak Derivatives of u , we shall show that $\exists C_2 > 0$ such that

$$\|D^2 u\|_{L^2(V)} \leq C_2 \left(\|f\|_{L^2(U)} + \|u\|_{L^2(U)} \right).$$

(3) Then we complete the proof.

(1). Fix any open set $V \subset\subset U$ and choose an open set W such that $V \subset\subset W \subset\subset U$. We shall choose $\xi \in C_1^\infty(\mathbb{R}^n)$ such that

$$\xi(x) = \begin{cases} 1, & x \in V \\ 0 \leq \xi \leq 1, & x \in W - V \\ 0, & x \in \mathbb{R}^n - W. \end{cases}$$

In particular, we call ξ a *cutoff* function to restrict calculations to W (away from ∂U). We choose h such that $\text{dist}(V, \partial U) > 4|h|$ and $\text{dist}(W, \partial U) > 3|h|$.

(2). For a weak solution $u \in H^1(U)$ to the Poisson problem $Lu = f$, observe that

$$\int_U \left(\sum_{i,j=1}^n a^{ij} u_{x_i} v_{x_j} + \sum_{i=1}^n b^i u_{x_i} v + cuv \right) dx = \int_U f v dx \quad \forall v \in H_0^1(U).$$

Since u is known, we can define

$$\tilde{f} := f - \sum_{i=1}^n b^i u_{x_i} - cu \in L^2(U)$$

so that we have instead

$$\sum_{i,j=1}^n \int_U a^{ij} u_{x_i} v_{x_j} dx = (\tilde{f}, v) \quad \forall v \in H_0^1(U).$$

(3). Ideally, $v = u_{x_k x_k}$. However, we do not know if $u \in H_{\text{loc}}^2(U)$. Instead, we will use a different quotient approximation to $u_{x_k x_k}$. Choose

$$v := -D_k^{-h}(\xi^2 D_k^h u) \quad (k = 1, 2, \dots, n).$$

Observe that if $\xi = 1$ on V , we can ignore ξ^2 so that

$$-D_k^{-h}(D_k^h u) = \frac{u(x + h e_k) - 2u(x) + u(x - h e_k)}{h^2}$$

which is a standard difference approximation to $u_{x_k x_k}$. For our choice of h , and since $\text{spt}(\xi) \subset \bar{W}$, v vanishes for x close to ∂U . Indeed, for x away from ∂U , v is a linear combination of functions in $H^1(U)$. Thus, $v \in H_0^1(U)$.

(4). Let us define the following,

$$A := \sum_{i,j=1}^n \int_U a^{ij} u_{x_i} v_{x_j} dx, \quad B := (\tilde{f}, v).$$

We will make use of, without proof, the equalities

$$\begin{aligned} \int_U v D_k^{-h} w dx &= - \int_U w D_k^h v dx \quad \text{for } w = \xi^2(D_k^h u) \\ D_k^h(vw) &= v^h D_k^h w + w D_k^h v \quad \text{where } v^h(x) = v(x + h e_k). \end{aligned}$$

By definition, we have

$$\begin{aligned}
A &= - \sum_{i,j=1}^n \int_U a^{ij} u_{x_i} \left[D_k^{-h} (\xi^2 D_k^h u) \right]_{x_j} dx \\
&= \sum_{i,j=1}^n \int_U D_k^h (a^{ij} u_{x_i}) (\xi^2 D_k^h u)_{x_j} dx \\
&= \sum_{i,j=1}^n \int_U \left[a^{ij,h} (D_k^h u_{x_i}) (\xi^2 D_k^h u)_{x_j} + (D_k^h a^{ij}) u_{x_i} (\xi^2 D_k^h u)_{x_j} \right] dx \\
&= \sum_{i,j=1}^n \int_U a^{ij,h} D_k^h u_{x_i} D_k^h u_{x_j} \xi^2 dx \\
&\quad + \sum_{i,j=1}^n \int_U \left[a^{ij,h} D_k^h u_{x_i} D_k^h u_{x_j} 2\xi \xi_{x_j} + (D_k^h a^{ij}) u_{x_i} D_k^h u_{x_j} \xi^2 + (D_k^h a^{ij}) u_{x_i} D_k^h u_{x_j} 2\xi \xi_{x_j} \right] dx \\
&=: A_1 + A_2.
\end{aligned}$$

We will bound both A_1 and A_2 to show that A is bounded. With A_1 , observe by uniform ellipticity, that

$$A_1 \geq \theta \int_U \xi^2 \left| D_k^h Du \right|^2 dx \quad (\theta > 0).$$

For A_2 , observe that

$$|A_2| \leq \int_U \left(\left| a^{ij,h} \right| \left| D_k^h u_{x_i} \right| \left| D_k^h u \right| 2\xi |\xi_{x_j}| + \left| D_k^h a^{ij} \right| |u_{x_i}| \left| D_k^h u_{x_j} \right| \xi^2 + \left| D_k^h a^{ij} \right| |u_{x_i}| \left| D_k^h u \right| 2\xi |\xi_{x_j}| \right) dx.$$

In particular, we can see that

$$\left| a^{ij,h} \right| \leq C, \quad |x_{x_j}| \leq C, \quad \left| D_k^h a^{ij} \right| \leq C$$

so that we have the simplified bound

$$|A_2| \leq C \int_U \left(\xi \left| D_k^h Du \right| \left| D_k^h u \right| + \xi |Du| \left| D_k^h Du \right| + \xi |Du| \left| D_k^h u \right| \right) dx.$$

By Cauchy's inequality with ϵ , we see explicitly

$$\begin{aligned}
\xi \left| D_k^h Du \right| \left| D_k^h u \right| &\leq \epsilon \xi^2 \left| D_k^h Du \right|^2 + \frac{1}{4\epsilon} \left| D_k^h u \right|^2 \\
\xi |Du| \left| D_k^h Du \right| &\leq \epsilon \xi^2 \left| D_k^h Du \right|^2 + \frac{1}{4\epsilon} |Du|^2 \\
\xi |Du| \left| D_k^h u \right| &\leq \xi \left(\frac{1}{2} |Du|^2 + \frac{1}{2} \left| D_k^h u \right|^2 \right) \quad \left(\epsilon = \frac{1}{2} \right).
\end{aligned}$$

Then,

$$|A_2| \leq 2\epsilon C \int_U \xi^2 \left| D_k^h Du \right|^2 dx + C_\epsilon \int_W \left(\left| D_k^h u \right|^2 + |Du|^2 \right) dx \quad \left(C_\epsilon = \frac{C}{\epsilon} + C \right).$$

By using the theorem on Difference Quotients and Weak Derivatives, since $u \in H^1(U)$ and $W \subset\subset U$, we know $\exists C_2$ such that

$$\left\| D_k^h u \right\|_{L^2(W)} \leq C_2 \|Du\|_{L^2(U)}$$

for h small enough. Thus, we find that

$$|A_2| \leq 2\epsilon C \int_U \xi^2 |D_k^h Du|^2 dx + \tilde{C}_\epsilon \int_U |Du|^2 dx.$$

Choosing ϵ such that $2\epsilon C = \theta/2$ and combining the bounds in A_1 and A_2 , we see that

$$A \geq A_1 - |A_2| \geq \frac{\theta}{2} \int_U \xi^2 |D_k^h Du|^2 dx - \hat{C} \int_U |Du|^2 dx$$

which is the lower bound for A .

(5). We will now bound B . Observe that

$$|B| = \left| \int_U \left(f - \sum_{i=1}^n b^i u_{x_i} - cu \right) v dx \right| \leq C \int_U (|f| + |Du| + |u|) |v| dx$$

where we have assumed that $b^i, c \in L^\infty(U)$. Then, by Cauchy's inequality with ϵ , we have

$$|B| \leq 3\epsilon C \int_U |v|^2 dx + \frac{1}{4\epsilon} \int_U (|f|^2 + |Du|^2 + |u|^2) dx.$$

For the first integral, recall that $v := -D_k^{-h}(\xi^2 D_k^h u)$, so that we have

$$\int_U |v|^2 dx = \int_U |D_k^{-h}(\xi^2 D_k^h u)|^2 dx.$$

Recall that $\text{spt}(\xi) \subset \bar{W}$ and that $\text{dist}(W, \partial U) > 3|h|$. However, $-h$ in D_k^{-h} shifts ξ by $\pm h$ and so shifts $\text{spt}(\xi)$ out of $\bar{W}$. To fix this, we will define Z such that $\text{dist}(Z, \partial U) > 2|h|$. Then, $\text{spt}(\xi(x \pm he)) \subset \bar{Z}$. Indeed, we have

$$\int_U |v|^2 dx = \int_Z D_k^{-h}(\xi^2(D_k^h u)) dx.$$

In (3), we showed that $v = -D_k^{-h}(\xi^2 D_k^h u) \in H_0^1(U)$. Similarly, we also found that $\xi^2 D_k^h u \in H_0^1(U)$. Then, by the Difference Quotients (and Weak Derivatives) theorem, we see that

$$\int_U |v|^2 dx \leq \int_U |D(\xi^2 D_k^h u)|^2 dx = \int_W |(D\xi^2)D_k^h u + \xi^2 D D_k^h u|^2 dx.$$

Then, by the triangle inequality,

$$\begin{aligned} \int_U |v|^2 dx &\leq C \int_W \left(|D_k^h u|^2 + \xi^4 |D_k^h Du|^2 \right) dx \\ &\leq C \int_U \left(|Du|^2 + \xi^2 |D_k^h Du|^2 \right) dx \end{aligned}$$

where ξ^2 in ξ^4 is absorbed into the constant since ξ has compact support and is bounded. Note that we have also invoked the Difference Quotients Theorem. Combining the bound for $|v|^2$ with our estimate for B , we thus find that

$$|B| \leq C\epsilon \int_U \xi^2 |D_k^h Du|^2 dx + \bar{C}_\epsilon \int_U (|f|^2 + |Du|^2 + |u|^2) dx.$$

(6). We have $A = B$. So, combining the estimates for A and B (and choosing $C\epsilon = \theta/4$), we see that

$$0 = A - B \geq A - |B| \geq \frac{\theta}{4} \int_U \xi^2 \left| D_k^h Du \right|^2 dx - \bar{C} \int_U \left(|f|^2 + |Du|^2 + |u|^2 \right) dx.$$

This tells us that $\exists C > 0$ such that

$$\int_U \xi^2 \left| D_k^h Du \right|^2 dx \leq C \int_U \left(|f|^2 + |Du|^2 + |u|^2 \right) dx$$

or, since $\xi = 1$ on V , observe that

$$\int_V \left| D_k^h Du \right|^2 dx \leq C \int_U \left(|f|^2 + |Du|^2 + |u|^2 \right) dx$$

which is true $\forall h$ such that $0 < |h| < \text{dist}(V, \partial U)/4$. Indeed, by the definition of the Sobolev norms,

$$\left\| D_k^h Du \right\|_{L^2(V)} \leq C \left(\|f\|_{L^2(U)} + \|Du\|_{L^2(U)} + \|u\|_{L^2(U)} \right) = C \left(\|f\|_{L^2(U)} + \|u\|_{H^1(U)} \right)$$

proving inequality (1). Observe that the estimate we just derived holds for all k . We know that $u \in H^1(U)$ and so $u_{x_j} \in L^2(U)$ for $j = 1, 2, \dots, n$; so, the RHS of the estimate is finite. From the Difference Quotients Theorem, we know that if $\|D^h u\|_{L^p(V)} \leq \tilde{C}$, then $\|Du\|_{L^p(V)} \leq \tilde{C}$. Applying this result to $u_{x_j} \in L^2(V)$ and using the estimate derived, we have that $u_{x_j} \in H^1(V)$ for $j = 1, 2, \dots, n$; and, that

$$\|D^2 u\|_{L^2(V)}^2 \leq C \left(\|f\|_{L^2(U)}^2 + \|u\|_{H^1(U)}^2 \right)$$

proving inequality (2). We need only to conclude. Adding $\|u\|_{H^1(V)}^2$ to both sides of inequality (2) and using that

$$\|u\|_{H^1(V)} \leq \|u\|_{H^1(U)}^2$$

we have that

$$\|u\|_{H^2(V)}^2 \leq C \left(\|f\|_{L^2(U)}^2 + \|u\|_{H^1(U)}^2 \right).$$

Since $V \subset\subset U$ is arbitrary, we can conclude that $u \in H_{\text{loc}}^2(U)$.

(7). To complete the proof, we must show that we can replace $\|u\|_{H^1(U)}^2$ in the above inequality by $\|u\|_{L^2(U)}^2$.

We have that $V \subset\subset W \subset\subset U$. Introducing X such that $V \subset\subset X \subset\subset W$, we repeat steps (1)-(6) (where we replace V, W, U by V, X, W) to find that

$$\|u\|_{H^2(V)} \leq C \left(\|f\|_{L^2(W)}^2 + \|u\|_{H^1(W)}^2 \right).$$

We introduce now Y such that $W \subset\subset Y \subset\subset U$, and $\zeta \in C_c^\infty(\mathbb{R}^n)$ such that

$$\zeta(x) = \begin{cases} 1, & x \in W \\ 0 \leq \zeta \leq 1, & x \in Y - W \\ 0, & x \in \mathbb{R}^n - Y. \end{cases}$$

In (3), we found that

$$\sum_{i,j=1}^n \int_U a^{ij} u_{x_i} v_{x_j} dx = (\tilde{f}, v) \quad \forall v \in H_0^1(U).$$

If we set $v = \zeta^2 u$, we can find that

$$\|Du\|_{L^2(W)}^2 \leq C \left(\|f\|_{L^2(U)}^2 + \|u\|_{L^2(U)}^2 \right).$$

and by adding $\|Du\|_{L^2(W)}^2$ on both sides, we find

$$\|u\|_{H^1(W)}^2 \leq C \left(\|f\|_{L^2(U)}^2 + \|u\|_{L^2(U)}^2 \right).$$

We combine both inequalities on $\|u\|_{H^1(W)}$ and $\|u\|_{H^2(V)}$ to find that

$$\|u\|_{H^2(V)} \leq C \left(\|f\|_{L^2(U)} + \|u\|_{L^2(U)} \right)$$

to which we can replace $\|u\|_{H^1(U)}^2$ with the inequality above. $\square$

Remark. Observe that $u \in H^1(U)$ is a weak solution to $Lu = f$ in U implies that $B[u, v] = (f, v)$ for all $v \in H_0^1(U)$. If the assumptions of Theorem 3.3.5 indeed hold, then we have also that $u \in H_{\text{loc}}^2(U)$. Consequently, since $C_c^\infty(U) \subset H_0^1(U)$,

$$(f, v) = B[u, v] = \int_U \left(\sum_{i,j=1}^n a^{ij} u_{x_i} v_{x_j} + \sum_{i=1}^n b^i u_{x_i} v + cuv \right) dx =_{\text{IBP}} (Lu, v) \quad \forall v \in C_c^\infty(U).$$

Indeed, we have that

$$(Lu - f, v) = 0 \implies Lu = f \text{ a.e. in } U.$$

for all $v \in C_c^\infty(U)$. A solution to this problem is called a **strong solution**.

Theorem 3.3.6 (Higher Interior Regularity). Assume $U \subset \mathbb{R}^n$ is open and bounded, $a^{ij}, b^i, c \in C^{m+1}(U)$, $f \in H^m(U)$, and $u \in H^1(U)$ is a weak solution to $Lu = f$ in U , for $m \in \{0, 1, 2, \dots\}$. Then, we have the following conclusions:

- (i) $u \in H_{\text{loc}}^{m+2}(U)$.
- (ii) $\forall V \subset\subset U$, $\exists C > 0$ such that

$$\|u\|_{H^{m+2}(V)} \leq C \left(\|f\|_{H^m(U)} + \|u\|_{L^2(U)} \right).$$

Proof. The proof is by induction using the Interior H^2 -Regularity theorem. $\square$

Remark. The consequences of Higher Interior Regularity show that $u \in H_{\text{loc}}^{m+2}(V)$ ($H^{m+2}(V) = W^{m+2,2}(V)$). We have that by the general Sobolev inequalities that if $m+2 > n/2$, then for $V \subset\subset U$ with ∂V in C^1 ,

$$u \in C^{\ell, \gamma}(\bar{V}), \quad \ell = m+2 - \left\lfloor \frac{n}{2} \right\rfloor - 1, \quad \gamma = \begin{cases} \left\lfloor \frac{n}{2} \right\rfloor + 1 - \frac{n}{2}, & \text{if } \frac{n}{2} \notin \mathbb{Z} \\ \text{any positive number} < 1, & \text{if } \frac{n}{2} \in \mathbb{Z}. \end{cases}$$

If $u \in C^{\ell, \gamma}(\bar{V})$, then $u \in C^\ell(\bar{V})$, and so since $V \subset\subset U$ is arbitrary, we can locally say that $u \in C^\ell(U)$.

Example 3.2. For $n = 3$, $m = 2$, we have $\ell = 2$ and so $u \in C^2(U)$, which is the local "calculus" solution.

Theorem 3.3.7 (The Smooth Solution). Assume $U \subset \mathbb{R}^n$ is open and bounded, $a^{ij}, b^i, c \in C^\infty(U)$, $f \in C^\infty(U)$, and $u \in H^1(U)$ is the weak solution to $Lu = f$ in U , then $u \in C^\infty(U)$.

3.3.1 Boundary Regularity

Theorem 3.3.8 (C^2 Boundary Regularity). Assume $U \subset \mathbb{R}^n$ is open, bounded, and ∂U is C^2 . Assume $a^{ij} \in C^1(\bar{U})$, $b^i, c \in L^\infty(U)$, $f \in L^2(U)$, and $u \in H_0^1(U)$ is a weak solution to the BVP

$$\begin{cases} Lu = f, & \text{in } U \\ u = 0, & \text{on } \partial U. \end{cases}$$

Then, $u \in H^2(U)$ and $\exists C > 0$ such that the following estimate holds:

$$\|u\|_{H^2(U)} \leq C \left(\|f\|_{L^2(U)} + \|u\|_{L^2(U)} \right).$$

Proof. A sketch of the proof will be given since all techniques have already been seen.

(1)-(5). First, study the locally flat boundary case. Assume $U = B^0(0, 1) \cap \mathbb{R}_+^n$, and $V = B^0(0, 1/2) \cap \mathbb{R}_+^n$. With appropriate cutoff functions ζ and difference quotients (similar to the proof of Interior H^2 -Regularity), we first obtain a bound for derivatives in the direction not normal to the flat boundary. This nets

$$\sum_{\substack{k, l=1 \\ k+l < 2n}}^n \|u_k u_l\|_{L^2(V)} \leq C \left(\|f\|_{L^2(U)} + \|u\|_{H^1(U)} \right).$$

Similar to step (7) of the proof on Interior H^2 -Regularity, we can replace $\|u\|_{H^1(U)}$ by $\|u\|_{L^2(U)}$.

Then, for $u_{x_n x_n}$, note from the proof of Interior H^2 -Regularity that on the interior, the solution is a *strong* solution. Thus, we can work with $Lu = f$ a.e on U together with the ellipticity assumption ($a^m \geq \theta > 0$). Indeed, we find

$$|u_{x_n x_n}| \leq C \left(\sum_{\substack{i, j=1 \\ i+j < 2n}}^n |u_{x_i x_j}| + |Du| + |u| + |f| \right)$$

from which we can arrive at

$$\|u_{x_n x_n}\|_{L^2(V)} \leq C \left(\sum_{\substack{i, j=1 \\ i+j < 2n}}^n \|u_{x_i x_j}\|_{L^2(V)} + \|Du\|_{L^2(V)} + \|u\|_{L^2(V)} + \|f\|_{L^2(V)} \right).$$

Combining bounds for second derivatives, we find

$$\|u\|_{H^2(U)} \leq C \left(\|f\|_{L^2(U)} + \|u\|_{L^2(U)} \right).$$

(6)-(7). Now study the curved boundary case. Take any $x^0 \in \partial U$. Let $\eta = \phi(x)$ be a straightening map. Then, since ∂U is C^2 , we have for sufficiently small r that $\Phi(x)$ is C^2 on $B^0(x^0, r)$ with C^2 -inverse $x = \Psi(\eta)$.

Choose s sufficiently small such that

$$\begin{aligned} U' &= B^0(0, s) \cap \{\eta_n \geq 0\} \subset \Phi(U \cap B(x^0, r)) \\ V' &= B^0(0, s/2) \cap \{\eta_n \geq 0\}. \end{aligned}$$

(8)-(9). Now transform the PDE to $L'u' = f'$ in U' and check that the weak formulation on U' is well-defined, and that L' satisfies the same assumptions as L (all fine since ∂U is C^2).

(10). Now we apply steps (1)-(5) to L' . Observe that U' is a half-ball, so from steps (1)-(5), we have that

$$\|u'\|_{H^2(V')} \leq C \left(\|f'\|_{L^2(U')} + \|u'\|_{L^2(U')} \right).$$

Mapping back to the original coordinate system using $V := \Phi(V')$ and that $\Psi(U') \subset U$, we find that

$$\|u\|_{H^2(V)} \leq C \left(\|f\|_{L^2(U)} + \|u\|_{L^2(U)} \right).$$

(11). The above result is for a single point C^0 on the boundary, but we know ∂U is compact, so we can find a finite number of sets $V_1, \dots, V_N$ as above so we find $V_0 \subset\subset U$ such that $U = \bigcup_{i=1}^N V_i$. Then, by previous steps, on the "boundary sets" V_i ($i = 1, \dots, N$), we have the inequality

$$\|u\|_{H^2(V_i)} \leq C \left(\|f\|_{L^2(U)} + \|u\|_{L^2(U)} \right).$$

On the "interior set" V_0 , we find also

$$\|u\|_{H^2(V_0)} \leq C \left(\|f\|_{L^2(U)} + \|u\|_{L^2(U)} \right)$$

by the Interior H^2 -Regularity Theorem. By adding each inequality for i along with the interior estimate, we find

$$\|u\|_{H^2(U)} \leq \sum_{i=1}^N \|u\|_{H^2(V_i)} \leq C \left(\|f\|_{L^2(U)} + \|u\|_{L^2(U)} \right).$$

□

Remark. Observe that if $u \in H_0^1(U)$ is the *unique* weak solution, then it can be shown that $\|u\|_{L^2(U)} \leq C\|f\|_{L^2(U)}$. Then, estimate in Theorem 3.3.8 can be shown to reduce to

$$\|u\|_{H^2(U)} \leq C\|f\|_{L^2(U)}.$$

Theorem 3.3.9 (Higher Boundary Regularity). Assume $U \subset \mathbb{R}^n$ is open and bounded, and ∂U is C^{m+2} ($m = 0, 1, \dots$). Assume also $a^{ij}, b^i, c \in C^{m+1}(\bar{U})$, $f \in H^m(U)$, and $u \in H_0^1(U)$ is a weak solution to the BVP

$$\begin{cases} Lu = f, & \text{in } U \\ u = 0, & \text{on } \partial U. \end{cases}$$

Then, $u \in H^{m+2}(U)$ and $\exists C > 0$ such that

$$\|u\|_{H^{m+2}(U)} \leq C \left(\|f\|_{H^m(U)} + \|u\|_{L^2(U)} \right)$$

and, if u is unique, we have the tighter bound

$$\|u\|_{H^{m+2}(U)} \leq C\|f\|_{H^m(U)}.$$

Remark. Observe that if $u \in W_2^{m+2}(U)$, ∂U is C^2 , and if $m + 2 > n/2$, then by the General Sobolev Inequality,

$$u \in C^\ell(\bar{U}) \quad \text{where } \ell = m + 2 - \left\lfloor \frac{n}{2} \right\rfloor - 1.$$

Example 3.3. Consider, for example, $n = 3$, $m = 2$. Then, $u \in C^2(\bar{U})$ and so we have the classical "calculus" solution.

Theorem 3.3.10 (Infinite Differentiability Until the Boundary). Assume $U \subset \mathbb{R}^n$ is open, bounded, and ∂U is C^∞ . Assume also that $a^{ij}, b^i, c \in C^\infty(\bar{U})$, $f \in H^\infty(U)$, and $u \in H_0^1(U)$ is the weak solution to the BVP

$$\begin{cases} Lu = f, & \text{in } U \\ u = 0, & \text{on } \partial U. \end{cases}$$

Then, $u \in C^\infty(\bar{U})$.

4 Second-Order Parabolic Equations

Typically, parabolic equations deal with time-dependent functions. In particular, we previously assumed that $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Now, we will assume Banach-space valued functions $f : [0, T] \rightarrow X$, where X is a Banach space with norm $\|\cdot\|_X$.

4.0.1 Spaces Involving Time

Definition 4.0.1. A function $u : [0, T] \rightarrow X$ is continuous at a point $r \in (0, T)$ if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } \forall t \in (0, T) \text{ with } |t - r| < \delta \implies \|u(t) - u(r)\|_X < \epsilon.$$

Definition 4.0.2.

$$\|u\|_{C([0, T]; X)} := \max_{0 \leq t \leq T} \|u(t)\|_X.$$

Definition 4.0.3. $C([0, T]; X)$ is the space of all continuous functions $u : [0, T] \rightarrow X$ such that

$$\|u\|_{C([0, T]; X)} < \infty.$$

In particular, we define

$$\|u\|_{L^p(0, T; X)} := \begin{cases} \left(\int_0^T \|u(t)\|_X^p dt \right)^{1/p}, & \text{for } 1 \leq p < \infty \\ \text{ess sup}_{0 \leq t \leq T} \|u(t)\|_X, & \text{for } p = \infty. \end{cases}$$

Further, $L^p(0, T; X)$ is the space of (strongly) measurable functions $u : [0, T] \rightarrow X$ such that $\|u\|_{L^p(0, T; X)} < \infty$.

And, for $1 \leq p \leq \infty$, $L_{\text{loc}}^p(0, T; X)$ is the space of (strongly) measurable functions $u : [0, T] \rightarrow X$ such that $u \in L^p(t_0, t; X)$ where $[t_0, t] \subset (0, T)$.

Definition 4.0.4. Let $u \in L^1(0, T; X)$. If $\exists v \in L^1(0, T; X)$ such that

$$\int_0^T \phi'(t) u(t) dt = - \int_0^T \phi(t) v(t) dt \quad \forall \phi \in C_c^\infty(0, T)$$

then u is called **weakly differentiable** on $(0, T)$ and v is the **weak derivative**. We, in particular, write $u' = v$.

Definition 4.0.5. The Sobolev space $W^{1,p}(0, T; X)$ consists of all functions $u \in L^1(0, T; X)$ such that u' exists in the weak sense and belong to $L^p(0, T; X)$. Furthermore, we define the norm

$$\|u\|_{W^{1,p}(0, T; X)} := \begin{cases} \left[\int_0^T (\|u(t)\|_X^p + \|u'(t)\|_X^p) dt \right]^{1/p}, & \text{if } 1 \leq p < \infty \\ \text{ess sup}_{0 \leq t \leq T} (\|u(t)\|_X + \|u'(t)\|_X), & \text{if } p = \infty. \end{cases}$$

Notation. In the special case $p = 2$, we write $H^1(0, T; X)$ instead of $W^{1,2}(0, T; X)$.

Remark. $W^{1,p}(0, T; X)$ is a Banach space for any $1 \leq p \leq \infty$ if X is a Banach space.

Definition 4.0.6 (Mollification in Time). Mollification of Banach-Valued Functions in $L^1_{\text{loc}}(0, T; X)$ are equivalent to the previous formulation:

$$\eta(t) = \begin{cases} C \exp\left(\frac{1}{t^2 - 1}\right), & |t| < 1 \\ 0, & |t| \geq 1, \end{cases} \quad C \text{ s.t. } \int_{-\infty}^{\infty} \eta(t) dt = 1.$$

$$\eta_\epsilon(t) = \frac{1}{\epsilon} \eta\left(\frac{t}{\epsilon}\right) \longrightarrow \text{spt}(\eta_\epsilon) \subset (-\epsilon, \epsilon).$$

$$f^\epsilon(t) = \eta_\epsilon * f(t) = \int_{-\infty}^{\infty} \eta_\epsilon(t - \tau) f(\tau) d\tau = \int_{t-\epsilon}^{t+\epsilon} \eta_\epsilon(t - \tau) f(\tau) d\tau.$$

Let X be a Banach space and let $f \in L^1_{\text{loc}}(0, T; X)$ with $T > 0$. For $f^\epsilon(t) = \eta_\epsilon * f(t)$ in $(\epsilon, T - \epsilon)$, we have by the Properties of Mollifiers that

- (i) $f^\epsilon \in C^\infty(\epsilon, T - \epsilon)$ and $\partial_t^k f^\epsilon(t) = (\partial_t^k \eta_\epsilon) * f(t)$ for $k = 1, 2, \dots$.
- (ii) $\lim_{\epsilon \rightarrow 0} \|f^\epsilon(t) - f(t)\|_X \rightarrow 0$ for a.e. $t \in (0, T)$.
- (iii) If $f \in C(0, T; X)$, then $f^\epsilon \rightarrow f$ uniformly in X as $\epsilon \rightarrow 0$ on compact subsets of $(0, T)$.
- (iv) If $1 \leq p < \infty$ and $f \in L^p_{\text{loc}}(0, T; X)$, then $f^\epsilon \rightarrow f$ in $L^p_{\text{loc}}(0, T; X)$.

Theorem 4.0.1 (Bochner's Theorem). A (strongly) measurable function $f : [0, T] \rightarrow X$ is **Bochner integrable** if and only if the mapping $t \rightarrow \|f(t)\|_X$ is integrable. In this case,

$$\left\| \int_0^T f(t) dt \right\|_X \leq \int_0^T \|f(t)\|_X dt.$$

Further, we have also the action of u^* is such that

$$\langle u^*, \int_0^T f(t) dt \rangle = \int_0^T \langle u^*, f(t) \rangle dt \quad \forall u^* \in X^*.$$

Theorem 4.0.2 (Calculus in an Abstract Space). Let $u \in W^{1,p}(0, T; X)$ for $1 \leq p \leq \infty$. Then, we have the following:

- (i) $u \in C(0, T; X)$
- (ii)

$$u(t) = u(s) + \int_s^t u'(\tau) d\tau \quad \text{for all } 0 \leq s \leq t \leq T.$$

- (iii) $\exists C > 0$ such that

$$\max_{0 \leq t \leq T} \|u(t)\|_X \leq C \|u\|_{W^{1,p}(0, T; X)}.$$

Proof. We will use the following:

- (a) $u \in L^p(0, T; X) \Rightarrow u \in L^q(0, T; X)$ for $1 \leq q \leq p$.
- (b) If u is Lebesgue integrable, then $\int_s^t u(\tau) d\tau$ can be shown to be a continuous function in both s and t .

(1). Let us extend u to 0 on $(-\infty, 0)$ and (T, ∞) . Set also $u^\epsilon(t) = \eta_\epsilon * u(t)$ in $(\epsilon, T - \epsilon)$. By "mollification in time", we know that $u^\epsilon \in C^\infty(\epsilon, T - \epsilon; X)$ and $(u^\epsilon)'(t) = \eta'_\epsilon * u(t)$ in $(\epsilon, T - \epsilon)$.

We are given that $u \in W^{1,p}(0, T; X) \Rightarrow u \in L^p(0, T; X)$ and $u' \in L^p(0, T; X) \Rightarrow y' \in L^p_{\text{loc}}(0, T; X)$. Again, by mollification in time", observe that

$$\|u^\epsilon(t) - u(t)\|_X \rightarrow 0 \quad \text{a.e. } t \in (0, T).$$

Then, since $(u^\epsilon)'(t) = \eta_\epsilon * u'(t)$ and⁸ $u'(t) \in L^p_{\text{loc}}(0, T; X)$, by "mollification in time", we also have that $(u^\epsilon)' \rightarrow u'$ in $L^p_{\text{loc}}(0, T; X)$. By (a), it follows that $(u^\epsilon)' \rightarrow u'$ in $L^1_{\text{loc}}(0, T; X)$, which means that

$$\lim_{\epsilon \rightarrow 0} \int_0^t \|(u^\epsilon)'(\tau) - u'(\tau)\|_X d\tau = 0 \quad \forall [s, t] \subset (0, T).$$

By Bochner's Theorem, observe that

$$\lim_{\epsilon \rightarrow 0} \left\| \int_0^t [(u^\epsilon)'(\tau) - u'(\tau)] d\tau \right\| = 0 \quad \forall [s, t] \subset (0, T).$$

Since $u^\epsilon(t) \in C^\infty(\epsilon, T - \epsilon; X)$, we have that

$$u^\epsilon(t) = u^\epsilon(s) + \int_s^t (u^\epsilon)'(\tau) d\tau \quad \text{for any } 0 < s < t < T.$$

Thus, we have now

$$\begin{aligned} & \left\| u^\epsilon(t) - u(t) - u^\epsilon(s) + u(s) - \int_s^t [(u^\epsilon)' - u'] d\tau \right\|_X \\ & \leq \|u^\epsilon(t) - u(t)\|_X + \|u^\epsilon(s) - u(s)\|_X + \left\| \int_s^t [(u^\epsilon)' - u'] d\tau \right\|_X. \end{aligned}$$

However, in the limit that $\epsilon \rightarrow 0$, the bound from above all approach zero, and so we should find that

$$\lim_{\epsilon \rightarrow 0} \left\| u^\epsilon(t) - u(t) - u^\epsilon(s) + u(s) - \int_s^t [(u^\epsilon)' - u'] d\tau \right\|_X \leq 0.$$

Then, by the result using smoothness of $u^\epsilon(t)$, we find

$$u(t) = u(s) + \int_s^t u'(\tau) d\tau \quad \text{in } X \text{ for a.e. } 0 < s < t < T.$$

We must show that u is continuous on $[0, T]$. By (b), since $u' \in L^1(0, T; X)$, we know that $\int_s^t u'(\tau) d\tau$ is continuous in both s and t . This means that given $\epsilon > 0$, $\exists \delta > 0$ such that

$$|t - r| < \delta \implies \left\| \int_s^t u'(\tau) d\tau - \int_s^r u'(\tau) d\tau \right\|_X < \epsilon.$$

Indeed, given $\epsilon > 0$, $\exists \delta > 0$ such that

$$|t - r| < \delta \implies \|u(t) - u(r)\|_X = \left\| u(s) + \int_s^t u'(\tau) d\tau - u(s) - \int_s^r u'(\tau) d\tau \right\|_X < \epsilon$$

⁸We are able to shift the derivative from the mollifier to the function u . This was proved in a previous section.

and similarly,

$$\|s - r\| < \delta \implies \|u(s) - u(r)\|_X < \epsilon.$$

In other words, u is continuous on $[0, T]$ (i) and

$$u(t) = u(s) + \int_s^t u'(\tau) d\tau \quad \forall 0 \leq s \leq t \leq T.$$

□

Definition 4.0.7. A real-valued function is **absolutely continuous** provided that for all $\epsilon > 0$ there exists $\delta > 0$ such that for every finite disjoint collection $\{(a_k, b_k)\}_{k=1}^n$ of open intervals in (a, b) :

$$\text{if } \sum_{k=1}^n (b_k - a_k) < \delta \implies \sum_{k=1}^n |f(b_k) - f(a_k)| < \epsilon.$$

Theorem 4.0.3. A real-valued function f on a compact domain $[a, b]$ is absolutely continuous if and only if there exists a Lebesgue integrable function on $[a, b]$ such that

$$f(x) = f(a) + \int_a^x g(t) dt \quad \forall x \in [a, b].$$

When we discuss parabolic PDEs, the two most important function spaces are $u \in L^2(0, T; H_0^1(U))$ and $u' \in L^2(0, T; H^{-1}(U))$.

Proposition 4.0.3.1. Let $u, v \in L^2(U, 0, T; H_0^1(U))$ and $u', v' \in L^2(0, T; H^{-1}(U))$. Then,

$$\frac{d}{dt}(u(t), v(t))_{L^2(U)} = (u'(t), v(t))_{L^2(U)} + (u(t), v'(t))_{L^2(U)} \quad \text{a.e. } 0 \leq t \leq T.$$

Theorem 4.0.4 (More Calculus). Suppose $u \in L^2(0, T; H_0^1(U))$ with $u' \in L^2(0, T; H^{-1}(U))$. Then,

(i) $u \in C(0, T; L^2(U))$.

(ii) The mapping $t \rightarrow \|u(t)\|_{L^2(U)}^2$ is absolutely continuous with

$$\frac{d}{dt}\|u(t)\|_{L^2(U)}^2 = 2\langle u'(t), u(t) \rangle \quad \text{for a.e. } 0 \leq t \leq T$$

(iii) $\exists C > 0$ such that

$$\max_{0 \leq t \leq T} \|u(t)\|_{L^2(U)} \leq C \left(\|u\|_{L^2(0, T; H_0^1(U))} + \|u'\|_{L^2(0, T; H^{-1}(U))} \right).$$

Proof. (1). Let us extend u by reflection and cutoff functions to $[-\sigma, T + \sigma]$. For $0 < \epsilon, \delta < \sigma$ define $u^\epsilon = \eta_\epsilon * u$ and $u^\delta = \eta_\delta * u$. Then, for any $0 < t < T$,

$$\frac{d}{dt} \|u^\epsilon(t) - u^\delta(t)\|_{L^2(U)}^2 = 2(u^\epsilon(t), u^\delta(t), u^{\epsilon'}(t) - u^{\delta'}(t))_{L^2(U)}.$$

Integrating, we find for all $0 \leq s, t \leq T$,

$$\|u^\epsilon(t) - u(t)\|_{L^2(U)}^2 = \|u^\epsilon(s) - u^\delta(s)\|_{L^2(U)}^2 + 2 \int_s^t (u^\epsilon(\tau) - u^\delta(\tau), u^{\epsilon'}(\tau) - u^{\delta'}(\tau))_{L^2(U)} d\tau.$$

To bound this, we require the following two results:

- (a) $u^\epsilon(t) = \eta_\epsilon(t) * u(t) \in H_0^1(U)$.
 (b) $u^{\epsilon'}(t) = (\eta'_\epsilon * u)(t) \in H_0^1(U) \subset L^2(U) \subset H^{-1}(U)$.

Thus, we can show that

$$(u^\epsilon(\tau) - u^\delta(\tau), u^{\epsilon'}(\tau) - u^{\delta'}(\tau))_{L^2(U)} = \langle u^\epsilon(\tau) - u^\delta(\tau), u^{\epsilon'}(\tau) - u^{\delta'}(\tau) \rangle.$$

Now, let $u \in X$, $u^* \in X^*$. Then,

$$|\langle u^*, u \rangle| = \left| \frac{\langle u^*, u \rangle}{\|u\|_X} \|u\|_X \right| \leq \sup_{z \in X} \frac{|\langle u^*, z \rangle|}{\|z\|_X} \|u\|_X = \|u^*\|_{X^*} \|u\|_X.$$

Taking the supremum over $t \in [0, T]$ and using what we have just showed, we find that

$$\begin{aligned} S(u_{\epsilon\delta}(t)) &:= \sup_{0 \leq t \leq T} \|u^\epsilon(t) - u^\delta(t)\|_{L^2(U)} \leq \|u^\epsilon(s) - u^\delta(s)\|_{L^2(U)} \\ &\quad + 2 \int_0^T \|u^{\epsilon'}(\tau) - u^{\delta'}(\tau)\|_{H^{-1}(U)} \|u^\epsilon(\tau) - u^\delta(\tau)\|_{L^2(U)} d\tau \end{aligned}$$

Thus, by Cauchy's inequality with ϵ , we have the result

$$\begin{aligned} S_{\epsilon\delta}(t) &\leq \|u^\epsilon(s) - u^\delta(s)\|_{L^2(U)}^2 + \int_0^T \|u^{\epsilon'}(\tau) - u^{\delta'}(\tau)\|_{H^{-1}(U)}^2 d\tau + \int_0^T \|u^\epsilon(\tau) - u^\delta(\tau)\|_{H^1(U)}^2 d\tau \\ &\leq \|u^\epsilon(s) - u^\delta(s)\|_{L^2(U)}^2 + \|u^{\epsilon'} - u^{\delta'}\|_{L^2(0,T;H^{-1}(U))}^2 + \|u^\epsilon - u^\delta\|_{L^2(0,T;H^1(U))}^2. \end{aligned}$$

(2). Applying the same steps as in (1), we should eventually find

$$\|u^\epsilon(t)\|_{L^2(U)}^2 = \|u^\epsilon(s)\|_{L^2(U)}^2 + 2 \int_s^t \langle u^{\epsilon'}(\tau), u^\epsilon(\tau) \rangle d\tau \quad \forall (s, t) \text{ s.t. } 0 \leq s, t \leq T.$$

As in (1), since $u \in L^2(0, T; H_0^1(U))$, we find that $u \in L^2(0, T; L^2(U))$, by def. of mollification (2) so that $\|u^\epsilon(t)\|_{L^2(U)} \rightarrow \|u(t)\|_{L^2(U)}$ for a.e. $0 \leq t \leq T$ as $\epsilon \rightarrow 0$. Now we shall claim that the following is true:

$$\lim_{\epsilon \rightarrow 0} \int_s^t \langle u^{\epsilon'}(\tau), u^\epsilon(\tau) \rangle d\tau = \int_s^t \langle u'(\tau), u(\tau) \rangle d\tau.$$

Observe, on the right hand side, that

$$\begin{aligned} &\left| \int_s^t \langle u^{\epsilon'}(\tau), u^\epsilon(\tau) \rangle d\tau - \int_s^t \langle u'(\tau), u(\tau) \rangle d\tau \right| \\ &\leq \left| \int_s^t \langle u^{\epsilon'}(\tau), u^\epsilon(\tau) \rangle d\tau - \int_s^t \langle u^{\epsilon'}(\tau), u(\tau) \rangle d\tau \right| + \left| \int_s^t \langle u^{\epsilon'}(\tau), u(\tau) \rangle d\tau - \int_s^t \langle u'(\tau), u(\tau) \rangle d\tau \right| \\ &= \left| \int_s^t \langle u^{\epsilon'}(\tau), u^\epsilon(\tau) - u(\tau) \rangle d\tau \right| + \left| \int_s^t \langle u^{\epsilon'}(\tau) - u'(\tau), u(\tau) \rangle d\tau \right| \\ &\leq \int_0^T \|u^{\epsilon'}(\tau)\|_{H^{-1}(U)} \|u^\epsilon(\tau) - u(\tau)\|_{H^1(U)} d\tau + \int_0^T \|u^{\epsilon'}(\tau) - u'(\tau)\|_{H^{-1}(U)} \|u(\tau)\|_{H^1(U)} d\tau \\ &:= I_1 + I_2. \end{aligned}$$

For I_1 , notice that

$$\begin{aligned} I_1 &\leq \left(\int_0^T \|u^{\epsilon'}(\tau)\|_{H^{-1}(U)}^2 d\tau \right)^{1/2} \left(\int_0^T \|u^{\epsilon}(\tau) - u(\tau)\|_{H^1(U)}^2 d\tau \right)^{1/2} \\ &= \|u^{\epsilon'}\|_{L^2(0,T;H^{-1}(U))} \|u^{\epsilon} - u\|_{L^2(0,T;H_0^1(U))} \end{aligned}$$

by Hölder's inequality. However, we have from (1) that $\|u^{\epsilon} - u\|_{L^2(0,T;H_0^1(U))} \rightarrow 0$ as $\epsilon \rightarrow 0$, so $I_1 \rightarrow 0$ also. For I_2 , we apply simply the same methods as for I_1 to find that $I_2 \rightarrow 0$ as well since $\|u^{\epsilon'} - u'\|_{L^2(0,T;H^{-1}(U))} \rightarrow 0$.

Now, take $\epsilon \rightarrow 0$ in (*). Consequently, we find that

$$\|u(t)\|_{L^2(U)}^2 = \|u(s)\|_{L^2(U)}^2 + 2 \int_s^t \langle u'(\tau), u(\tau) \rangle d\tau \quad \text{for a.e. } 0 \leq s, t \leq T.$$

If we identify u with its continuous version, then this holds for all $0 \leq s, t \leq T$. Note also that

$$\int_0^T \langle u'(\tau), u(\tau) \rangle d\tau \leq \int_0^T \|u'(\tau)\|_{H^{-1}(U)} \|u(\tau)\|_{H^1(U)} d\tau < \infty \quad \text{so that } \langle u'(\tau), u(\tau) \rangle \in L^1(0, T).$$

Thus, by Absolute Continuity, we have that $\|u(t)\|_{L^2(U)}^2$ is absolutely continuous. Differentiating $\|u(t)\|_{L^2(U)}^2$ with respect to t implies the proof of (ii).

(3). To show (iii), we integrate $\|u(t)\|_{L^2(U)}^2$ for $0 \leq s \leq T$.

$$T\|u(t)\|_{L^2(U)}^2 = \int_0^t \|u(s)\|_{L^2(U)}^2 ds + 2 \int_0^T \int_s^t \langle u'(\tau), u(\tau) \rangle d\tau ds.$$

Cauchy's inequality with $\epsilon = 1/2$ shows that

$$\begin{aligned} T\|u(t)\|_{L^2(U)}^2 &\leq \|u\|_{L^2(0,T;L^2(U))}^2 + 2 \int_0^T \int_0^t \|u'(\tau)\|_{H^{-1}(U)} \|u(\tau)\|_{H^1(U)} d\tau ds \\ &\leq \|u\|_{L^2(0,T;L^2(U))}^2 + T \int_0^T \left[\|u'(\tau)\|_{H^{-1}(U)}^2 + \|u(\tau)\|_{H^1(U)}^2 \right] d\tau \\ &= \|u\|_{L^2(0,T;L^2(U))}^2 + T \left(\|u'\|_{L^2(0,T;H^{-1}(U))}^2 + \|u\|_{L^2(0,T;H_0^1(U))}^2 \right) \end{aligned}$$

which implies (iii). □

4.1 Second Order Parabolic Equations

4.1.1 Parabolic Equations

In studying these PDEs, we shall assume that $U \subset \mathbb{R}^n$ is open and bounded, and we set

$$U_T = U \times (0, T] \quad \text{for } T > 0.$$

We consider following the initial/boundary value problem (IBVP):

Given $f : U_T \rightarrow \mathbb{R}$, $g : U \rightarrow \mathbb{R}$, find $u : \bar{U}_T \rightarrow \mathbb{R}$ such that

$$\begin{cases} u_t + Lu = f, & \text{in } U_T \\ u = 0, & \text{on } \partial U \times [0, T] \\ u = g, & \text{on } U \times \{t = 0\} \end{cases}$$

where

$$Lu := - \sum_{i,j=1}^n (a^{ij}(x, t) u_{x_i})_{x_j} + \sum_{i=1}^n b^i(x, t) u_{x_i} + c(x, t) u.$$

Definition 4.1.1. $\partial_t + L$ is **uniformly parabolic** if $\exists \theta > 0$ (constant) such that

$$\sum_{i,j=1}^n a^{ij}(x, t) \xi_i \xi_j \geq \theta |\xi|^2 \quad \forall (x, t) \in U_T, \xi \in \mathbb{R}^n.$$

Example 4.1. Consider $a^{ij} = \delta_{ij}$, $b^i = c = f = 0$. Then, we arrive at $\partial_t u - \Delta u = 0$ which is the heat equation.

4.1.2 Weak Solutions

Our parabolic assumptions are the same as those we made in the previous subsection. Additionally, we shall assume that $\partial_t + L$ is always uniformly parabolic. Further, $a^{ij}, b^i, c \in L^\infty(U_T)$, $a^{ij} = a^{ji}$, $f \in L^2(U_T)$ and $g \in L^2(U)$.

Notation. We use $(\cdot, \cdot) = (\cdot, \cdot)_{L^2(U)}$ to implicitly assume the L^2 inner product. Similarly, we use $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_{H^{-1}(U), H_0^1(U)}$ denotes the H_0^1 duality pairing.

Weak Formulation. Assume u is a smooth solution to the IBVP. Then, multiplying the PDE by a test function $v \in C_c^\infty(U)$ and integrating by parts on U shows that

$$(u_t, v) + B[u, v; t] = (f, v) \quad \text{for } 0 \leq t \leq T$$

where

$$B[u, v; t] = \int_U \left[\sum_{i,j=1}^n a^{ij}(x, t) u_{x_i} v_{x_j} + \sum_{i=1}^n b^i(x, t) u_{x_i} v + c(x, t) uv \right] dx.$$

We wish to weaken the assumptions on the function space. However, to do so, we will not consider u as a function of x and t , but rather as a map $t \rightarrow u(\cdot, t)$ with $u(\cdot, t) \in H_0^1(U)$ for $0 \leq t \leq T$. We write this map as:

$$\mathbf{u} : [0, T] \rightarrow H_0^1(U) \text{ defined by } [\mathbf{u}(t)](x) := u(x, t) \quad \text{for } x \in U, 0 \leq t \leq T.$$

Likewise, we have also

$$\mathbf{f} : [0, T] \rightarrow L^2(U) \text{ is defined by } [\mathbf{f}(t)](x) := f(x, t) \quad \text{for } x \in U, 0 \leq t \leq T.$$

Observe that assuming $f \in L^2(U_T)$, we have

$$\|\mathbf{f}\|_{L^2(0,T;L^2(U))}^2 = \int_0^T \|\mathbf{f}(t)\|_{L^2(U)}^2 dt = \int_0^T \int_U |f(x, t)|^2 dx dt = \|f\|_{L^2(U_T)}^2 < \infty.$$

We thus can conclude that $\mathbf{f} \in L^2(0, T; L^2(U))$. Now, we write the weak formulation as

$$(\mathbf{u}', v) + B[\mathbf{u}, \mathbf{v}; t] = (\mathbf{f}, v) \quad \forall v \in H_0^1(U), \text{ for a.e. } t \in (0, T).$$

We must ask what space we need for $\mathbf{u}$. We know that $\mathbf{u}(t) \in H_0^1(U)$ for $0 \leq t \leq T$, but we also have to take care of $\mathbf{u}'$. Observe that

$$\begin{aligned} (\mathbf{u}', v) &= -B[\mathbf{u}, \mathbf{v}; t] + (\mathbf{f}, v) \\ &= \int_U \left[\left(\mathbf{f} - \sum_{i=1}^n b^i \mathbf{u}_{x_i} - c\mathbf{u} \right) v + \sum_{j=1}^n \left(- \sum_{i=1}^n a^{ij} \mathbf{u}_{x_i} \right) v_{x_j} \right] dx \\ &:= \mathbf{g}^0(t) + \mathbf{g}^j(t) \end{aligned}$$

where $\mathbf{g}^0(t) \in L^2(U)$ and $\mathbf{g}^j(t) \in L^2(U)$. Indeed, we have that

$$(\mathbf{u}', v) = \int_U \left(\mathbf{g}^0 v + \sum_{j=1}^n \mathbf{g}^j v_{x_j} \right) dx = \langle \mathbf{g}, v \rangle$$

by Characterization of H^{-1} with $\mathbf{g} \in H^{-1}(U)$ and $v \in H_0^1(U)$. Furthermore, observe that

$$\|\mathbf{u}'(t)\|_{H^{-1}(U)} \leq \left(\sum_{j=0}^n \|\mathbf{g}^j\|_{L^2(U)}^2 \right)^{1/2} \leq C \left(\|\mathbf{u}(t)\|_{H^1(U)} + \|\mathbf{f}(t)\|_{H^{-1}(U)} \right).$$

It may be reasonable to look for a weak solution with $\mathbf{u}'(t) \in H^{-1}(U)$ for a.e. $t \in (0, T)$. Then we can write that $(\mathbf{u}'(t), v) = \langle \mathbf{u}'(t), v(t) \rangle$ by Characterization of $H^{-1}(U)$. We now define the weak solution.

Definition 4.1.2. $\mathbf{u} \in L^2(0, T; H_0^1(U))$ with $\mathbf{u}' \in L^2(0, T; H^{-1}(U))$ is a **weak solution** of the IBVP

$$\begin{cases} u_t + Lu = f, & \text{in } U_t \\ u = 0, & \text{on } \partial U \times [0, T] \\ u = f, & \text{on } U \times \{t = 0\}, \end{cases}$$

if

- (i) $\langle \mathbf{u}', v \rangle + B[\mathbf{u}, \mathbf{v}; t] = (\mathbf{f}, v) \quad \forall v \in H_0^1(U)$ and a.e. $0 \leq t \leq T$;
- (ii) the initial condition (IC) is satisfied in $L^2(U)$, i.e. $(\mathbf{u}(0), v) = (g, v) \quad \forall v \in L^2(U)$.

References